# MAT120 : PRACTICE PROBLEM
# DEPARTMENT OF MATHEMATICS AND NATURAL SCIENCES
# BRAC UNIVERSITY

# Integral

## BASIC INTEGRALS

I assume that you know the integrals (1-14) given below:

**1.)** $\int x^n dx = \frac{x^{n+1}}{n+1} + c, n \neq -1$

**2.)** $\int \frac{dx}{x} = \ln |x| + c$

**3.)** $\int e^x dx = e^x + c$

**4.)** $\int a^x dx = \frac{1}{\ln(a)} a^x + c \; a > 0$

**5.)** $\int \cos(x) dx = \sin(x) + c$

**6.)** $\int \sin(x) dx = -\cos(x) + c$

**7.)** $\int \sec^2(x) dx = \tan(x) + c$

**8.)** $\int \sec(x)\tan(x) dx = \sec(x) + c$

**9.)** $\int \csc^2(x) dx = -\cot(x) + c$

**10.)** $\int \csc(x)\cot(x) dx = \csc(x) + c$

**11.)** $\int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1}(x) + c$

**12.)** $\int \frac{1}{1+x^2} dx = \tan^{-1}(x) + c$

**13.)** $\int \cosh(x) dx = \sinh(x) + c$

**14.)** $\int \sinh(x) dx = \cosh(x) + c$

The integrals that follow are also basic, but I don't assume you have them memorized. If there was a question on a test concerning these then I would give a hint.

**15.)** $\int \operatorname{sech}^2(x) dx = \tanh(x) + c$

**16.)** $\int \frac{1}{\sqrt{x^2-1}} dx = \cosh^{-1}(x) + c$

**17.)** $\int \frac{1}{\sqrt{x^2+1}} dx = \sinh^{-1}(x) + c$

**18.)** $\int \frac{1}{1-x^2} dx = \tanh^{-1}(x) + c$

# Product Rule

Students often seem to invent new, but wrong, methods of integration on tests. For example, a very popular mistake goes as follows: $\int fg dx = \int f dx \int g dx$ [ incorrect !!!] This is almost never

true. However, there is a way to deal with products in integrals. This method is known as Integration By Parts (IBP): what follows is the method and its proof,

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx} \Rightarrow u\frac{dv}{dx} = \frac{d}{dx}(uv) - v\frac{du}{dx}$$

Now integrates,

$$\int u\frac{dv}{dx}dx = \int \frac{d}{dx}(uv)dx - \int v\frac{du}{dx}dx$$
$$\therefore \int udv = uv - \int vdu$$

In practice, what makes this difficult is seeing how to choose what should be the " $dv$ " and consequently what the " $u$" "ought to be. This often requires some trial and error before you arrive at a profitable path forward.

**Example 9.5.1(I draw the box to the side to organize mv thoughts)**
**E1**

$$\int xe^x dx = \int udv \quad = xe^x - \int e^x dx \; 4\, u = x \; dv = e^x dx \; du = dx \quad v = e^x$$
$$= xe^x - e^x + C$$

**Example 9.5.2**
**[E2]**
Examples E1 and E2 show that we can remove $x$ by a proper application of IBP.

**Example 9.5.3**
**E3**

$$\int \ln(x)dx \quad = x\ln(x) - \int x\frac{dx}{x}$$
$$= x\ln(x) - x + c$$
$$= x(\ln(x) - 1) + c$$

| $u = \ln(x)$ | $dv = dx$ |
|---|---|
| $du = \frac{dx}{x}$ | $v = x$ |

**Example 9.5.4**
**EU**

$$\int \tan^{-1}(x)dx = \quad x\tan^{-1}(x) - \int \frac{x}{1+x^2}dx = x\tan^{-1}(x) - \frac{1}{2}\int \frac{d\omega}{\omega}$$
$$= x\tan^{-1}(x) = \frac{1}{2}\ln|\omega| + c$$
$$4\begin{matrix} w = 1+x^2 \\ dw = 2xdx \end{matrix}$$
$$= x\tan^{-1}(x) - \frac{1}{2}\ln(1+x^2) + C$$

By now you may be wondering, is there any good way to anticipate what to choose for $u$ in setting up IBP? The answer is, L.I.A.T.E. this works pretty good for most examples.

- listed in order of choosing $u$.
  For many examples (if it appears to be $IBP$) we can choose $u$ by simply going down this list and making $U$ whatever we see first. Let's see how this worked
  E1) $\int xe^x dx$ has $x$ and $e^x$ which are algebraic
  $1 \gg$ no log or inverse
  $\frac{A}{E}$ IS our example hastens of $\Rightarrow u = x$
  [E3) $\int \ln(x)dx$ has a logarithm $\Rightarrow$ (L) $\Rightarrow$ chase $u = \ln(x)$
  A better answer, is to practice and then practice some more. Just as with u-substitution integration technique acreage both skill 39 calculation $Q$ di creative insight.

**Example 9.5.5**
**ES**

$$\int x^3e^x\,dx \quad = x^3e^x - 3\left[x^2e^x - 2\left(xe^x - \int e^x dx\right)\right] + \begin{matrix} u = x & dV = e^x dx \\ du = dx & v = e^x \end{matrix}$$
$$= x^3e^x - 3[x^2e^x - 2xe^x + 2e^x] + c$$
$$= x^3e^x - 3x^2e^x + 6xe^x - 6e^x + c$$
$$= e^x(x^3 - 3x^2 + 6x - 6) + c$$

I hope you can see how to do $\int' x^4e^x dx$ and so forth, it's just a matter of patience and persistence. People tend to find the next example a little unsettling upon first exposure.

**Example 9.5.6**

- $\int e^x(\cos(x)\,dx = e^x(\cos(x) - \sin(x)) - \int e^x\cos(x)dx$ : weave come full-circle
  $\Rightarrow 2\int e^x\cos(x)dx = e^x(\cos(x) - \sin(x))$
  $\Rightarrow \int e^x\cos(x)dx = \frac{1}{2}e^x(\cos(x) - \sin(x)) + c$

**Remark:** There is another way to calculate this if we allow ourselves the luxury of some complex variable calculus. Technically speaking, I probably should lay a little more ground work before showing you the calculation below, but you'll forgive me I think,

$$\begin{aligned}
&= \int e^x \frac{1}{2}\left(e^{ix} + e^{-ix}\right)dx \\
&= \frac{1}{2}\int \left(e^{(1+i)x} + e^{(1-i)x}\right)dx \\
&= \frac{1}{2}\left(\frac{1}{1+i}e^{(1+i)x} + \frac{1}{1-i}e^{(1-i)x}\right) + c \\
&= \frac{1}{2}\left(\frac{1-i}{1-i^2}e^{(1+i)x} + \frac{1+i}{1-i^2}e^{(1-i)x}\right) + c \\
&= \frac{1}{2}\left(\frac{1}{2}e^{(1+i)x} + \frac{1}{2}e^{(1-i)x}\right) + \frac{1}{2}\left(\frac{-i}{2}e^{(1+i)x} + \frac{i}{2}e^{(1-i)x}\right) + c \\
&= \frac{1}{2}\left(\frac{1}{2}\left(e^{ix} + e^{-ix}\right) + \frac{1}{2i}\left(e^{ix} - e^{-ix}\right)\right)e^x + c \\
&= \frac{1}{2}(\cos(x) + \sin(x))e^x + c
\end{aligned}$$

I made some major assumptions in the calculation above; I assumed that the integral of the complex exponential works according to the same calculus pattern as the real exponential. This is in fact true, and I will justify this in part in a later section. In your complex variables course you will learn the complete background to justify such integrations. In fact, the integration I just completed is not that exotic, no deep theorem of complex variables is required because the integrand is analytic everywhere, it has no poles in the complex plane.

**Example 9.5.7**

$$\begin{aligned}
&\int \sin^{-1}(x)\mathrm{dx} = \sin^{-1}(x) + \frac{1}{2}\int \frac{du}{\sqrt{u}} + \quad u = 1 - x^2 \\
&= \sin^{21}(x) + \sqrt{u} + c \qquad du = -2xdx \therefore \frac{du}{2} = -xdx \\
&= \sin^{-1}(x) + \sqrt{1 - x^2} + c
\end{aligned}$$

**Example 9.5.8**

$$\begin{aligned}
I &= \int cos(ln(x))dx = x(\cos\ln(x)) + \sin(\ln(x))) - I \\
\therefore I &= \frac{1}{2} \times (\cos(\ln(x)) + \sin(\ln(x)) + C
\end{aligned}$$

**Example 9.5.9**
Some people prefer these arguments over those which are based on trigonometric identities. Personally, I prefer the arguments we made in 9.3 .5 since they are a bit more direct. Also, I see a need for students to learn more complex variables early on because: 1.) it's not really that hard, 2.) they're really neat, 3.) there is a silly but undeniable bias against complex numbers simply because of their label "complex", this bias must be confronted since it may cause the student to ignore or dismiss the best solution to many problems,

$$\text{Er)} \int \cos^n(x)dx = \int \cos^{n-1}(x)\cos(x)dx$$

$$= \cos^{n-1}(x)\sin(x) - \int \sin(x)(n-1)\cos^n(x)(-\sin(x)) \qquad \begin{matrix} du = (n-1)\cos^{n-2}(x)(-\sin(x)) \\ dv = \cos(x)dx \\ v = \sin(x) \end{matrix}$$

$$= \cos^{n-1}(x)\sin(x) + (n-1)\int \sin^2(x)\cos^{n-1}(x)dx$$

$$= \cos^{n-1}(x)\sin(x) + (n-1)\int (1-\cos^2(x))\cos^{n-2}(x)dx$$

$$= \cos^{n-1}(x)\sin(x) + (n-1)\left(\int \cos^{n-2}(x)dx + \int \cos^n(x)dx\right)$$

$\therefore \int \cos^n(x)dx = \frac{1}{n}\cos^{n-1}(x)\sin(x) + \frac{n-1}{n}\int \cos^{n-2}(x)dx$ solving tod Let $n = 2$ then we find:

$$\int \cos^2(x)dx = \frac{1}{2}\cos(x)\sin(x) + \frac{1}{2}\int dx = \frac{1}{2}(\cos(x)\sin(x) + x) + C$$

Let $n = 3$ then:

$$\begin{aligned} \int \cos^2(x)dx &= \frac{1}{3}\cos^2(x)\sin(x) + \frac{2}{3}\int \cos(x)dx \\ &= \frac{1}{3}\cos^2(x)\sin(x) + \frac{2}{3}\sin(x) + C \\ &= \frac{1}{3}\sin(x)(\cos^2(x) + 1) + C \end{aligned}$$

**Remark:** this is the other popular anted to compute integrals of powers of cosine, there is also a similar "recurrence" relation for $\sin\ \theta$.

(Actually, this is problem 44a-b on page 494 of the version of Stewart we are using at the moment)

# Indefinite Integrals

1 Evaluate each of the following indefinite integrals.
(a) $\int 10x^9 - 12x^3 - 5dx$
(b) $\int 10x^9 - 12x^3 dx - 5$

2 Evaluate each of the following indefinite integrals.
(a) $\int t^7 + 33t^2 + 8tdt$
(b) $\int t^7 dt + 33t^2 + 8t$

3 Evaluate each of the following indefinite integrals.
(a) $\int 6x^5 - 7x^3 + 12x^2 - 10dx$
(b) $\int 6x^5 - 7x^3 dx + 12x^2 - 10$
(c) $\int 6x^5 dx - 7x^3 + 12x^2 - 10$

4 Evaluate each of the following indefinite integrals.
(a) $\int 21x^6 - 9x^5 - x^3 - xdx$
(b) $\int 21x^6 - 9x^5 - x^3 dx - x$
(c) $\int 21x^6 - 9x^5 dx - x^3 - x$

For problems 5 - 9 evaluate the indefinite integral.

5 $\int 8t^5 - 15t^2 - 1dt$

6 $\int 120y^9 - 24y^5 - 4y^3 dy$

7 $\int dw$

8 $\int x^9 + 14x^6 - 10x^3 + 13x\, dx$

9 $\int 8\mathrm{x}^6 - \mathrm{x}^4 - 7\mathrm{x}^2 + 11\mathrm{x} - 12\ \mathrm{dx}$

10 Determine f (x) given that $f'(x) = 16\mathrm{x}^4 - 9\mathrm{x}^2 - \mathrm{x}$

11 Determine g (t) given that $g'(t) = 4\mathrm{t}^5 + 16\mathrm{t}^2 - 18\mathrm{t} + 72$

12 Determine R (z) given that $R'(z) = 4\mathrm{z}^{15} + 121\mathrm{z}^{10} + 20\mathrm{z}^5 + \mathrm{z} - 4$

13 Determine f (x) given that $f''(x) = 8\mathrm{x}^3 - 12\mathrm{x} + 3$

# Computing Indefinite Integrals

For problems 1 - 43 evaluate the given integral.

1 $\int (7x^5 - 5x^4 + 6x^2 - 14x + 3)dx$

2 $\int (t^4 - 9t^3 + 12t^2 - 7t)dt$

3 $\int (4 - 18w^{11} - 9w^9 + 8w^7 + 2w^5)dw$

4 $\int (x^9 - 6x^4 - 21x^2 - 1 + 9x)\, dx$

5 $\int -7dz$

6 $\int 4dw$

7 $\int (10z^{-6} + 8z^{-5} - z^{-2} + 1)dz$

8 $\int (y^{-16} + 24y^{-12} - 14y^{-8} - 2y^{-4})dy$

9 $\int (2x^{-9} + 12x^{-5} + 7x^{-3} - x^{-2})dx$

10 $\int (5z^{-4} + 5z^4 - 9)dz$

11 $\int (6t^3 + 8t^{-6} + t^{-10})dt$

12 $\int (x^{-3} + 9x^2 + 11x^8 - 7x^{-12})dx$

13 $\int (\sqrt[7]{w^2} + 3 - 9\sqrt[3]{w^7})dw$

14 $\int (w^5 + \sqrt{w^5} - \sqrt[5]{w})dw$

15 $\int (6\sqrt[3]{v^2} - 7\sqrt[4]{v})dv$

16 $\int (\frac{6}{y^3} - \frac{1}{7y^6} + \frac{1}{y^2})dy$

17 $\int (8 + u^5 - \frac{1}{u^5} + \frac{1}{6u^5})du$

18 $\int (\frac{12}{x^5} + \frac{1}{4x^8} + \frac{6}{7x^2})dx$

19 $\int (\sqrt[3]{t^5} - \frac{1}{\sqrt{t^9}} + t^4)dt$

20 $\int (\frac{2}{z^6} - \frac{1}{5\sqrt[7]{z^8}} + 9)dz$

21 $\int (x^3 + \frac{1}{x^3} - \sqrt{x^3})dx$

22 $\int x^6(1 - 4x^2 + x^3)dx$

23 $\int (6 - 2u)^2 du$

24 $\int \{2 - (3 + y)(4 - y^3)\}dy$

25 $\int \sqrt{w}(\sqrt[3]{w} - \sqrt[4]{w})dw$

26 $\int 3v\left(v^2 - \frac{1}{6v^2} + \sqrt[3]{v^2}\right)dv$

27 $\int \frac{8x^5 - 2x^3 + 7}{x^2}dx$

28 $\int \frac{9 - z + 2z^4 + 10z^6}{z^4}dz$

29 $\int \frac{2\sqrt{t} - 4t + \sqrt[3]{t}}{t^2}dt$

30 $\int \frac{(1-x)(2+x)}{x}dx$

31 $\int 6\sin(t) - 2\cos(t)dt$

32 $\int \sec^2(u) + 7\sec(u)\tan(u)du$

33 $\int \csc^2(y) - \sec^2(y)dy$

34 $\int 8\cos(z) - 3\csc(z)\cot(z)dz$

35 $\int \tan(x)[\cot(x) - \cos(x)]dx$

36 $\int \frac{\cos^3(v) + \sin(v)}{\cos^2(v)}dv$

37 $\int (w^2 + 2\mathbf{e}^w)dw$

38 $\int (\mathbf{e}^t + \frac{2}{t})dt$

39 $\int \frac{14}{x} - \frac{3}{x^2}dx$

40 $\int \mathbf{e}^{-u}(\mathbf{e}^{2u} + \mathbf{e}^u)du$

41 $\int \frac{1}{7z} + \frac{1}{\mathbf{e}^{-z}} + \frac{1}{4z^8}dz$

42 $\int 1 + w^2 - \frac{6}{1+w^2}dw$

43 $\int \frac{5}{1+t^2} + \frac{1}{10\sqrt{1-t^2}}dt$

44 Determine $f(x)$ given that $f'(x) = 12x^5 + 30x^2$ and $f(4) = -23$.

45 Determine $h(z)$ given that $h'(z) = 12z^3 - 14z^2 + 10$ and $h(-1) = 8$.

46 Determine $g(v)$ given that $g'(v) = \frac{1}{2}v^{-\frac{1}{2}} - \frac{1}{4}v^{-\frac{3}{4}}$ and $g(16) = 1$.

47 Determine $P(t)$ given that $P'(t) = 6\mathbf{e}^t - 4 - 10t$ and $P(0) = -6$.

48 Determine $g(x)$ given that $g''(x) = 12x^2 - 30x + 4, g(-1) = 7$ and $g(2) = 3$.

49 Determine $f(u)$ given that $f''(u) = 60u^4 - 60u^2, f(-1) = 14$ and $f'(1) = 6$.

50 Determine $h(t)$ given that $h''(t) = 6t - 14 + 9\mathbf{e}^t, h(0) = 4$ and $h(3) = 9\mathbf{e}^3 + 8$.

# Substitution Rule for Indefinite Integrals

For problems $1 - 31$ evaluate the given integral.

1 $\int 12v(7 + 6v^2)^9 dv$

2 $\int (4x^3 - 12x)(x^4 - 6x^2)^{-3}dx$

3 $\int (z^2 - 4)(12z - z^3)^4 dz$

4 $\int 7z^2(14 + 8z^3)^{-5}dz$

5 $\int 3(y^6 - 4y^{-3})(y^7 + 14y^{-2} - 7)^6 dy$

6 $\int \left(\frac{1}{2}x^3 - 1\right)\sqrt{8x - x^4}dx$

7 $\int (6w^{-4} + 12w^{-7})\sqrt[4]{w^{-3} + w^{-6}}dw$

8 $\int \cos(7t)dt$

9 $\int (v - 2v^3)\cos(v^2 - v^4)dv$

10 $\int \sqrt{z}\sin\left(1 + \sqrt{z^3}\right)dz$

11 $\int \csc^2(1 + 2x)dx$

12 $\int 7w^{-5}\sec(w^{-4})\tan(w^{-4})dw$

13 $\int (2 - t^2)\mathbf{e}^{6t - t^3}dt$

14 $\int 12z^{-2}\mathbf{e}^{4+z^{-1}}dz$

15 $\int \frac{1}{4-9w}dw$

16 $\int \frac{9y}{y^2+3}dy$

17 $\int \frac{6x^2-10x^4}{x^5-x^3}dx$

18 $\int \frac{1}{t}\sin(1-\ln(t))dt$

19 $\int [6v-18\sin(6v)]\sqrt[5]{v^2+\cos(6v)}dv$

20 $\int \mathbf{e}^{-3z}\sec(\mathbf{e}^{-3z})\tan(\mathbf{e}^{-3z})dz$

21 $\int (\cos(x)+\sin(x))\mathrm{e}^{\sin(x)-\cos(x)}dx$

22 $\int \frac{[\ln(w^2)]^4}{w}dw$

23 $\int \cos(v)\cos(1+\sin(v))dv$

24 $\int \frac{y+\sin(2y)}{y^2-\cos(2y)}dy$

25 $\int \sec^7(t)\tan(t)dt$

26 $\int \mathbf{e}^z\sec^2(\mathbf{e}^z)[1+\tan(\mathbf{e}^z)]^{-3}dz$

27 $\int \frac{7}{1+5x^2}dx$

28 $\int \frac{2}{3+4t^2}dt$

29 $\int \frac{1}{\sqrt{16-y^2}}dy$

30 $\int \frac{3}{\sqrt{7-4v^2}}dv$

31 $\int \frac{x}{1+x^4}dx$

32 Evaluate each of the following integrals.
(a) $\int \frac{1}{3+x}dx$
(b) $\int \frac{x}{3+x^2}dx$
(c) $\int \frac{x}{(3+x^2)^7}dx$
(d) $\int \frac{1}{3+x^2}dx$

33 Evaluate each of the following integrals.
(a) $\int \frac{4w}{25+9w^2}dw$
(b) $\int \frac{4w}{(25+9w^2)^3}dw$
(c) $\int \frac{4}{25+9w^2}dw$

# More Substitution Rule

Evaluate each of the following integrals.

1 $\int 3x\cos(4-x^2)-8x\sqrt{4-x^2}dx$

2 $\int \frac{4}{(9+6t)^5}+\frac{13}{9+6t}dt$

3 $\int (6-5w)\mathbf{e}^{12w-5w^2}+(20w-24)\sec^2(12w-5w^2)dw$

4 $\int \frac{\sin(1+\ln(2x))-\sqrt{1+\ln(2x)}}{x}dx$

5 $\int 17(x\mathbf{e}^x+\mathbf{e}^x)\sin(x\mathbf{e}^x)-14\sin(x)dx$

6 $\int \frac{1}{3t}+\sec(9t)\tan(9t)e^{\sec(9t)}dt$

7 $\int 8w^2+\frac{\sin(w)+\cos(w)}{\sin(w)-\cos(w)}dw$

8 $\int 8+(3+x^6)\cos(21x+x^7)+9x^2-4\sqrt{x}dx$

9 $\int \sin(y)\cos(y)\sqrt{3+\sin^2(y)}+5\mathbf{e}^y dy$

10 $\int \sin(2-t)+8\cos(5t)-\mathbf{e}^{3t}dt$

11 $\int \frac{4x^2-1}{\sqrt[4]{6x-8x^3}}+9x\mathrm{e}^{x^2}dx$

12 $\int z^3+\sqrt{4-3z}-4\sec(8z)\tan(8z)dz$

13 $\int \frac{17}{6-w}+\sin(w)\sin[1+\cos(w)]dw$

14 $\int \frac{\sqrt{1+2\ln(7x)}}{x}+\frac{10x^3}{x^4+9}dx$

15 $\int x\sin(x^2)[\cos^4(x^2)+8\cos^2(x^2)-10]dx$

16 $\int \csc\left(\frac{t}{2}\right)\cot\left(\frac{t}{2}\right)\left[\csc^6\left(\frac{t}{2}\right)+3\csc^4\left(\frac{t}{2}\right)-8\csc\left(\frac{t}{2}\right)\right]dt$

17 $\int \frac{3+7y}{y^2+3}dy$

18 $\int \frac{15z+27}{100z^2+11}dz$

19 $\int \frac{8x+1}{\sqrt{16-x^2}}dx$

20 $\int \frac{2-w}{\sqrt{25-2w^2}}dw$

21 $\int \frac{9z^5}{2+3z^3}dz$

22 $\int 4t^{15}\sqrt{1-t^8}dt$

23 $\int \cot(x)dx$

24 $\int \csc(x)dx$

25 $\int \frac{x}{1+x^4}dx$

26 $\int \mathbf{e}^{8t}(4+\mathbf{e}^{4t})^{-3}dt$

27 $\int x^8\sqrt{2-x^3}dx$

# Area Problem

For problems $1-5$ estimate the area of the region between the function and the $x$-axis on the given interval using $n=6$ and using,

(a) the right end points of the subintervals for the height of the rectangles,
(b) the left end points of the subintervals for the height of the rectangles and,
(c) the midpoints of the subintervals for the height of the rectangles.

1 $f(x) = 15 + 4x - x^3$ on $[1,3]$

2 $g(x) = -3x^2 + 2x - 1$ on $[-4,0]$

3 $h(x) = 8\ln(x) - x$ on $[2,6]$

4 $f(x) = \sin^2\left(\frac{x}{2}\right)$ on $[0,3]$

5 $g(x) = \sin(x)\cos(x) - 1$ on $[-2,1]$

For problems $6 - 8$ estimate the net area between the function and the x-axis on the given interval using $n = 8$ and the midpoints of the subintervals for the height of the rectangles. Without looking at a graph of the function on the interval does it appear that more of the area is above or below the x-axis?

6 $h(x) = 8x - \sqrt{x+4}$ on $[-3,2]$

7 $g(x) = 5 + x - x^2$ on $[0,4]$

8 $f(x) = x\mathbf{e}^{-x^2}$ on $[-1,1]$

# Definition of the Definite Integral

For problems 1 - 4 use the definition of the definite integral to evaluate the integral. Use the right end point of each interval for $x_i^*$.

1 $\int_{-2}^{1} 7 - 4x dx$

2 $\int_{0}^{2} 3x^2 + 4x dx$

3 $\int_{-1}^{1} (x-3)^2 dx$

4 $\int_{0}^{3} 8x^3 + 3x - 2 dx$

5 Evaluate: $\int_{-123}^{-123} \cos^6(2x) - \sin^8(4x) dx$

For problems $6 - 8$ determine the value of the given integral given that $\int_{-2}^{5} f(x)dx = 1$ and $\int_{-2}^{5} g(x)dx = 8$

6 $\int_{-2}^{5} -3g(x)dx$

7 $\int_{-2}^{5} 7f(x) - \frac{1}{4}g(x)dx$

8 $\int_{5}^{-2} 12g(x) - 3f(x)dx$

9 Determine the value of $\int_7^{-1} f(x)dx$ given that $\int_{13}^{7} f(x)dx = -9$ and $\int_{13}^{-1} f(x)dx = -12$.

10 Determine the value of $\int_0^6 4f(x)dx$ given that $\int_0^5 f(x)dx = 10$ and $\int_5^6 f(x)dx = 3$.

11 Determine the value of $\int_2^{10} f(x)dx$ given that $\int_2^4 f(x)dx = -1, \int_4^7 f(x)dx = 3$ and $\int_{10}^{7} f(x)dx = -8$ 12. Determine the value of $\int_{-5}^{-1} f(x)dx$ given that $\int_2^{-5} f(x)dx = 56, \int_7^2 f(x)dx = -90$ and $\int_{-1}^{7} f(x)dx = 45$

For problems 13 - 17 sketch the graph of the integrand and use the area interpretation of the definite integral to determine the value of the integral.

13 $\int_{-2}^{1} 12 - 5x dx$

14 $\int_0^4 \sqrt{16 - x^2} dx$

15 $\int_{-3}^{3} 5 - \sqrt{9 - x^2} dx$

16 $\int_{-1}^{3} 8x - 3 dx$

17 $\int_1^6 |x - 3| dx$

For problems 18 - 23 differentiate each of the following integrals with respect to x.

18 $\int_{-8}^{x} \mathbf{e}^{\cos(t)} dt$

19 $\int_2^{x^2} \sqrt{\cos(t) + 3} dt$

20 $\int_0^{\mathbf{e}^{3x}} \frac{1}{t^4 + t^2 + 1} dt$

21 $\int_{\sin(9x)}^{8} \frac{\mathbf{e}^t}{7t} dt$

22 $\int_{x^3}^{x} \cos^4(t) - \sin^2(t) dt$

23 $\int_{9x}^{\tan(x)} \frac{\cos(t) + 2}{\sin(t) + 4} dt$

24 Evaluate the limit : $\lim_{x \to 0} \frac{\int_0^x \mathbf{e}^{t^2} dt}{x}$

# Computing Definite Integrals

1 Evaluate each of the following integrals.
(a) $\int 3z^2 - 4 + \frac{4}{z^2} dz$

(b) $\int_1^4 3z^2 - 4 + \frac{4}{z^2}\,dz$

(c) $\int_{-2}^1 3z^2 - 4 + \frac{4}{z^2}\,dz$

2 Evaluate each of the following integrals.

(a) $\int 6x + \frac{1}{3x}\,dx$

(b) $\int_0^7 6x + \frac{1}{3x}\,dx$

(c) $\int_3^7 6x + \frac{1}{3x}\,dx$

3 Evaluate each of the following integrals.

(a) $\int \sin(y) + \sec^2(y)dy$

(b) $\int_0^{\frac{\pi}{4}} \sin(y) + \sec^2(y)dy$

(c) $\int_0^{\frac{2\pi}{3}} \sin(y) + \sec^2(y)dy$

Evaluate each of the following integrals, if possible. If it is not possible clearly explain why it is not possible to evaluate the integral.

4 $\int_0^3 10t - 6t^2 + 9dt$

5 $\int_{-1}^4 24z^2 + 5z^4 dz$

6 $\int_1^0 9w - 3w^2 + 4w^3 dw$

7 $\int_{-3}^{-1} 15t^2 - 10t - 2dt$

8 $\int_{-2}^4 v^3 - 7v^2 + 3vdv$

9 $\int_0^{16} 9\sqrt{x} + 10\sqrt[4]{x}dx$

10 $\int_{-1}^2 8\sqrt[3]{z} - 12\sqrt[5]{z}dz$

11 $\int_1^4 \sqrt{y^5} - \frac{1}{\sqrt[3]{y}}\,dy$

12 $\int_1^4 \frac{6}{x^3} - \frac{1}{3x^2}\,dx$

13 $\int_6^{-3} 8w^3 - 25w^4 + \frac{4}{3w^5}\,dw$

14 $\int_{-1}^{-3} \frac{4}{3z^2} - \frac{6}{z^3}\,dz$

15 $\int_0^6 (3-t)(2t^2+3)dt$

16 $\int_4^1 \sqrt{x}(x - 2x^2 + 1)dx$

17 $\int_2^5 \frac{6z^5-8z^4+2z^2}{z^4}\,dz$

18 $\int_{-2}^{-4} \frac{9x^4-8x^3+x}{3x^2}\,dx$

19 $\int_{-8}^{2} \frac{7v^{10}+4v^6-3v^2}{v^5}\,dv$

20 $\int_1^2 \frac{(y-2)(y+2)}{y^2}\,dy$

21 $\int_0^{\frac{\pi}{4}} 8\sec^2(t) + 2\sec(t)\tan(t)dt$

22 $\int_{-\frac{\pi}{3}}^{\frac{\pi}{6}} 3\cos(w) + \sin(w)dw$

23 $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 12\sec^2(y) - 9\csc^2(y)dy$

24 $\int_{\frac{2\pi}{3}}^{\frac{\pi}{4}} 3\sin(v) + 8\csc(v)\cot(v)dv$

25 $\int_{-3}^{1} 4x - 7\mathbf{e}^x dx$

26 $\int_{-2}^{1} \frac{4\mathbf{e}^{2w}+4w\mathbf{e}^w}{\mathbf{e}^w}\,dw$

27 $\int_0^{\frac{1}{2}} \frac{3}{\sqrt{1-x^2}} + \frac{7}{x^2+1}\,dx$

28 $\int_{-2}^{3} 5\sin(t) + \frac{1}{\sqrt{1-t^2}}\,dt$

29 $\int_6^{10} \frac{4}{z} + \frac{1}{2z^2}\,dz$

30 $\int_1^6 2x^3 + \frac{3}{8x}\,dx$

31 $\int_{-4}^{-1} f(t)dt$ where $f(t) = \begin{cases} 9+6t^2 & t > -3 \\ 8t & t \le -3 \end{cases}$

32 $\int_{-2}^{4} g(x)dx$ where $g(x) = \begin{cases} 9-2\mathbf{e}^x & x > 0 \\ 8\sin(x) & x \le 0 \end{cases}$

33 $\int_4^9 h(w)dw$ where $h(w) = \begin{cases} 4 & w > 6 \\ 3w+1 & w \le 6 \end{cases}$

34 $\int_{-1}^{7} f(x)dx$ where $f(x) = \begin{cases} 9x^2 & x > 5 \\ -7 & 1 < x \le 5 \\ 3-8x & x \le 1 \end{cases}$

35 $\int_{-3}^{1} |8 + 4x| dx$

36 $\int_{2}^{8} |3v - 12| dv$

37 $\int_{0}^{6} |10 - 2z| dz$

38 $\int_{-3}^{6} |t^2 - 4| dt$

# Substitution Rule for Definite Integrals

**Example 9.2.1:** Consider $\int xe^{x^2} dx$. Choose $u = x^2$ which produces $du = 2xdx \Longrightarrow xdx = \frac{du}{2}$. Thus, the given integral simplifies as follows:

$$\int xe^{x^2} dx = \int e^u \frac{du}{2} = \frac{1}{2} e^u + c = \frac{1}{2} e^{x^2} + c$$

**Example 9.2.2**: Consider $\int \cos(x) \cos(\sin(x)) dx$. Choose $u = \sin(x)$ which yields $du = \cos(x) dx$. Thus the given integral simplifies as follows:

$$\int \cos(x) \cos(\sin(x)) dx = \int \cos(u) du = \sin(u) + c = \sin(\sin(x)) + c$$

**Example 9.2.3**: Consider $\int x\sqrt{x+3} dx$. Choose $u = x + 3$ so that $x = 3 - u$ and $du = dx$. Thus the integral simplifies as follows:

$$\begin{aligned} \int x\sqrt{x+3} dx &= \int (3-u)\sqrt{u} du \\ &= \int \left(3\sqrt{u} - u^{\frac{3}{2}}\right) du \\ &= 2u^{\frac{3}{2}} - \frac{2}{5} u^{\frac{5}{2}} + c \\ &= 2(x+3)^{\frac{3}{2}} - \frac{2}{5}(x+3)^{\frac{5}{2}} + c \end{aligned}$$

**Example 9.2.4:** Consider $\int \sec(x) dx$. Choose $u = \sec(x) + \tan(x)$ so that $\sec(x) dx = \frac{du}{u}$ (this is not really obvious; you should check this is true):

$$\int \sec(x) dx = \int \frac{du}{-} = \ln|u| + c = \ln|\sec(x) + \tan(x)| + c$$

This integral is worth remembering, we'll see it again. The examples given here are not meant to be comprehensive. You might need to go review my calculus I notes if you are rusty on u-substitution.

Evaluate each of the following integrals, if possible. If it is not possible clearly explain why it is not possible to evaluate the integral.

1 $\int_{-2}^{3} \frac{4}{(5+2x)^3}\,dx$

2 $\int_{1}^{0} 10(1-2w^2)\sqrt[4]{7-3w+2w^3}\,dw$

3 $\int_{-1}^{4} (t-2)\mathbf{e}^{t^2-4t}\,dt$

4 $\int_{1}^{6} 7\cos\left(\frac{\pi z}{2}\right)\left(4+\sin\left(\frac{\pi z}{2}\right)\right)^5\,dz$

5 $\int_{0}^{1} \frac{w^3}{6w^4+3}\,dw$

6 $\int_{-1}^{1} x^2\cos(x^3+2) - x^2\mathbf{e}^{x^3+2}\,dx$

7 $\int_{0}^{\frac{\pi}{3}} \frac{4\sin(3t)}{2+\cos(3t)} + \frac{7\sin(3t)}{(2+\cos(3t))^2}\,dt$

8 $\int_{0}^{\pi} \sec^2(y)\sqrt{2+\tan(y)}\,dy$

9 $\int_{1}^{9} \sqrt{x^5} + \frac{\sin(\sqrt{x})}{\sqrt{x}}\,dx$

10 $\int_{0}^{1} \sec^2(w) - \frac{2}{4w^2+1}\,dw$

11 $\int_{3}^{0} \mathbf{e}^{-4t}\sqrt{2+\mathbf{e}^{-4t}} + 8\mathbf{e}^{t}\,dt$

12 $\int_{3}^{7} \frac{9\mathbf{e}^x}{\mathbf{e}^x+4} + \frac{[\ln(2x)]^2}{x}\,dx$

13 $\int_{0}^{\pi} \sin\left(\frac{v}{2}\right)\left[6+3\cos^2\left(\frac{v}{2}\right) - 4\cos^4\left(\frac{v}{2}\right)\right]dv$

14 $\int_{1}^{2} \mathbf{e}^{-t} + 3t\mathbf{e}^{5-t^2}\,dt$

15 $\int_{0}^{6} \frac{8t^3}{2t^4+1} - \frac{7t}{t^2-9}\,dt$

16 $\int_{2}^{6} \sqrt{1+2y} + (4-y)(y^2-8y+5)^4\,dy$

17 $\int_{0}^{1} \mathbf{e}^{2z}\sin(\mathbf{e}^{2z}-1) + \sin(z)\mathbf{e}^{2-\cos(z)}\,dz$

# Integration by Parts

1 $\int 8t\mathbf{e}^{7t}dt$

2 $\int_{\pi}^{2\pi}(1-3x)\sin\left(\frac{1}{2}x\right)dx$

3 $\int_{-1}^{2}w^2\mathbf{e}^{4w}dw$

4 $\int_{1}^{3}(2-x)^2\ln(4x)dx$

5 $\int(6+3z)\cos(1+4z)dz$

6 $\int 2y^2\cos(9y)dy$

7 $\int(3z+z^2)\sin(z)dz$

8 $\int\sqrt{x^3}\ln(\sqrt[3]{x})dx$

9 $\int(2w^2-w)\mathbf{e}^{7w-1}dw$

10 $\int 9t\sec^2(2t)dt$

11 $\int_{0}^{\frac{\pi}{8}}\mathbf{e}^{-x}\sin(4x)dx$

12 $\int 8\tan^{-1}(2y)dy$

13 $\int\mathbf{e}^{6t}\cos(2t)dt$

14 $\int -3\sin^{-1}(10x)dx$

15 $\int\mathbf{e}^{3-z}\sin(2+z)dz$

16 $\int_{-1}^{0}2x^{17}\mathbf{e}^{1+x^9}dx$

17 $\int 9t^{11}\cos(1-t^6)dt$

18 $\int\frac{x^7}{\sqrt{x^4+1}}dx$

19 $\int(5+x^4)\sin\left(\frac{1}{2}x\right)dx$

20 $\int 2z^5\mathbf{e}^{1-z}dz$

21 $\int 2z^5\mathbf{e}^{1-z}dz$

# Integrals Involving Trig Functions

**Example 9.3.5.1:** (all of these use $u=\cos(x)$ )

$$\int \sin^3(x)dx = \int \sin^2(x)\sin(x)dx$$
$$= \int (1-\cos^2(x))\sin(x)dx$$
$$= \int (u^2-1)du$$
$$= \frac{1}{3}\cos^3(x) - \cos(x) + c$$
$$\int \sin^5(x)dx = \int \sin^4(x)\sin(x)dx$$
$$= \int (1-\cos^2(x))^2\sin(x)dx$$
$$= -\int (1-2u^2+u^4)du$$
$$= -\cos(x) + \frac{2}{3}\cos^3(x) - \frac{1}{5}\cos^5(x) + c$$

$$= \int (1-\cos^2(x))^3\sin(x)dx$$

$$\begin{aligned}\int \sin^7(x)dx &= \int \sin^6(x)\sin(x)dx \\ &= \int (1-\cos^2(x))^3\sin(x)dx \\ &= -\int (1-3u^2+3u^4-u^6)du \\ &= -\cos(x) + \frac{3}{4}\cos^3(x) - \frac{3}{5}\cos^5(x) + \frac{1}{7}\mathrm{c}\end{aligned}$$

It should be obvious how to calculate $\int \sin^9(x)dx$ in view of the calculations above. Moreover, I hope you can see how to calculate $\int \cos^3(x)dx$, $\int$ cos**5(x)dx and so forth. For odd powers of cosine a substitution of $u = \sin(x)$ will prove useful.

**Example 9.3.5.2:** I'll use $\cos^2(x) = \frac{1}{2}(1+\cos(2x))$ in what follows:

$$\begin{aligned}\int \cos^2(x)dx &= \int \frac{1}{2}(1+\cos(2x))dx \\ &= \frac{1}{2}\int (1+\cos(2x))dx \\ &= \frac{x}{2} + \frac{1}{4}\sin(2x) + c\end{aligned}$$

The integral of $\cos^4(x)$ follows a similar pattern,

$$\begin{aligned}\int \cos^4(x)dx &= \int \frac{1}{2}(1+\cos(2x))\frac{1}{2}(1+\cos(2x))dx \\ &= \frac{1}{4}\int (1+2\cos(2x)+\cos^2(2x))dx \\ &= \frac{1}{4}\int \left[1+2\cos(2x)+\frac{1}{2}(1+\cos(4x))\right]dx \\ &= \int \left[\frac{3}{8}+\frac{1}{2}\cos(2x)+\frac{1}{4}\cos(4x)\right]dx \\ &= \frac{3x}{8}+\frac{1}{4}\sin(2x)+\frac{1}{16}\sin(4x)+c\end{aligned}$$

If you understand this example then $\int \cos^6(x)dx$ shouldn't be much more trouble. Integrals of $\sin^2(x), \sin^4(x), \sin^6(x)$ follow from very similar calculations.

Essentially the calculations in this subsection follow from the identity $1+\tan^2(x)=\sec^2(x)$ as well as the derivatives $\frac{d(\sec(x))}{dx}=\sec(x)\tan(x)$ and $\frac{d(\tan(x))}{dx}=\sec^2(x)$ which suggest substitutions of $u=\sec(x)$ or $u=\tan(x)$ when the opportunity presents itself.

**Example 9.3.6.1:**

$$\int \tan^2(x)dx = \int (\sec^2(x)-1)dx = \tan(x)-x+c$$

**Example 9.3.6.2:**

$$\begin{aligned}\int \tan^4(x)dx &= \int (\sec^2(x)-1)^2dx \\ &= \int (\sec^4(x)-2\sec^2(x)+1)dx \\ &= \int \big((1+\tan^2(x))\sec^2(x)-2\sec^2(x)+1\big)dx \\ &= x-2\tan(x)+\int (1+u^2)du, \text{ where } u=\tan(x) \text{ so } du=\sec^2(x)dx \\ &= x-\tan(x)+\frac{1}{3}\tan^3(x)+c\end{aligned}$$

**Problems**

1 $\int \cos^5(2t)\sin^2(2t)dt$

2 $\int \cos^3(12x)dx$

3 $\int \cos^2(z)\sin^4(z)dz$

4 $\int_{\frac{\pi}{3}}^{\pi} \sin^5\left(\frac{3}{4}w\right)\cos^6\left(\frac{3}{4}w\right)dw$

5 $\int_0^{\pi} \cos^{11}(5z)\sin^3(5z)dz$

6 $\int \sin^2(7x)dx$

7 $\int_0^{\frac{\pi}{6}} \tan^3(8x)\sec^3(8x)dx$

8 $\int \sec^8\left(\frac{1}{2}t\right)\tan^5\left(\frac{1}{2}t\right)dt$

9 $\int \sec^2(9z)\tan^3(9z)dz$

10 $\int_{\frac{3\pi}{4}}^{\pi} \sec^6(10t)\tan^4(10t)dt$

11 $\int \tan^{12}(2w)\sec^6(2w)dw$

12 $\int \cot^2(3x)\csc^6(3x)dx$

13 $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \csc^3\left(\frac{1}{4}w\right)\cot^3\left(\frac{1}{4}w\right)dw$

14 $\int \csc^4(6w)dw$

15 $\int \csc^{12}(x)\cot^5(x)dx$

16 $\int \cot(x)dx$

17 $\int \cot^3(x)dx$

18 $\int \csc(x)dx$

19 $\int \csc^3(x)dx$

20 $\int_{-2}^{4} \sin(8x)\cos(15x)dx$

21 $\int \cos(2x)\cos(24x)dx$

22 $\int \sin(13z)\sin(9z)dz$

23 $\int \frac{\cos^5(2t)}{\sin^3(2t)}dt$

24 $\int \frac{\sin^3(2-x)}{\cos^2(2-x)}dx$

25 $\int \frac{\sec^6\left(\frac{1}{2}z\right)}{\tan^8\left(\frac{1}{2}z\right)}dz$

26 $\int \frac{\tan^5(x)}{\sec^2(x)}dx$

27 $\int \frac{1+9\cos^5(8w)}{\sin^2(8w)}dw$

28 $\int (3+7\cos^3(x))\sin^2(x)dx$

29 $\int \sin^3(9y)\sec^2(9y)dy$

30 $\int \tan^5(z)\cos^5(z)dz$

31 $\int [\tan(2t)-\sin^3(2t)]\sec^3(2t)dt$

# Trig Substitutions

In principle these are really nothing more than an implicit u-substitution. If you keep the identities and the technique of trig substitution is very natural. The principle aim of trig-substitution is to remove square roots, however the method is useful for a wider class of examples. Let us proceed by example, we'll summarize the cases after we've played a little.

Examples 9.4.1 and 9.4.2: (can you see the typo in E1? Its missing a 2 somewhere)

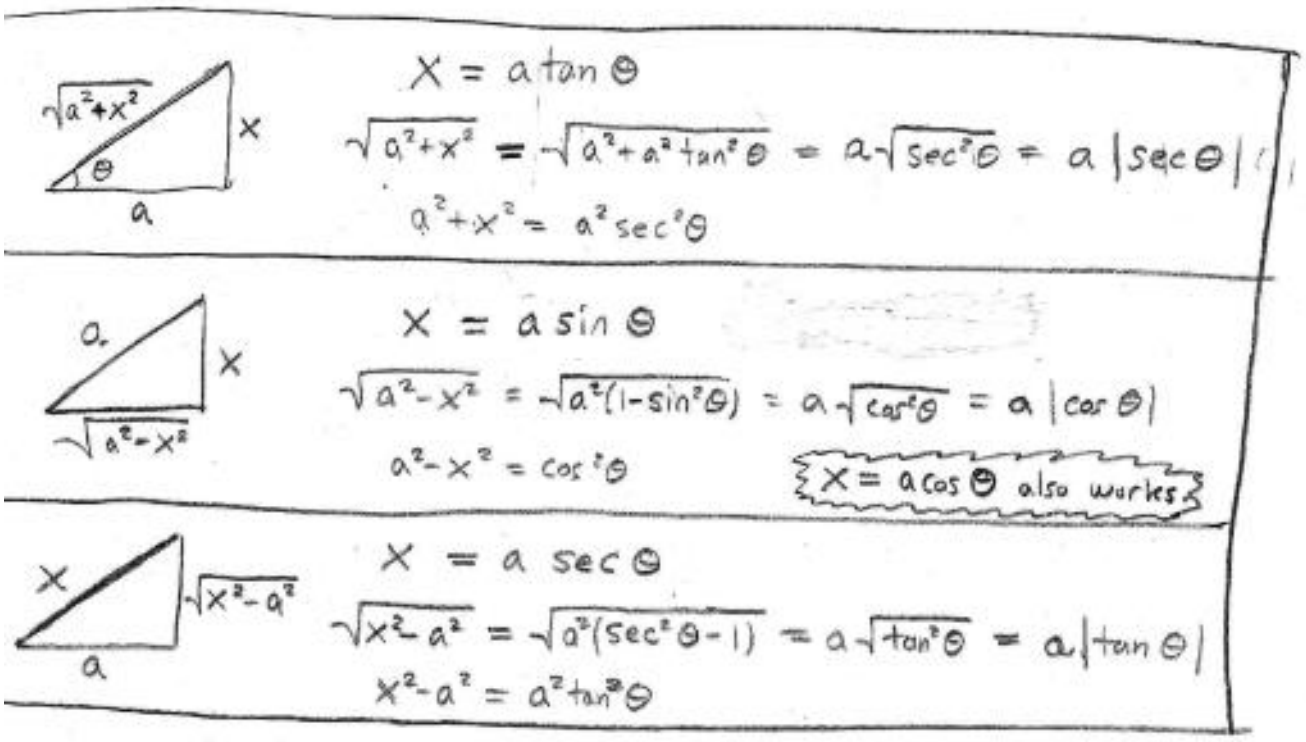
$$\sqrt{16-x^2}$$

**E2**

$$\begin{aligned}\int \frac{\sqrt{9-x^2}}{x^2}dx &= 9\int \frac{\cos\theta\cos\theta d\theta}{9\sin^2\theta} + \left[\begin{matrix} x = 3\sin\theta \\ dx = 3\cos\theta d\theta \\ \sqrt{9-x^2} = \sqrt{9-9\sin^2\theta} = 3\cos\theta \end{matrix}\right. \\ &= \int \cot^2\theta d\theta \\ &= \int (\csc^2\theta - 1)d\theta \\ &= -\cot\theta - \theta + c \\ &= \frac{-3\cos\theta}{3\sin\theta} - \theta + c \\ &= \frac{-\sqrt{9-x^2}}{x} - \sin^{-1}\left(\frac{x}{3}\right) + c\end{aligned}$$

I like to draw the triangle to illustrate the substitution. It also helps reverse the substitution. For example, given $x = 3\sin(\theta)$ it might not have been immediately obvious that $\frac{-3\cos(\theta)}{3\sin(\theta)} = -\cot(\theta) = -\frac{\sqrt{9-x^2}}{x^2}$. That fact should be clear from the triangle. In other cases, the triangle cannot remove the ugliness, like in E1 where $\theta = \sin^{-1}\left(\frac{x}{4}\right)$. I know of no particular way to clean that up.

**Three Patterns we can reduce via Trigonometric Substitution:**




You can use could also use $x = a\cos(\theta)$ in middle case. In each substiution the squareroot is eliminated. We trade an integral with a square root for a new integral of some trigonometric function. We know how to integrate a large variety of trigonometric functions so this is a good bargain for most examples. There are more advanced trigonometric substitutions, but we will focus on just these three basic cases.

**Examples 9.4.3**

[3] Evaluate $\int \frac{dx}{\sqrt{4+x^2}}$. Well try the $/S^2$. subsist.

$$x = 2\tan\ \theta \Rightarrow 4 + x^2 = 4 + 4\tan^2\ \theta = 4(1 + \tan^2\ \theta) = 45$$
$$dx = 2\sec^2\ \theta d\theta \& 4 + x^2 = 4\sec^2\ \theta \text{ hance,}$$
$$\int \frac{dx}{\sqrt{4+x^2}} = \int \frac{2\sec^2\ \theta d\theta}{\sqrt{4\sec^2\ \theta}}$$
$$= \int \sec(\theta) d\theta \cdots \text{ let } u = \sec\ \theta + \tan\ \theta$$

**Examples 9.4.4**

**E4** $\int \frac{x^3 dx}{\sqrt{9-x^2}}$ for $-3 < x < 3$.

$\frac{3}{x}$

$x\sqrt{9-x^2}\ 3\cos\ \theta\ dx = -3\sin\ \theta d\theta$

$9 - x^2 = 9 - 9\cos^2\ \theta = 9\sin^2\ \theta$

$$\int \frac{x^3 dx}{\sqrt{9-x^2}} = \int \frac{(3\cos\ \theta)^3(-3\sin\ \theta d\theta)}{\sqrt{9\sin^2\ \theta}} \text{ since } \frac{\sin\ \theta}{|\sin\ \theta|} = 1 \text{ for } 0 < \theta < 90^\circ$$

$$= -\int 27\cos^3\ \theta d\theta$$

$$= -27\int (1 - \sin^2\ \theta)\cos\ \theta d\theta$$

$$= 27 \cdot (u^2 - 1)du$$

$$= 27\left[\frac{u^3}{3} - u\right] + c$$

$$= 9\sin^3\ \theta - 27\sin\ \theta + C$$

$$= 9\left(\frac{\sqrt{9-x^2}}{3}\right)^3 - 27\left(\frac{\sqrt{9-x^2}}{3}\right) + C$$

$$= -9\sqrt{9-x^2} + \frac{1}{3}(9-x^2)^{3/2} + C$$

Subtle Remark: I used $0 < \theta < 90^\circ \Rightarrow |\sin\ \theta| = \sin\ \theta > 0$ implicitly when I simplified $\sqrt{\sin^2\ \theta} = \sin\ \theta$. In principle you might find $\sqrt{\sin^2\ \theta} = -\sin\ \theta$, but that is not the case here, because $\sin\ \theta > 0$ and square roots are by convention positive. Notice that $0 < \theta < 90^\circ$ follows from $-3 < x = 3\sin\ \theta < 3 \Rightarrow -1 < \sin\ \theta < 1 \Rightarrow 0 < \theta < 90^\circ$. From the beginning I knew that $|x| < 3$ because otherwise the integrate is not real-valued.

**Examples 9.4.5**

**ES** $\int \frac{dx}{\sqrt{25x^2-4}}$ for $x > 2/5$

look's like $\sqrt{x^2 - a^2}$ but not suite yet, *do* algebra on it.

$$\sqrt{25x^2 - 4} = \sqrt{25(x^2 - 4/25)}$$

$$= 5\sqrt{x^2 - (2/5)^2} \text{ suggoots we use } x = \frac{2}{5}\sec\ \theta$$

$$\frac{x}{2/5}\int \sqrt{x^2-(2/6)^2}$$

$$x=\frac{2}{5}\sec(\theta)$$

$$dx=\frac{2}{5}\sec(\theta)\tan\theta d\theta=$$

$$x^2-(2/5)^2=(2/5)^2[\sec^2\theta-1]=\left(\frac{2}{5}\tan\theta\right)^2$$

$$\int \frac{dx}{\sqrt{25x^2-4}}=\frac{1}{5}\int \frac{dx}{\sqrt{x^2-2/5}}$$

$$=\frac{1}{5}\int \frac{\frac{2}{5}\sec(\theta)\tan(\theta)d\theta}{\sqrt{\left(\frac{2}{5}\tan\theta\right)^2}}$$

$$=\frac{1}{5}\int \sec\theta d\theta$$

$$=\frac{1}{5}\ln|\sec\theta+\tan\theta|+C \text{ using } E3$$

$$=\frac{1}{5}\ln\left|\frac{5x}{2}+\frac{\sqrt{x^2-(2/6)^3}}{2/5}\right|+C$$

$$=\frac{1}{5}\ln\left|\frac{5x}{2}+\frac{\sqrt{25x^2-4}}{2}\right|+C$$

I should remind you that last semester when I calculated the area of the circle, I had to use a trigonometric substitution. You might go back and look at Example 7.3.6 of page 166 in my notes. Perhaps the integration I did there will seem easy now.

**Problem**

1 $\sqrt{64t^2+1}$

2 $\sqrt{4z^2-49}$

3 $\sqrt{7-w^2}$

4 $(16-81x^2)^{\frac{7}{2}}$

5 $\sqrt{6+9y^2}$

6 $(1-8z^2)^{\frac{3}{2}}$

7 $\sqrt{9-16(3x-1)^2}$

8 $(11+(t^2+1)^2)^{\frac{5}{2}}$

9 $\sqrt{144(z+8)^2-3}$

10 $\sqrt{4x^2-24x+43}$

11 $(2z^2-24z+36)^{\frac{11}{2}}$

12 $\sqrt{-4-10t-5t^2}$

13 $\sqrt{9\sin^2(4t)-1}$

14 $\sqrt{36-9\mathbf{e}^{3z}}$

15 $\sqrt{x+16}$

16 $\int 3x^5\sqrt{16-x^2}dx$

17 $\int t^3(25+81t^2)^{\frac{5}{2}}dt$

18 $\int_0^{\frac{1}{4}} \frac{w^3}{\sqrt{1-9w^2}}dw$

19 $\int \frac{z^5}{(9z^2-25)^{\frac{3}{2}}}dz$

20 $\int_{-3}^{-1} y^3\sqrt{49y^2-4}dy$

21 $\int_1^5 \frac{5}{x^2\sqrt{x^2+4}}dx$

22 $\int \frac{\sqrt{3-4t^2}}{t^2}dt$

23 $\int \frac{w^5}{\sqrt{8w^2+1}}dw$

24 $\int \frac{\sqrt{x^2-15}}{x^3}dx$

25 $\int \frac{2}{(x-3)^6\sqrt{-x^2+6x-5}}dx$

26 $\int \frac{1}{(z+1)^2(2z^2+4z-34)^{\frac{3}{2}}}dz$

27 $\int \frac{\sqrt{4y^2-16y+19}}{(y-2)^6}dy$

28 $\int_9^{12} \frac{(t-4)^3}{\sqrt{t^2-8t+7}}dt$

29 $\int_0^6 \sqrt{5x^2+10x+6}dx$

30 $\int x^7\sqrt{9-x^4}dx$

31 $\int \frac{\mathbf{e}^{12t}}{\sqrt{4\mathbf{e}^{6t}-1}}dt$

32 $\int \sin(z)\cos^3(z)\sqrt{16+\cos^2(z)}dz$

# Integrating Rational Functions

We can integrate any polynomial, it's easy, just use linearity and the power rule. What about rational functions? Is there some method to integrate an arbitrary rational function? It is just the quotient of two polynomials, how bad can it be? Pretty bad actually, pretty bad. However, the difficulty is not insurmountable.
The key is realizing we can undo the algebraic maneuver of making a common denominator. If I have a product $XY$ in the denominator then it can be split into a sum of a term with denominator X and another term with denominator Y. $I$ call this idea reverse common "denominatoring", but ok that's not really word.

**Example 9.6.0**
EO How to integrate $\frac{x+5}{x^2+5x+6}$ ? Upto now none of the previous techniques seem to help with this one. So, notice that we can break up this fraction into two fractions following from the factorization of $x^2+5x+6=(x+2)(x+3)$,

$$\frac{x+5}{(x+2)(x+3)}=\frac{A}{x+3}+\frac{8}{x+2}-\left(\begin{array}{l}\text{this guess is verified}\\ \text{by algebra below}\end{array}\right.$$

We con figure outs the now unknown $A \notin B$ ar follows, first multiply by denominator to obtain,

$$(x+2)(x+3)\frac{x+5}{(x+2)(x+3)}=(x+2)(x+3)\left[\frac{A}{x+3}+\frac{B}{x+2}\right]$$

Which gives the (simple to solve) eq",

$$x+5=(x+2)A+(x+3)B$$

So plug in the roots $x=-2$ then $x=-3$ to get

$$\begin{aligned}-2+5=3&=(-2+2)A+(-2+3)B=B\ \therefore\ B=3\\ -3+5=2&=(-3+2)A+(-3+3)B=-A\ \therefore A=-2\end{aligned}$$

Thus, we have by the algebra above,

$$\frac{x+5}{(x+2)(x+3)}=\frac{-2}{x+3}+\frac{3}{x+2}$$

Now we can integrate the RHS of the above, remember how?

$$\int \frac{x+5}{(x+2)(x+3)}dx = -2\int \frac{1}{x+3}dx + 3\int \frac{1}{x+2}dx$$
$$= -2\int \frac{du}{u} + 3\int \frac{dw}{w} \quad u = x+3 \quad w = x+2$$
$$= -2\ln|x+3| + 3\ln|x+2| + C$$

**Integrals that motivate the algebra for partial fractions**

You will learn over the course of many examples that to do partial fractions means we rewrite a given rational function as a sum of other rational functions. It is not terribly surprising this alone is possible, what is perhaps surprising is that there are certain types of rational functions which integrate nicely. I call these special rational functions the "basic rational functions". A basic rational function is one which cannot be further reduced into a sum of other basic rational functions, loosely speaking.
We can integrate the basic rational functions: (you'll prove these in one of you homework's, it's a combination of u-substitution and trig-subst.)

**Known Integrals of Basic Rational Functions:**

**1.)** Let $a_n, a_{n-1}, \dots a_1, a_0 \in \mathbb{R}$, then $P(x) = a_n x^n + \cdots + a_2 x^2 + a_1 x + a_0$ is a polynomial. It can be shown

$$\int P(x)dx = \frac{a_n}{n+1}x^{n+1} + \cdots + \frac{a_2}{3}x^3 + \frac{a_1}{2}x^2 + a_0 x + C$$

**2.)** Let $a \in \mathbb{R}$ then $R(x) = \frac{1}{x+a}$ is a basic rational function. It is the reciprocal of a linear factor. It can be shown

$$\int \frac{1}{x+a}dx = \ln|x+a| + C$$

**3.)** Let $a \in \mathbb{R}$ and $n \in \mathbb{N}$ such that $n \geq 2$ then $R(x) = \frac{1}{(x+a)^n}$ is a basic rational function. It is the reciprocal of a repeated linear factor.

$$\int \frac{1}{(x+a)^n}dx = \frac{-1}{n(x+a)^n} + C$$

**4.)** Let $b, c \in \mathbb{R}$ such that $b^2 - 4c < 0$ then $R(x) = \frac{1}{x^2+bx+c}$ is a basic rational function. It is the reciprocal of an irreducible quadratic. Let $\alpha = -b/2$ and $\beta = \sqrt{c - b^2/4}$ then the integral is given as follows:

$$\int \frac{1}{x^2 + bx + c} dx = \frac{1}{\beta} \tan^{-1} \left[\frac{1}{\beta}(x - \alpha)\right] + C$$

**5.)** Let $b, c \in \mathbb{R}$ such that $b^2 - 4c < 0$ then $R(x) = \frac{x}{x^2+bx+c}$ is a basic rational function. It is $x$ over an irreducible quadratic. Let $\alpha = -b/2$ and $\beta = \sqrt{c - b^2/4}$ then the integral is given as follows:

$$\int \frac{x}{x^2 + bx + c} dx = \frac{1}{2} \ln (x^2 + bx + c) - \frac{b}{2\beta} \tan^{-1} \left[\frac{1}{\beta}(x - \alpha)\right] + C$$

**Cases 4.) and 5.)** are considerably simplified in the case $\alpha = 0$. For example,

$$\int \frac{dx}{x^2 + 4} = \frac{1}{2} \tan^{-1} \left(\frac{x}{2}\right) + C \quad \int \frac{x dx}{x^2 + 4} = \frac{1}{2} \ln (x^2 + 4) + C$$

Many examples have this simplified form. It wouldn't be unreasonable to ask you to integrate $\frac{1}{(x^2+t)^2}$, this is a basic rational function. It is not possible to break it down further. The substitution $x = 2\tan(\theta)$ will make that integration work out nicely. Once you understand that example then you (by "you" I actually just mean me in this case) can do the general case for a repeated quadratic in the denominator:

$$\begin{aligned}
\int \frac{1}{(x^2 + bx + c)^2} dx &= \int \frac{1}{[(\alpha + b/)^2 + \alpha]^2} dx \\
&= \int \frac{1}{\left[\alpha^2 \left(\left(\frac{1}{\alpha}(x + b/k)\right)^2 + 1\right)\right]^2} dx \\
&= \int \frac{1}{(\alpha^2)^2 \left[\left(\frac{1}{\alpha}(x + b/k)\right)^2 + 1\right]^2} dx \\
&= \int \frac{1}{\alpha^4 [u^2 + 1]^2} \alpha du : \begin{matrix} u = \frac{1}{\alpha}(x + b/2) \\ du = \frac{1}{\alpha} dx < a dx = \alpha du \end{matrix} \\
&= \frac{1}{\alpha^3} \int \frac{du}{(1 + u^2)^2}
\end{aligned}$$

Let's consider this integral separately for a moment,

$$\int \frac{du}{1+u^2} = \frac{1}{2}\theta + \frac{1}{4}\sin(2\theta) + c$$
$$= \frac{1}{2}\tan^{-1}(u) + \frac{1}{4}\sin(2\tan^{-1}(u)) + C$$

Hence returning to our original portrait,

$$\int \frac{1}{(x^2+bx+c)^2}dx = \frac{1}{\alpha^3}\left\{\frac{1}{2}\tan^{-1}(u) + \frac{1}{4}\sin(2\tan^{-1}(u))\right\} + c$$
$$= \left(c - \frac{b^2}{4}\right)^{-\frac{2}{2}}\left\{\frac{1}{2}\tan^{-1}\left(\frac{1}{\sqrt{c-b/4}}(x+b/2)\right)\right.$$
$$\left. + \frac{1}{4}\sin\left(2\tan^{-1}\left(\frac{1}{\sqrt{c-b/4}}(x+b/2)\right)\right)\right\} + C$$

(If you take a close look you'll see lama using " $\alpha''$ in the place $\beta$ resided previously oops) It is not too hard to see that irreducible quadratics raised to higher powers will also be manageable by an appropriate trig-subst. There is also an iterative formula known to

tackle the integral of $f(x) = (x^2+bx+c)^{-n}$ for $n \in \mathbb{N}$ such that $n > 1$. If you present and derive that iterative formula, I will award 4 bonus points if you do a good job. Ok, enough about all that, by in large we will only need to know 1-5.) and even then, most often we have $\alpha = 0$ in our examples.

**Example 9.6.1( *I'd* call the given function an improper rational function)**
**PARTIAL Fractions:**

**Goal:** given some rational function $f(x) = \frac{P(x)}{Q(x)}$ integrate it.

**Outline:** Long %, Partial Fraction Decompose it, integrate the "Basic" rational functions that result.

**EI)** $f(x) = \frac{x^4+x^7+2x^2+3x-2}{x^2-3x+2}$ ← need to do long%

$$= x^2 - 3x + 2\sqrt{x^4 + 4x + 12}$$
$$= \frac{-(x^4 - 3x^3 + 2x^2)}{4x^3 + 0x^2 + 3x - 2}$$
$$= \frac{-(4x^3 - 12x^2 + 8x)}{12x^2 - 5 - 2}$$
$$= \frac{-(12x^2 - 36x + 24)}{0[31x - 26]} \text{ remainder}$$
$$\therefore f(x) = \underbrace{x^2 + 4x + 12}_{\text{can integrate}} + \frac{31x - 26}{\underbrace{x^2 - 3x + 2}_{\text{weill use portal}}}$$

this no problem fractions to traak it 4.

$\frac{31x - 26}{(x-2)(x-1)} = \frac{A}{x-2} + \frac{B}{x-1}$ \& Partial Fructal Decump. factor.

$\therefore\ 31x - 26 = A(x-1) + B(x-2)$: Malitillying by $(x-2)(x-1)$

$$x = 11\ 5 = -8$$
$$x = 2]36 = A$$
$$f(x) = x^2 + 4x + 12 + \frac{36}{x-2} - \frac{5}{x-1}$$
$$\int f(x)dx = \frac{x^3}{3} + 2x^2 + 12x + 36\ln|x-2| - 5\ln|x-1| + C$$

Long division, thought you'd gotten away from it didn't you. There is a small typo in E1 can you find it?

**Example 9.6.2**

$x = 2111 = 5A + 48 + 2C \Rightarrow 11 = 15 - 4c + 2C \Rightarrow 2C = 4 \Rightarrow c = 2$

i. $A = 3, B = -2, C = 2$

$$\int f(x)dx = \int \frac{3}{x}dx + \int \frac{2}{x^2+1}dx - \int \frac{2x}{x^2+1}dx$$
$$= 3\ln|x| + 2\tan^{-1}(x) - \int \frac{du}{u}$$
$$= 3\ln|x| + 2\tan^{-1}(x) - \ln|x^2+1| + c$$

**Example 9.6.3**

$x = 11 - 1 = 4C: C = -1/4.$
$x = 0 \text{ f } -3 = -A - B + C = -A - 3/4 \Rightarrow -12 = -4A - 3$
$\Rightarrow -9 = -4A$
$\Rightarrow A = 9/4$

$$\int f(x)dx = \frac{9}{4}\int \frac{dx}{x+1} - \frac{1}{4}\int \frac{dx}{x-1} + \frac{1}{2}\int \frac{dx}{(x+1)^2}$$

$$= \frac{9}{4}\int \frac{du}{u} - \frac{1}{4}\int \frac{dw}{w} + \frac{1}{2}\int \frac{du}{u^2} \quad \begin{matrix} u = x+1 & du = dx \\ w = x-1 & dw = dx \end{matrix}$$

$$= \frac{9}{4}\ln|u| - \frac{1}{4}\ln|w| - \frac{1}{2}\frac{1}{u} + c$$

$$= \frac{9}{4}\ln|x+1| - \frac{1}{4}\ln|x-1| - \frac{1}{2}\frac{1}{(x+1)} + C$$

**Remark:** you are probably curious; how do I know how to break it up into pieces? Why is there $A, B$ for one problem then $A, B, C$ for another? What is the algorithm? The examples that follow probably have enough variety for you to see the algorithm.

**Example 9.6.4 and 9.6.5**
**EU**

$$f(x) = \frac{1}{(x+1)(x-2)^2(x^2+1)(x^2+2)^2(x^2+3)^3}$$

$$= \frac{A}{(x+1)} + \frac{B}{(x-2)} + \frac{C}{(x-2)^2} + \frac{Dx+E}{x^2+1} + 2$$
$$C + \frac{Fx+G}{x^2+2} + \frac{Hx+I}{(x^2+2)^2} + \frac{Jx+K}{x^2+3} + \frac{Lx+M}{(x^2+3)^2} + \frac{Nx+O}{(x^2+3)^2}$$

I'm not going to find $A, B, \ldots, N, O$ for you but it should be clear now you would go about o it.
**ES**

$$f(x) = \frac{x^2+2x-7}{(x-3)^5}$$
$$= \frac{A}{(x-3)} + \frac{B}{(x-3)^2} + \frac{C}{(x-3)^3} + \frac{D}{(x-4)^4} + \frac{E}{(x-5)^5}$$

**Example 9.6.6**
Now we can $d \cdot$ partial fractions on $\frac{4x-1}{(x-2)^2} = \frac{A}{x-2} + \frac{B}{(x-2)^2}$

$$
\begin{aligned}
4x-1 &= A(x-2)+B \\
\frac{x}{x}=0\Big)-1 &= -2A+B \\
7 &= B \\
f(x)dx &= \int \left(1+\frac{4}{x-2}+\frac{7}{(x-2)^2}\right)dx \\
&= x+4\int \frac{1}{u}du+7\int \frac{1}{u^2}du\ A=4 \\
&= x+4\ln|u|-7\frac{1}{u}+c \\
&= x+4\ln|x-2|-\frac{7}{x-2}+C
\end{aligned}
$$

I hope you see the idea now. There are additional examples in the suggested homework solutions posted on the course website. Let me sketch the general pattern now that we have a good sampling of examples:

The question is how to integrate any rational function $f(x)=\frac{P(x)}{Q(x)}$ This reduces by polynomial lay division to the problem of integrate $Q(x)$

$$f(x)=S(x)+\frac{R(x)}{Q(x)}$$

Where $R(x)$ is the remainder so $\deg(A(x))<\deg(Q(x))$. So how to integrate $P(x)/Q(x)$ in general? (Note $S(x)$ is easy! if a pronominal we can intaglio those, no problem). Then firm algebra we know that any Polynomial $f(x)$ can be fractured to:
$f(x)=$ (linear factors) (erred. quad. factor)
Hence $R(x)\neq Q(x)$ can be factored likewise, (erred means irreducible)

$$
\begin{aligned}
\frac{R(x)}{Q(x)} &= \frac{(\text{linear } factors)\ (erred.\ \text{quad. } foldout)}{(lin\text{ear factor})\ (erred.\ \text{quad factors})} \\
&= \frac{\frac{A}{x-r_1}+\frac{B}{(x-r_1)^2}+\cdots+\frac{Cx+D}{x^2+6x+c}+\frac{Ex+F}{(x^2+bx+C)^t}+\cdots}{}
\end{aligned}
$$

which and how many of theae "Basic Rational" terms depends on the details of $R(x)\&Q(x)$.

Personally, I think this belongs at the beginning of the section, but my students tell me examples first then the general story, you can thank them for this section being less than logically ordered. In any event, it should be clear we can integrate any rational function via these methods. In practice it would be wise to use Mathematical or a TI-89 for complicated examples. I often use my TI-89 to check my partial fractions decomposition for silly errors. If you are going to buy a calculator the TI-89 is hard to beat, I probably shouldn't tell you, but it can do most everything we learn this semester. Of course, the same is true for Mathematical, but that is just inconvenient

enough to keep you doing your homework in-class

**Exercise 9.6.7: integrate the function below.**

$$f(x) = \frac{1}{x^3 + x}$$

# Partial Fractions

1 $\int \frac{9}{z^2-12z}\,dz$

2 $\int \frac{7x}{x^2+14x+40}\,dx$

3 $\int_0^4 \frac{8y-1}{2y^2-15y-8}\,dy$

4 $\int \frac{9-w^2}{(w+1)(3w-5)(w+4)}\,dw$

5 $\int_1^8 \frac{12}{z^3-2z^2-63z}\,dz$

6 $\int \frac{7x+2x^2}{(x-4)(2x+3)(2x+1)}\,dx$

7 $\int \frac{4x+10}{(x-2)(x-1)^2}\,dx$

8 $\int_1^2 \frac{24}{t^4-6t^3}\,dt$

9 $\int \frac{10z+2}{(z+1)^2(z-3)^2}\,dz$

10 $\int \frac{8w+w^2}{(w-7)(w^2+16)}\,dw$

11 $\int \frac{6y-7}{(2y+1)(4y^2+1)}\,dy$

12 $\int \frac{8t^3-5t^2+72t-10}{(t^2+2)(t^2+9)}\,dt$

13 $\int \frac{16w^3+6w^2+12w+21}{(w^2+9)(4w^2+3)}\,dw$

14 $\int \frac{x^4+5x^3+20x+16}{x(x^2+4)^2}\,dx$

15 $\int \frac{6-z^2}{2z^2+z-21}\,dz$

16 $\int \frac{4x^3-x}{x^2-x-30}\,dx$

17 $\int \frac{8-t^3}{(t-3)(t+1)^2}\,dt$

18 $\int \frac{x^6-6x^5+3x^4-10x^3-9x^2+12x-27}{x^4+3x^2}\,dx$

# Integrals Involving Roots

1 $\int \frac{5}{4-\sqrt{3-z}}\,dz$

2 $\int \frac{1}{x+4\sqrt{x-3}}\,dx$

3 $\int \frac{4}{t-\sqrt{t+7}+1}\,dt$

4 $\int \frac{z}{2z+3\sqrt{2z+1}-9}\,dz$

5 $\int \frac{w}{w-2\sqrt{6-3w}-5}\,dw$

6 $\int \cos(\sqrt{x})dx$

# Integrals Involving Quadratics

1 $\int \frac{x+1}{x^2+10x+18}\,dx$

2 $\int \frac{15}{3y^2-4y+4}\,dy$

3 $\int \frac{12-9z}{1-4z-4z^2}\,dz$

4 $\int \frac{3-7t}{(t^2+12t+40)^2}\,dt$

5 $\int \frac{11w+4}{(3+6w-w^2)^2}\,dw$

6 $\int \frac{3}{(2x^2+10x+4)^{\frac{7}{2}}}\,dx$

# Integration Strategy

# Improper Integrals

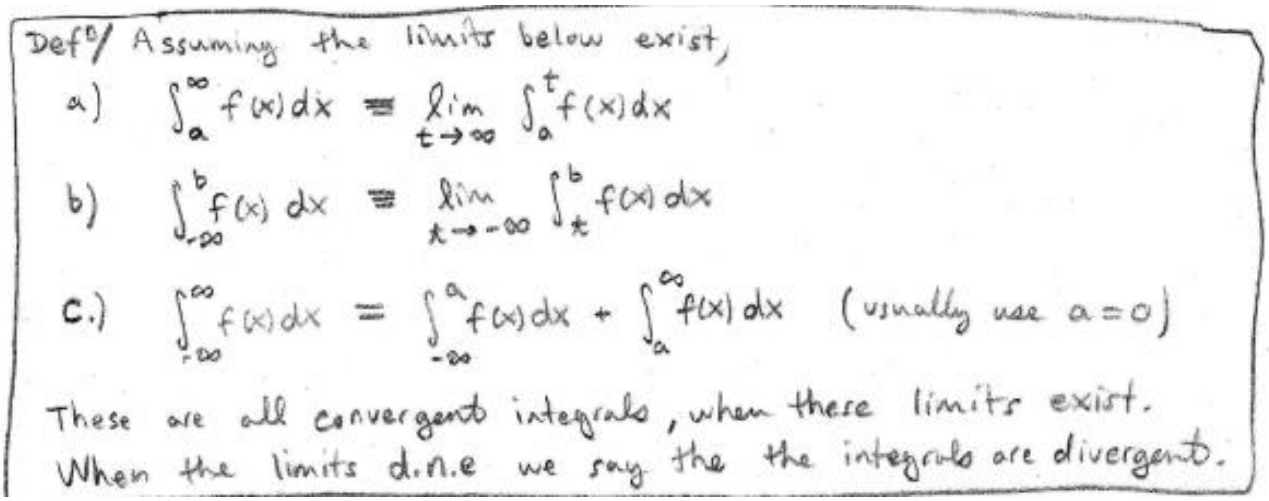
Don't worry. The Liberty Way will not be violated in this section. The "improper" here refers to one of two possibilities:

**1.)** The integrand in $\int_n^b f(x)dx$ has a vertical asymptote at some $c \in [a, b]$.

**2.)** The integral is something like $\int_a^\infty f(x)dx$.

In both cases the integral is defined in terms of a natural limiting process. In both cases the integrals may converge or diverge depending on the details of the limiting process. We will need

to recall the various tricks and common sense observations we made about limits in calculus I. On occasion L'Hopital's Rule may be necessary.
Begin with case 2.).




Def"/ Assuming the knits below exist,

**a)** $\int_a^{\infty} f(x)dx = \lim_{t\to\infty} \int_a^t f(x)dx$

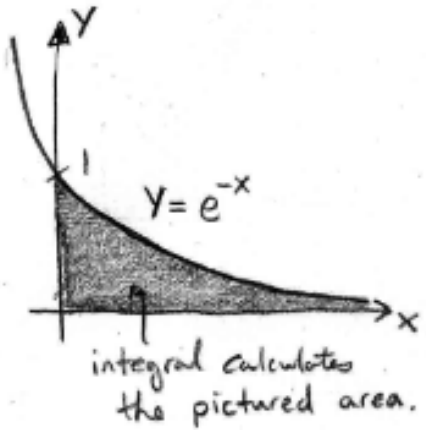
**b)** $\int_{-\infty}^{b} f(x)dx = \lim_{x\to-\infty} \int_t^b f(x)dx$

**c.)** $\int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{a} f(x)dx + \int_a^{\infty} f(x)dx$ (usually use $a = 0$

These are all divergent integrals, when there limits exist. When the limits din.e we sang the the integrates are divergent.

**Example 9.7.1**

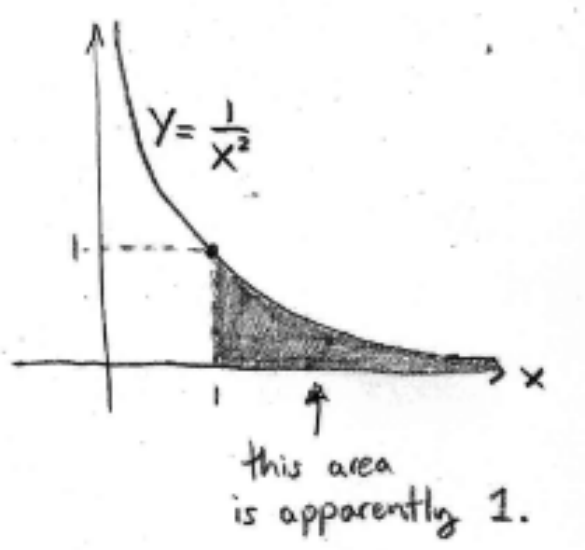
**E**1




$$\begin{aligned}\int_0^\infty e^{-x}dx &\equiv \lim_{t\to\infty}\int_0^t e^{-x}dx \\ &= \lim_{t\to\infty}(-e^{-x}|_0^t) \\ &= \lim_{t\to\infty}(-e^{-t}+1) \\ &= 1\end{aligned}$$

I suppose it may be surprising that we can have a shape with an infinite length on one side yet in total a finite area. The non-intuitive feature that makes this possible is that the height of the object gets very small very quickly so the net area does not blow up. It is not enough that the height goes to zero, we'll see that soon.

**Example 9.7.2**
**E2**

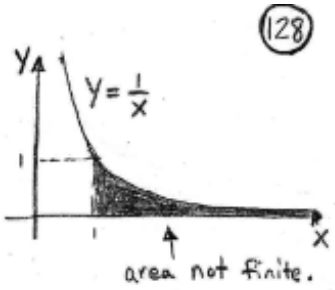



$$\begin{aligned}\int_1^\infty \frac{1}{x^2}dx &= \lim_{t\to\infty}\int_1^t \frac{1}{x^2}dx \\ &= \lim_{t\to\infty}\left(\frac{-1}{x}\Big|_1^t\right) \\ &= \lim_{t\to\infty}\left(\frac{-1}{t_2}+1\right) \\ &= 1\end{aligned}$$

this area is apparently 1.

**Example 9.7.3**

***E3***



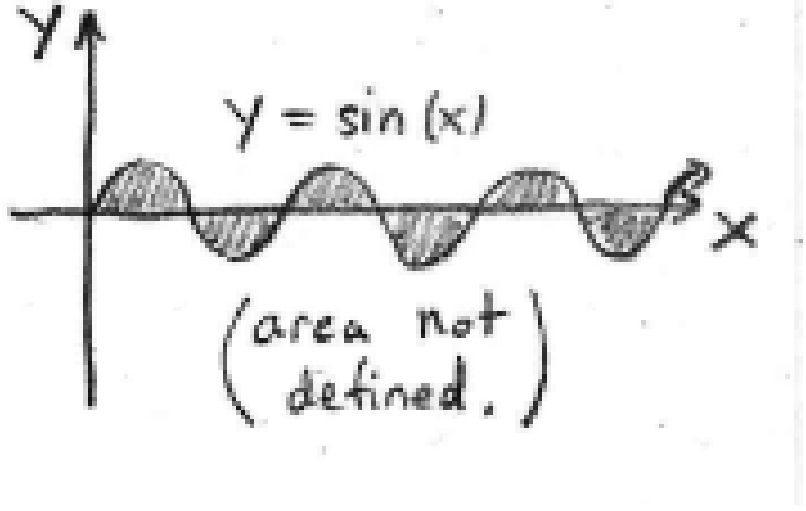

$$\begin{aligned}\int_1^\infty \frac{1}{x}dx &= \lim_{t\to\infty}\int_1^t \frac{1}{x}dx \\ &= \lim_{t\to\infty}(\ln(x)|_1^t) \\ &= \lim_{t\to\infty}(\ln(t)) \\ &= \infty \text{ (divergent)}\end{aligned}$$

Now here is the example I was referring to after E1. The integrated $\frac{1}{x}$ goes to zero as $x \to \infty$ yet the total area under the curve diverges to infinity. What does that mean? It means that as we take larger and larger upper bounds the area under the curve from one up to those bounds keeps changing, it never settles down to just one value. Instead with each higher bound we'd find the area gets larger and larger without any end in sight. In E1 and E2 we found the limits converged to one, geometrically this means that if we actually calculate those integrals for very large upper bounds we would find the values got very close to one. The next example has a different kind of divergence.

**Example 9.7.4**

E4

$$\begin{aligned}\int_{-\infty}^0 \sin(x)dx &= \lim_{t\to-\infty}\int_t^0 \sin(x)dx \\ &= \lim_{t\to-\infty}(-\cos(x)|_t^0) \\ &= \lim_{t\to-\infty}(-1+\cos(t)) \\ &= d, n_1 e \text{ ( divergent because cos(t)" oscilater at } \infty \text{ (")}\end{aligned}$$



**Example 9.7.5**

$$\text{ES} \int_0^{\infty} \frac{2x}{1+x^4}\,dx = \lim_{t\to\infty}\left(\int_0^{t} \frac{2x}{1+x^4}\,dx\right)$$
$$= \lim_{t\to\infty}\left(\int_0^{t^2} \frac{du}{1+u^2}\right) \begin{matrix} u = x^2\ u(0) = 0 \\ du = 2xdx\ u(t) = t^2 \end{matrix}$$
$$= \lim_{t\to\infty}\left(\tan^{-1}(u)|_0^{t^2}\right)$$
$$= \lim_{t\to\infty}\left(\tan^{-1}(t^2) - \tan^{-1}(0)\right)$$
$$= \pi/2$$

**Example 9.7.6**

Sometimes we need to use some integration technique before applying the appropriate limiting process. Here I do a $u$-substitution before applying the definition of improper integration for the given limits.

EG

$$\int x^2 e^{-x^3}dx = \int \frac{1}{3}e^u du + \begin{matrix} u = -x^3 \\ du = 3x^2dx \end{matrix}$$
$$= \frac{-1}{3}e^4 + c$$
$$= -\frac{1}{3}e^{-x^3} + c$$
$$\int_{-\infty}^{\infty} x^2 e^{-x^3}dx = \lim_{t\to-\infty}\int_t^0 x^2 e^{-x^3}dx + \lim_{t_2\to\infty}\int_0^{t_2} x^2 e^{-x^3}dx$$
$$= \lim_{t_1\to-\infty}\left(-\frac{1}{3}e^{-x^3}\Big|_{t_1}^{0}\right) + \lim_{t_2\to\infty}\left(\frac{-1}{3}e^{-x^3}\Big|_0^{t_2}\right)$$
$$= \lim_{t_1\to-\infty}\left(\frac{-1}{3} + \frac{e^-}{3}\downarrow t_1^3\right) + \lim_{t_2\to\infty}\left(\frac{-1}{3}e_0^{-t_2^3} + \frac{1}{3}\right)$$
$$= \text{ divergent}$$

**Example 9.7.7**

***E*7**

$$\int \frac{\ln(x)}{x^3}dx = \frac{-\ln(x)}{2x^2} - \int \frac{-1}{2x^2}\frac{dx}{x}$$
$$= \frac{-1}{2x^2}\ln(x) + \frac{1}{2}\frac{-1}{2x^2} + c$$
$$= \frac{-1}{2x^2}\left(\ln(x) + \frac{1}{2}\right) + c$$

$$u = \ln(x) \quad dV = dx/x^3$$
$$du = \frac{dx}{x} \quad V = \frac{-1}{2x^2}$$

$$\int_1^{\infty} \frac{\ln(x)}{x^3} dx = \lim_{t\to\infty} \int_1^t \frac{\ln(x)dx}{x^3}$$
$$= \lim_{x\to\infty} \left( \frac{-1}{2x^2}\left(\ln(x) + \frac{1}{2}\right)\right)_t^t$$
$$= \lim_{t\to\infty} \left( \frac{\ln(t) + 1/2}{-2t^2} + \frac{\ln(1) + 1/3}{2}\right)$$
$$f_{(15)}^{f} \lim_{t\to\infty} \left(\frac{1/t}{-4t}\right) + \frac{1}{4} \ll \text{ pulled the out before}$$
$$= \lim_{t\to\infty} \left(\frac{-1}{4t^2}\right)_0 + \frac{1}{4}$$
$$= 1/4$$

In E6 and E7 I decided to complete the integration then apply the limiting process to the appropriate integral. There is also notation to do this all at once. Care must be taken to change bounds in the other notation, 1 avoid the issue by just completing the anti derivative separately in E6 and E7.

Lets continue on to case 2.).

Def"/ Provide the limits below exist (are real numbers)

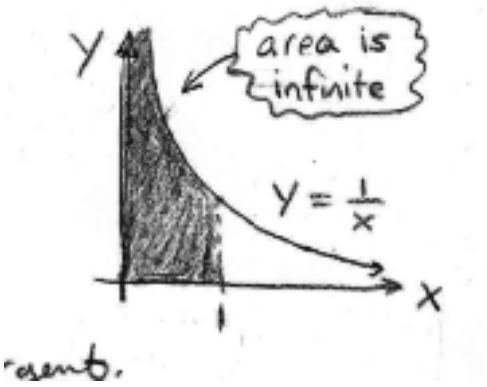
**a.)** $\int_a^b f(x)dx = \lim_{t\to 6^-} \int_a^t f(x)dx \ (btdom(f))$

**b.)** $\int_a^b f(x)dx = \lim_{t\to a^+} \int_t^b f(x)dx$ (a $\notin$ dom $(f)$)

**c.)** $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx \ (cqd\text{dom}\ (f))$

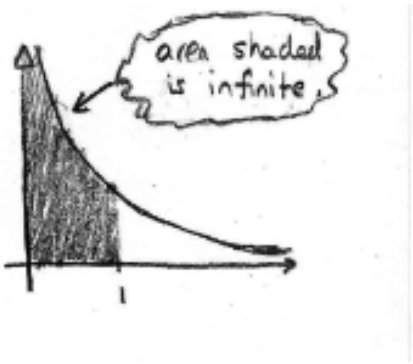
If the limit in $a$; $b$ or $c$ dine we say the integral diverge.

**Example 9.7.8**




$$\int_0^1 \frac{1}{x} dx = \lim_{t\to 0^+} \int_t^1 \frac{1}{x} dx$$
$$= \lim_{t\to 0^+} (\ln(x)|_t^1)$$
$$= \lim_{t\to 0^+} (\ln(1) - \ln(t)) = \infty$$

**Example 9.7.9**




$$\int_0^1 \frac{1}{x^2}dx = \lim_{t\to 0^+}\int_t^1 \frac{1}{x^2}dx$$

EI] $$= \lim_{t\to 0^+} \frac{-1}{x}\Big|_t^1$$

$$= \lim_{t\to 0^+}\left(-1+\frac{1}{t}\right) = \infty \text{ divergant}$$

**Example 9.7.10**

$$\int_0^1 \frac{1}{\sqrt{x}}dx = \lim_{t\to 0^+}\int_t^1 \frac{1}{\sqrt{x}}dx$$

[E10] $$= \lim_{t\to 0^+}\left(2\sqrt{x}\Big|_t^1\right)$$

$$= \lim_{t\to 0^+}(2-2\sqrt{t}) = 2 \text{ convergent}$$

**Example 9.7.11**

**EII**

$$\int_0^{\pi/4}\csc^2(x)dx = \lim_{t\to 0^+}\int_t^{\pi/4}\csc^2(x)dx$$

$$= \lim_{t\to 0^+}\left[-\cot(x)\Big|_t^{\pi/4}\right]$$

$$= \lim_{t\to 0^+}\left[-\cot(]_1^{\pi/4}\right) + \cot(t)\Big] \text{ divergeo}$$

**Example 9.7.12**

It is possible for the answer to be negative in examples like this one. The integral calculates the signed-area which can be negative in certain cases.

E12

$$\int_1^9 \frac{1}{\sqrt[3]{x-9}}dx = \lim_{t\to 9^-}\int_1^t \frac{1}{\sqrt[2]{x-9}}dx$$

$$= \lim_{t\to 9^-}\left(\frac{3}{2}(x-9)^{2/3}\int_1^t\right)$$

$$= \lim_{t\to 9^-}\left(\frac{3}{2}(t-9)^{2/3} - \frac{3}{2}(-8)^{2/3}\right)$$

$$= -6$$

**Example 9.7.13**
Our final example combines cases 1.) and 2.)

E13 $\int_2^{\infty} \frac{dx}{x\sqrt{x^2-4}} = \int_2^{3} \frac{dx}{x\sqrt{x^2-4}} + \int_3^{\infty} \frac{dx}{x\sqrt{x^2-4}} +$ $c\begin{cases} \text{follows from the trig- subtract, } x = 2\sec\,\theta. \\ \text{it tales a little work, try it.} \end{cases}$

Both Types of Impropriety Here. $\int \frac{dx}{x\sqrt{x^2\cdot 4}} = \sec^{-1}\left(\frac{1}{2}x\right) +$ (V.A. at $x = 2$ )

$$\int_2^3 \frac{dx}{x\sqrt{x-4}} = \lim_{x\to 2^+} \int_q^3 \frac{dx}{x\sqrt{x^2-4}} = \lim_{x\to 2^+} \left(\sec^{-1}\left(\frac{3}{2}\right) - \sec^{-1}\left(\frac{t}{2}\right)\right)$$
$$= \sec^{x\to(3/2)-\sec^{-1}(1)}$$
$$\int_3^{\infty} \frac{dx}{x\sqrt{x^2-4}} = \lim_{t\to\infty} \left(\sec^{-1}\left(\frac{1}{2}x\right)\Big|_3^t\right)$$
$$= \lim_{x\to-\infty} \left(\sec^{-1}\left(\frac{t}{2}\right) - \sec^{-1}(3/2)\right)$$
$$= \pi/2 - \sec^{-1}(3/2)$$
$$\int_2^{\infty} \frac{dx}{x\sqrt{x^2-4}} = \sec^{-1}(3/2) + \frac{\pi}{2} - \sec^{-1}(3/2) = \frac{\pi}{2}$$

**In-Class Exercise 9.7.14:** Calculate $\int_0^{\pi} \tan^2(x)dx$.

**In-Class Exercise 9.7.15:** Calculate $\int_{\infty}^{\infty} \frac{1}{x}dx$.

**Problems**

1 $\int_4^{\infty} 2 - 4x + 6x^2 dx$

2 $\int_0^5 \frac{1}{4w-20}dw$

3 $\int_{-1}^2 \frac{3}{\sqrt[6]{4-2z}}dz$

4 $\int_{-\infty}^0 x\mathbf{e}^{2+3x}dx$

5 $\int_0^{\infty} x\mathbf{e}^{2+3x}dx$

6 $\int_2^{\infty} \frac{1}{x^2+1}dx$

7 $\int_0^3 \frac{1}{z^2-4z}dz$

8 $\int_{-\infty}^1 \frac{x}{x^2+1}dx$

9 $\int_{-1}^{2} \frac{1}{y^2-2y-3}\,dy$

10 $\int_{-\infty}^{0} \cos(w)\,dw$

11 $\int_{10}^{\infty} \frac{1}{(5-2z)^2}\,dz$

12 $\int_{-\infty}^{\infty} \frac{z^3}{z^4+1}\,dz$

13 $\int_{1}^{4} \frac{1}{2y-6}\,dy$

14 $\int_{1}^{5} \frac{1}{\sqrt[3]{w-2}}\,dw$

15 $\int_{-2}^{1} \frac{\mathbf{e}^{\frac{1}{x}}}{x^2}\,dx$

16 $\int_{-\infty}^{\infty} x^2 \mathbf{e}^{x^3}\,dx$

17 $\int_{-\infty}^{\infty} \frac{y}{(y^2+1)^3}\,dy$

18 $\int_{0}^{3} \frac{w^3}{\sqrt{9-w^2}}\,dw$

19 $\int_{-3}^{1} \frac{1}{w^2+2w}\,dw$

20 $\int_{0}^{\infty} \frac{\mathbf{e}^{\frac{1}{x}}}{x^2}\,dx$

21 $\int_{0}^{\infty} \frac{1}{z[\ln(z)]^2}\,dz$

22 $\int_{0}^{\infty} \frac{1}{w-1}\,dw$

# Comparison Test for Improper Integrals

1 $\int_{4}^{\infty} \frac{1}{\sqrt[5]{z-2}}\,dz$

2 $\int_{0}^{\infty} \frac{w}{\sqrt{w^6+2}}\,dw$

3 $\int_{2}^{\infty} \frac{1}{(2w+3)^4}\,dw$

4 $\int_{12}^{\infty} \frac{y^2-4y+2}{y-7}\,dy$

5 $\int_{2}^{\infty} \frac{1}{\ln(x)}\,dx$

Hint : Sketch the graph of $y = x$ and $y = \ln(x)$ on the same axis system.

6 $\int_2^{\infty} \frac{\sqrt{z}-4\sin^2(z)}{z^3}\,dz$

7 $\int_{20}^{\infty} \frac{\sqrt[3]{2x}+\sin^2(x)}{\sqrt{x}-\cos^2(x)}\,dx$

8 $\int_0^{\infty} \frac{z\mathrm{e}^{-z}}{z^3+1}\,dz$

# Approximating Definite Integrals

**Numerical Integration: an Introduction:**

Questions: Of we use $L_n, R_n, M_n$ then how close is our approx. to the veal answer $\int_0^b f(x)dx \equiv I$

**(2)** Are there any other approx. estimation schemes?

**(3)** How do the different approx. compare (usually)

Lets begin with question (2)

Trapezoidal Rule with $n$-trapezoids: $T_n \cong \int_a^b f(x)dx$

$$T_n = \frac{\Delta x}{2}[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)]$$

$$\Delta x = \frac{b-a}{n} \text{ and } x_i = a + i\Delta x$$

"Proof

$\frac{\text{Illustration of } T_3}{\text{we can geometrically}}$ see 6 triangle d to gate area by

$$t_1 = \frac{1}{2}f(x_0)\Delta x + \frac{1}{2}f(x_1)\Delta x$$
$$t_2 = \frac{1}{2}f(x_1)\Delta x + \frac{1}{2}f(x_2)\Delta x$$
$$t_3 = \frac{1}{2}f(x_2)\Delta x + \frac{1}{2}f(x_3)\Delta x$$

$$T_3 = t_1 + t_2 + t_3 = \frac{1}{2}\Delta x\big(f(x_0) + 2f(x) + 2f(x_3) + f(x_3)\big)$$

**Some limitations and comprises:**

**(1)** As $n$ gets large the accuracy increases for each approx. modules the limiting accuracy of the computation.

**(2)** $L_n$ and $R_n$ have errors with opposite sign, and are halved when we double $n$.

**(3)** $T_n$ & $M_n$ are better approx. then $L_n$ or $R_n$

**(4)** Errors in $T_n$ and $M_n$ are appropriate in sign and are quarter when we double $n$

**(5)** $E_{M_n} \nRightarrow \frac{-1}{2}E_{T_n}$

Besides these rules of thum 6 we can add some were concrete limiting comments,

The (3) Error Bounds. Suppose $|f''(x)| \leq K$ fur $a \leq x \leq b$. If $E_T$ and $E_m$ are ercors in trap, and midst. fule them

$$|E_T| \leq \frac{K(b-a)^3}{12n^2} \; |E_m| \leq \frac{K(b-a)^3}{24n^2}$$

I expect you to be able to apply these on de given them, if there is some question on the test, I would supply the relevant inequality. It's up to you to use them,

(Example: Put an upper bound on $E_{T,\text{ where}}\, T_n$ is used to approximate $\int^{11} e^x dx$. Notion $f(x) = e^x \Rightarrow f''(x) = e^x$ which is increasing "sins $f'''(x) > 0$ for $5 \leq x \leq 11$ thus $f''(5) \leq f''(x) \leq f(11) \Rightarrow f''(ii)$ is an upper bound, specifically $f''(n) = e'' = K \geq |f''(x)|$

$$|E_T| \leq \frac{e^{11}(11-5)^3}{12n^2}$$

-Crucial to using the $T^2(3)$ is finding the $K$.

For the most part this section closely mirrors Stewart's text. I also feel no need to justify these wild claims about the error bounds. Proof by example is not proof. A good graduate course in numerical methods would derive these results. Error bounds are important since they tell us the worst-case scenario when we replace the true integral with a much easier to calculate finite sum. However, math aside, beware the machine

epsilon seeks to devour and corrupt. Just because the mathematics says the error should be smaller than a particular value does not mean the machine, we are using is capable of the precision we assume. And, I haven't even bothered to factor in the robot holocaust. That said, I don't plan on doing Simpson's rule without a computer.
Simpson's Rule
(125) Simpson's Rite approximation the area by intestine the area under a parabolic approx. of the function
$S_3$ for a linear function
Simpson $R$ unk

$$S_n = \frac{\Delta x}{3}[f(x_0) + 4f(x_1) + 2f(x_n) + 4f(x_0) + \cdots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)]$$

Where $n$ must be even and $\Delta x = \frac{b-a}{n}$.

Proof See book.

Remote No trice put torn 1,4,2,4,2, … ,2,4,1
Hew accurate is Simpson's role? Calculators base their numeric int an it, Tho (4) Error **Bunne Fo Simone's Rule:** Let $\left|f''^{ilr}(x)\right| \leq K$ for $a \leq x \leq b$ and let $E_s$ be aerier in imposition bare. $S_n$

$$\left|E_{s_n}\right| \leq \frac{K(b-a)^5}{180n^4}$$

**Remark:** If we dead $n$ then the error is $\frac{1}{16}$ of what is war.
Example If we approximate $\int_2^4 \sin\ \theta d\theta$ with $S_n$ then find $\left|E_{s_1}\right| \leq$ ?
Motive than $f(\theta) = \sin(\theta) \Rightarrow f''''(\theta) = \sin(\theta)$ which has the property $|\sin(\theta)| \leq 1 \Rightarrow |f''''(\theta)| \leq 1$. Thus

$$|E_{in}| \leq \frac{1 \cdot (4-2)^5}{180n^4}$$

If we demand accuracy within $0.01 \Rightarrow |E_{5n}| \approx 0.01$ thus

$$0.01 = \frac{32}{110n^4} \Rightarrow n = \left(\frac{32}{(0.011)180}\right)^{1/4} = 2.05 \Rightarrow S_{S_3} \text{ will certainty}$$

Perhaps this section has left you a little dazed. Why are we doing this anyway? Didn't the methods we used earlier in this chapter give better, nice exact results? The answer

is that yes those methods do give exact results but there are many problems which simply do not admit a nice closed-form answer in terms of elementary functions. For example,
**1.)** $\int \frac{\sin(x)}{x} dx$ arises in study of optics
**2.)** $\int e^{-x^2} dx$ the bell curve from probability
**3.)** $\int \sqrt{a^2\cos^2(t) + b^2\cos^2(t)}$ elliptical function
In a few special cases 3.) does permit an solution in terms of elementary functions. It can be shown that the integral converges for piece wise continuous functions. However, there is no guarantee that the anti-derivative for the given integrate has a nice formula. We will find approximations to the anti-derivative in a later chapter using power series arguments. This section has dealt only with the less challenging question of how to approximate a definite integral. Some definite integrals require we take a brute-force like approach. That is in essence what this section is. When we can't find the anti-derivative for the integrate, or when there is no formula given for the integrate, in such cases we have no alternative but to use brute force. This discussion in this section is just quibbling over which type of brute we want to be. In practice, the thing most people do is to use Mathematical or the Wolfram Integrator (it's online) to calculate

definite integrals which defy closed form solutions. If you could look inside Mathematical odds are you'd find something like Simpson's Rule being used to find the answer.
Finally, there is a nice summary section in Stewart giving some grand advice about how to integrate. It might help you gather your thoughts. Try reading section 8.5. I don't plan to formally cover integration tables. Who needs a table when you can create the table? And if you can't do the integral then Mathematical beats the table 99.9% of the time. I would strongly caution over using Mathematical, you need to suffer when doing the homework from this chapter. The burning sensation in your brain may be needed for you to level-up mathematically speaking.

As usual there are some additional examples in the suggested homework solutions.
**Remark:** we are skipping Chapter 9 for now. I'd like to do some of the sections in Chapter 9 from a "parametric viewpoint". That's hard if we haven't yet discussed what the "parametric viewpoint" is. Stewart takes a purely Cartesian viewpoint so he avoids the problem. I'd like to do it right the first time around so we'll wait on Chapter 9.

**Problems**

1 $\int_{-2}^{4} \sin(x^2+2)dx$ using $n=6$

2 $\int_{0}^{4} \sqrt[3]{x^4+6}dx$ using $n=6$

3 $\int_{1}^{5} \mathbf{e}^{\cos(x)}dx$ using $n=8$

4 $\int_{3}^{5} \frac{1}{1-\ln(x)}dx$ using $n=6$

5 $\int_{-3}^{1} \sin(x)\cos(x^2)dx$ using $n=8$

6 $\int_{-2}^{4} \sin(x^2+2)dx$ using $n=6$

7 $\int_{0}^{4} \sqrt[3]{x^4+6}dx$ using $n=6$

8 $\int_{1}^{5} \mathbf{e}^{\cos(x)}dx$ using $n=8$

9 $\int_{3}^{5} \frac{1}{1-\ln(x)}dx$ using $n=6$

10 $\int_{-3}^{1} \sin(x)\cos(x^2)dx$ using $n=8$

# EVALUATE DEFINITE INTEGRALS BY SUBSTITUTION

Example 1 Use the two methods above to evaluate $\int_0^2 \quad x(x^2+1)^3dx$.

Example 2 Evaluate

$$(a)\int_0^{\frac{\pi}{8}} \quad sin^5 2xcox2xdx \qquad (b)\int_2^5 \quad (2x-5)(x-3)^9dx$$

Example 3 Evaluate

(a) $\int_0^{3/4} \frac{dx}{1-x}$ (b) $\int_0^{ln3} e^x(1+e^x)^{1/2}dx$

1-4 Express the integral in terms of the variable $u$, but do not evaluate it.

1. (a) $\int_1^3 (2x-1)^3 dx; u = 2x-1$

   (b) $\int_0^4 3x\sqrt{25-x^2}dx; u = 25-x^2$

   (c) $\int_{-1/2}^{1/2} cos\,(\pi\theta)d\theta; u = \pi\theta$

   (d) $\int_0^1 (x+2)(x+1)^3 dx;\ u = x+1$

2. (a) $\int_{-1}^4 (5-2x)^8 dx; u = 5-2x$

   (b) $\int_{-\pi/3}^{2\pi/3} \frac{sin\ x}{\sqrt{2+cos\ x}} dx; u = 2+cos\ x$

   (c) $\int_0^{\pi/4} tan^2\ x sec^2\ x dx; u = tan\ x$

   (d) $\int_0^1 x^3\sqrt{x^2+3}dx; u = x^2+3$

3. (a) $\int_0^1 e^{2x-1}dx; u = 2x-1$

   (b) $\int_e^{e^2} \frac{ln\ x}{x} dx; u = ln\ x$

4. (a) $\int_1^{\sqrt{3}} \frac{\sqrt{tan^{-1}\ x}}{1+x^2} dx; u = tan^{-1}\ x$

   (b) $\int_1^{\sqrt{e}} \frac{dx}{x\sqrt{1-(ln\ x)^2}}; u = ln\ x$

5-18 Evaluate the definite integral two ways: first by a $u$ substitution in the definite integral and then by a $u$-substitution in the corresponding indefinite integral.

5. $\int_0^1 (2x+1)^3 dx$
6. $\int_1^2 (4x-2)^3 dx$
7. $\int_0^1 (2x-1)^3 dx$
8. $\int_1^2 (4-3x)^8 dx$
9. $\int_0^8 x\sqrt{1+x}dx$
10. $\int_{-3}^0 x\sqrt{1-x}dx$
11. $\int_0^{\pi/2} 4sin\,(x/2)dx$
12. $\int_0^{\pi/6} 2cos\ 3x dx$

13. $\int_{-2}^{-1} \frac{x}{(x^2+2)^3}\,dx$

14. $\int_{1-\pi}^{1+\pi} sec^2\left(\frac{1}{4}x - \frac{1}{4}\right)dx$

15. $\int_{-ln\ 3}^{ln\ 3} \frac{e^x}{e^x+4}\,dx$

16. $\int_{0}^{ln\ 5} e^x(3-4e^x)dx$

17. $\int_{1}^{3} \frac{dx}{\sqrt{x}(x+1)}$

18. $\int_{ln\ 2}^{ln\ (2/\sqrt{3})} \frac{e^{-x}dx}{\sqrt{1-e^{-2x}}}$

19-22 Evaluate the definite integral by expressing it in terms of $u$ and evaluating the resulting integral using a formula from geometry.

19. $\int_{-5/3}^{5/3} \sqrt{25-9x^2}dx; u = 3x$

20. $\int_{0}^{2} x\sqrt{16-x^4}dx; u = x^2$

21. $\int_{\pi/3}^{\pi/2} sin\ \theta\sqrt{1-4cos^2\ \theta}d\theta; u = 2cos\ \theta$

22. $\int_{e^{-3}}^{e^3} \frac{\sqrt{9-(ln\ x)^2}}{x}dx; u = ln\ x$

23. A particle moves with a velocity of $v(t) = sin\ \pi t\ m/s$ along an $s$-axis. Find the distance traveled by the particle over the time interval $0 \leq t \leq 1$.

24. A particle moves with a velocity of $v(t) = 3cos\ 2t\ m/s$ along an $s$-axis. Find the distance traveled by the particle over the time interval $0 \leq t \leq \pi/8$.

25. Find the area under the curve $y = 9/(x+2)^2$ over the interval $[-1,1]$.

26. Find the area under the curve $y = 1/(3x+1)^2$ over the interval $[0,1]$.

27. Find the area of the region enclosed by the graphs of $y = 1/\sqrt{1-9x^2}, y = 0, x = 0$, and $x = \frac{1}{6}$.

28. Find the area of the region enclosed by the graphs of $y = sin^{-1}\ x, x = 0$, and $y = \pi/2$.

29. Find the average value of $f(x) = x/(5x^2+1)^2$ over the interval $[0,2]$.

30. Find the average value of $f(x) = e^{3x}/(1+e^{6x})$ over the interval $[-(ln\ 3)/6, 0]$.

31-50 Evaluate the integrals by any method

31. $\int_1^5 \frac{dx}{\sqrt{2x-1}}$

32. $\int_1^2 \sqrt{5x-1}dx$

33. $\int_{-1}^1 \frac{x^2dx}{\sqrt{x^3+9}}$

34. $\int_{\pi/2}^{\pi} 6sin\ x(cos\ x+1)^5 dx$

35. $\int_1^3 \frac{x+2}{\sqrt{x^2+4x+7}}dx$

36. $\int_1^2 \frac{dx}{x^2-6x+9}$

37. $\int_0^{\pi/4} 4sin\ xcos\ xdx$

38. $\int_0^{\pi/4} \sqrt{tan\ x}sec^2\ xdx$

39. $\int_0^{\sqrt{x}} 5xcos\ (x^2)dx$

40. $\int_{\pi^2}^{4\pi^2} \frac{1}{\sqrt{x}}sin\ \sqrt{x}dx$

41. $\int_{\pi/12}^{\pi/9} sec^2\ 3\theta d\theta$

42. $\int_0^{\pi/6} tan\ 2\theta d\theta$

43. $\int_0^1 \frac{y^2dy}{\sqrt{4-3y}}$

44. $\int_{-1}^4 \frac{xdx}{\sqrt{5+x}}$

45. $\int_0^e \frac{dx}{2x+e}$

46. $\int_1^{\sqrt{2}} xe^{-x^2}dx$

47. $\int_0^1 \frac{x}{\sqrt{4-3x^4}}dx$

48. $\int_1^2 \frac{1}{\sqrt{x}\sqrt{4-x}}dx$

49. $\int_0^{1/\sqrt{3}} \frac{1}{1+9x^2}dx$

50. $\int_1^{\sqrt{2}} \frac{x}{3+x^4}dx$

# Exercise

## 7.1 Exercise

1 $\int (4-2x)^3 dx$

2 $\int 3\sqrt{4+2x}dx$

3 $\int x\sec^2(x^2)dx$

4 $\int 4x\tan(x^2)dx$

5 $\int \frac{\sin 3x}{2+\cos 3x}dx$

6 $\int \frac{1}{9+4x^2}dx$

7 $\int e^x \sinh(e^x)dx$

8 $\int \frac{\sec(\ln x)\tan(\ln x)}{x}dx$

9 $\int e^{\tan x}\sec^2 x dx$

10 $\int \frac{x}{\sqrt{1-x^4}}dx$

11 $\int \cos^5 5x \sin 5x dx$

12 $\int \frac{\cos x}{\sin x\sqrt{\sin^2 x+1}}dx$

13 $\int \frac{e^x}{\sqrt{4+e^{2x}}}dx$

14 $\int \frac{e^{\tan^{-1}x}}{1+x^2}dx$

15 $\int \frac{e^{\sqrt{x-1}}}{\sqrt{x-1}}dx$

16 $\int (x+1)\cot(x^2+2x)dx$

17 $\int \frac{\cosh\sqrt{x}}{\sqrt{x}}dx$

18 $\int \frac{dx}{x(\ln x)^2}$

19 $\int \frac{dx}{\sqrt{x}3\sqrt{x}}$

20 $\int \sec(\sin\theta)\tan(\sin\theta)\cos\theta d\theta$

21 $\int \frac{\operatorname{csch}^2(2/x)}{x^2}dx$

22 $\int \frac{dx}{\sqrt{x^2-4}}$

23 $\int \frac{e^{-x}}{4-e^{-2x}}dx$

24 $\int \frac{\cos(\ln x)}{x}dx$

25 $\int \frac{e^x}{\sqrt{1-e^{2x}}} dx$

26 $\int \frac{\sinh\left(x^{-1/2}\right)}{x^{3/2}} dx$

27 $\int \frac{x}{\csc\left(x^2\right)} dx$

28 $\int \frac{e^x}{\sqrt{4-e^{2x}}} dx$

29 $\int x4^{-x^2} dx$

30 $\int 2^{\pi x} dx$

## Chapter 7.2

Example 1: Use integration by parts to evaluate $\int x \cos x \, dx$

Example 2: Evaluate $\int xe^x \, dx$

Example 3: Evaluate $\int \ln x$

Example 4: Evaluate $\int x^2 e^{-x} \, dx$

Example 5: Evaluate $\int e^x \cos dx$

Example 6: In Example 11 of Section 5.3 we evaluated $\int x^2\sqrt{x-1} \, dx$ using u-substitution. Evaluate this integral using tabular integration by parts.

Example 7: Evaluate $\int_0^1 tan^{-1}(x) \, dx$

Example 8: Evaluate $\int \cos^4 x \, dx$

1-38 Evaluate the integral.

1 $\int xe^{-2x} dx$

2 $\int xe^{3x} dx$

3 $\int x^2 e^x dx$

4 $\int x^2 e^{-2x} dx$

5 $\int x\sin\ 3x dx$

6 $\int x\cos\ 2x dx$

7 $\int x^2\cos\ x dx$

8 $\int x^2\sin\ x dx$

9 $\int x\ln\ x dx$

10 $\int \sqrt{x}\ln\ x dx$

11 $\int (\ln\ x)^2 dx$

12 $\int \frac{\ln\ x}{\sqrt{x}} dx$

13 $\int \ln(3x-2) dx$

14 $\int \ln(x^2+4) dx$

15 $\int \sin^{-1}\ x dx$

16 $\int \cos^{-1}(2x) dx$

17 $\int \tan^{-1}(3x) dx$

18 $\int x\tan^{-1}\ x dx$

19 $\int e^x \sin\ x dx$

20 $\int e^{3x}\cos\ 2x dx$

21 $\int \sin(\ln\ x) dx$

22 $\int \cos(\ln\ x) dx$

23 $\int x\sec^2\ x dx$

24 $\int x\tan^2\ x dx$

25 $\int x^3 e^{x^2} dx$

26 $\int \frac{xe^x}{(x+1)^2} dx$

27 $\int_0^2 xe^{2x} dx$

28 $\int_0^1 xe^{-5x} dx$

29 $\int_1^e x^2 \ln\ x dx$

30 $\int_{\sqrt{e}}^{e} \frac{\ln\ x}{x^2} dx$

31 $\int_{-1}^{1} \ln(x+2) dx$

32 $\int_0^{\sqrt{3}/2} \sin^{-1}\ x dx$

33 $\int_2^4 \sec^{-1}\ \sqrt{\theta} d\theta$

34 $\int_1^2 x\sec^{-1}\ x dx$

35 $\int_0^{\pi} x\sin\ 2x dx$

36 $\int_0^{\pi} (x + x\cos\ x) dx$

37 $\int_1^3 \sqrt{x}\tan^{-1}\ \sqrt{x} dx$

38 $\int_0^2 \ln(x^2+1)dx$

## Chapter:7.3

Formula's:

1) $\int \sin^n x dx = -\frac{1}{n}\sin^{n-1} x\cos x + \frac{n-1}{n}\int \sin^{n-2} x dx$

2) $\int \cos^n x dx = \frac{1}{n}\cos^{n-1} x\sin x + \frac{n-1}{n}\int \cos^{n-2} x dx$

3) $\sin^2 x = \frac{1}{2}(1-\cos 2x)$ and 4) $\cos^2 x = \frac{1}{2}(1+\cos 2x)$

5) $\int \sin^2 x dx = \frac{1}{2}\int (1-\cos 2x)dx = \frac{1}{2}x - \frac{1}{4}\sin 2x + C$

6) $\int \cos^2 x dx = \frac{1}{2}\int (1+\cos 2x)dx = \frac{1}{2}x + \frac{1}{4}\sin 2x + C$

7) $\int \sin^3 x dx = \frac{1}{3}\cos^3 x - \cos x + C$

8) $\int \cos^3 x dx = \sin x - \frac{1}{3}\sin^3 x + C$

9) $\int \sin^4 x dx = \frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$

10) $\int \cos^4 x dx = \frac{3}{8}x + \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$

11) $\int \tan^n x dx = \frac{\tan^{n-1} x}{n-1} - \int \tan^{n-2} x dx$

12) $\int \sec^n x dx = \frac{\sec^{n-2} x\tan x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2} x dx$

13) $\int \tan x dx = \ln|\sec x| + C$

14) $\int \sec x dx = \ln|\sec x + \tan x| + C$

15) $\int \tan^2 x dx = \tan x - x + C$

16) $\int \sec^2 x dx = \tan x + C$

17) $\int \tan^3 x dx = \frac{1}{2}\tan^2 x - \ln|\sec x| + C$

2. Evaluate:

(a) $\int \sin^4 x \cos^5 x dx$
(b) $\int \sin^4 x \cos^4 x dx$

3. Evaluate:

$\int \sin 7x \cos 3x \, dx$

4. Evaluate:

a) $\int \tan 2x \sec 4x \, dx$

b) $\int \tan 3x \sec 3x \, dx$

c) $\int \tan 2x \sec x \, dx$

5. Evaluate the Integral:

1 $\int \cos^3 x \sin x dx$
2 $\int \sin^5 3x \cos 3x dx$
3 $\int \sin^2 5\theta d\theta$
4 $\int \cos^2 3x dx$
5 $\int \sin^3 a\theta d\theta$
6 $\int \cos^3 at dt$
7 $\int \sin ax \cos ax dx$
8 $\int \sin^3 x \cos^3 x dx$
9 $\int \sin^2 t \cos^3 t dt$
10 $\int \sin^3 x \cos^2 x dx$
11 $\int \sin^2 x \cos^2 x dx$
12 $\int \sin^2 x \cos^4 x dx$
13 $\int \sin 2x \cos 3x dx$
14 $\int \sin 3\theta \cos 2\theta d\theta$
15 $\int \sin x \cos(x/2) dx$
16 $\int \cos^{1/3} x \sin x dx$

17 $\int_0^{\pi/2} \cos^3 x dx$

18 $\int_0^{\pi/2} \sin^2 \frac{x}{2} \cos^2 \frac{x}{2} dx$

19 $\int_0^{\pi/3} \sin^4 3x \cos^3 3x dx$

20 $\int_{-\pi}^{\pi} \cos^2 5\theta d\theta$

21 $\int_0^{\pi/6} \sin 4x \cos 2x dx$

22 $\int_0^{2\pi} \sin^2 kx dx$

23 $\int \sec^2 (2x-1) dx$

24 $\int \tan 5x dx$

25 $\int e^{-x} \tan (e^{-x}) dx$

26 $\int \cot 3x dx$

27 $\int \sec 4x dx$

28 $\int \frac{\sec (\sqrt{x})}{\sqrt{x}} dx$

29 $\int \tan^2 x \sec^2 x dx$

30 $\int \tan^5 x \sec^4 x dx$

31 $\int \tan 4x \sec^4 4x dx$

32 $\int \tan^4 \theta \sec^4 \theta d\theta$

33 $\int \sec^5 x \tan^3 x dx$

34 $\int \tan^5 \theta \sec \theta d\theta$

35 $\int \tan^4 x \sec x dx$

36 $\int \tan^2 x \sec^3 x dx$

37 $\int \tan t \sec^3 t dt$

38 $\int \tan x \sec^5 x dx$

39 $\int \sec^4 x dx$

40 $\int \sec^5 x dx$

41 $\int \tan^3 4x dx$

42 $\int \tan^4 x dx$

43 $\int \sqrt{\tan x} \sec^4 x dx$

44 $\int \tan x \sec^{3/2} x dx$

45 $\int_0^{\pi/8} \tan^2 2x dx$

46 $\int_0^{\pi/6} \sec^3 2\theta \tan 2\theta d\theta$

47 $\int_0^{\pi/2} \tan^5 \frac{x}{2} dx$

48 $\int_0^{1/4} \sec \pi x \tan \pi x dx$

49 $\int \cot^3 x \operatorname{cosec}^3 x \, dx$

50 $\int \cot^2 3t \sec 3t \, dt$

51 $\int \cot^3 x \, dx$

52 $\int \operatorname{cosec}^4 x \, dx$

## Chapter 7.4

Evaluate the integral.

1. $\int \sqrt{4-x^2} dx$
2. $\int \sqrt{1-4x^2} dx$
3. $\int \frac{x^2}{\sqrt{16-x^2}} dx$
4. $\int \frac{dx}{x^2\sqrt{9-x^2}}$
5. $\int \frac{dx}{(4+x^2)^2}$
6. $\int \frac{x^2}{\sqrt{5+x^2}} dx$
7. $\int \frac{\sqrt{x^2-9}}{x} dx$
8. $\int \frac{dx}{x^2\sqrt{x^2-16}}$
9. $\int \frac{3x^3}{\sqrt{1-x^2}} dx$
10. $\int x^3\sqrt{5-x^2} dx$
11. $\int \frac{dx}{x^2\sqrt{9x^2-4}}$
12. $\int \frac{\sqrt{1+t^2}}{t} dt$
13. $\int \frac{dx}{(1-x^2)^{3/2}}$
14. $\int \frac{dx}{x^2\sqrt{x^2+25}}$
15. $\int \frac{dx}{\sqrt{x^2-9}}$
16. $\int \frac{dx}{1+2x^2+x^4}$
17. $\int \frac{dx}{(4x^2-9)^{3/2}}$
18. $\int \frac{3x^3}{\sqrt{x^2-25}} dx$
19. $\int e^x\sqrt{1-e^{2x}} dx$
20. $\int \frac{\cos\theta}{\sqrt{2-\sin^2\theta}} d\theta$
21. $\int_0^1 5x^3\sqrt{1-x^2} dx$
22. $\int_0^{1/2} \frac{dx}{(1-x^2)^2}$
23. $\int_{\sqrt{2}}^2 \frac{dx}{x^2\sqrt{x^2-1}}$
24. $\int_{\sqrt{2}}^2 \frac{\sqrt{2x^2-4}}{x} dx$
25. $\int_1^3 \frac{dx}{x^4\sqrt{x^2+3}}$
26. $\int_0^3 \frac{x^3}{(3+x^2)^{5/2}} dx$

**EXERCISE SET 7.4**

Evaluate the integral.

2. $\int \sqrt{4-x^2}dx$
3. $\int \frac{x^2}{\sqrt{16-x^2}}dx$
5. $\int \frac{dx}{(4+x^2)^2}$
7. $\int \frac{\sqrt{x^2-9}}{x}dx$
9. $\int \frac{3x^3}{\sqrt{1-x^2}}dx$
11. $\int \frac{dx}{x^2\sqrt{9x^2-4}}$
13. $\int \frac{dx}{(1-x^2)^{3/2}}$
15. $\int \frac{dx}{\sqrt{x^2-9}}$
17. $\int \frac{dx}{(4x^2-9)^{3/2}}$
19. $\int e^x\sqrt{1-e^{2x}}dx$
21. $\int_0^1 5x^3\sqrt{1-x^2}dx$
23. $\int_{\sqrt{2}}^2 \frac{dx}{x^2\sqrt{x^2-1}}$
25. $\int_1^3 \frac{dx}{x^4\sqrt{x^2+3}}$

2. $\int \sqrt{1-4x^2}dx$
4. $\int \frac{dx}{x^2\sqrt{9-x^2}}$
6. $\int \frac{x^2}{\sqrt{5+x^2}}dx$
8. $\int \frac{dx}{x^2\sqrt{x^2-16}}$
10. $\int x^3\sqrt{5-x^2}dx$
12. $\int \frac{\sqrt{1+t^2}}{t}dt$
14. $\int \frac{dx}{x^2\sqrt{x^2+25}}$
16. $\int \frac{dx}{1+2x^2+x^4}$
18. $\int \frac{3x^3}{\sqrt{x^2-25}}dx$
20. $\int \frac{\cos\theta}{\sqrt{2-\sin^2\theta}}d\theta$
22. $\int_0^{1/2} \frac{dx}{(1-x^2)^2}$
24. $\int_{\sqrt{2}}^2 \frac{\sqrt{2x^2-4}}{x}dx$
26. $\int_0^3 \frac{x^3}{(3+x^2)^{5/2}}dx$

Evaluate the integral.

3. $\int \sqrt{4-x^2}dx$
3. $\int \frac{x^2}{\sqrt{16-x^2}}dx$
5. $\int \frac{dx}{(4+x^2)^2}$
7. $\int \frac{\sqrt{x^2-9}}{x}dx$
9. $\int \frac{3x^3}{\sqrt{1-x^2}}dx$
11. $\int \frac{dx}{x^2\sqrt{9x^2-4}}$
13. $\int \frac{dx}{(1-x^2)^{3/2}}$

2. $\int \sqrt{1-4x^2}dx$
4. $\int \frac{dx}{x^2\sqrt{9-x^2}}$
6. $\int \frac{x^2}{\sqrt{5+x^2}}dx$
8. $\int \frac{dx}{x^2\sqrt{x^2-16}}$
10. $\int x^3\sqrt{5-x^2}dx$
12. $\int \frac{\sqrt{1+t^2}}{t}dt$
14. $\int \frac{dx}{x^2\sqrt{x^2+25}}$

15. $\int \frac{dx}{\sqrt{x^2-9}}$

16. $\int \frac{dx}{1+2x^2+x^4}$

17. $\int \frac{dx}{(4x^2-9)^{3/2}}$

18. $\int \frac{3x^3}{\sqrt{x^2-25}}dx$

19. $\int e^x\sqrt{1-e^{2x}}dx$

20. $\int \frac{\cos\theta}{\sqrt{2-\sin^2\theta}}d\theta$

21. $\int_0^1 5x^3\sqrt{1-x^2}dx$

22. $\int_0^{1/2} \frac{dx}{(1-x^2)^2}$

23. $\int_{\sqrt{2}}^2 \frac{dx}{x^2\sqrt{x^2-1}}$

24. $\int_{\sqrt{2}}^2 \frac{\sqrt{2x^2-4}}{x}dx$

25. $\int_1^3 \frac{dx}{x^4\sqrt{x^2+3}}$

26. $\int_0^3 \frac{x^3}{(3+x^2)^{5/2}}dx$

## Chapter 7.5 (INTEGRATING RATIONAL FUNCTIONS BY PARTIAL FRACTIONS)

Example 1

Evaluate $\int \frac{dx}{x^2+x-2}$

Example 2

Evaluate $\int \frac{2x+4}{x^3-2x^2}dx$

Example 3

Evaluate $\int \frac{x^2+x-2}{3x^2-x^2+3x-1}dx$

Example 4

Evaluate $\int \frac{3x^4+4x^3+16x^2+20x+9}{(x+2)(x^2+3)^2}dx$

Example 5

Evaluate $\int \frac{3x^4+3x^3-5x^2+x-1}{x^2+x-2}dx$

## Exercise Set (7.5)

1-8 Write out the form of the partial fraction decomposition. (Do not find the numerical values of the coefficients.)

1 $\frac{3x-1}{(x-3)(x+4)}$

2 $\frac{5}{x(x^2-4)}$

3 $\frac{2x-3}{x^3-x^2}$

4 $\frac{x^2}{(x+2)^3}$

5 $\frac{1-x^2}{x^3(x^2+2)}$

6 $\frac{3x}{(x-1)(x^2+6)}$

7 $\frac{4x^3-x}{(x^2+5)^2}$

8 $\frac{1-3x^4}{(x-2)(x^2+1)^2}$

9-34 Evaluate the integral.

9 $\int \frac{dx}{x^2-3x-4}$

10 $\int \frac{dx}{x^2-6x-7}$

11 $\int \frac{11x+17}{2x^2+7x-4}dx$

12 $\int \frac{5x-5}{3x^2-8x-3}dx$

13 $\int \frac{2x^2-9x-9}{x^3-9x}dx$

14 $\int \frac{dx}{x(x^2-1)}$

15 $\int \frac{x^2-8}{x+3}dx$

16 $\int \frac{x^2+1}{x-1}dx$

17 $\int \frac{3x^2-10}{x^2-4x+4}dx$

18 $\int \frac{x^2}{x^2-3x+2}dx$

19 $\int \frac{2x-3}{x^2-3x-10}dx$

20 $\int \frac{3x+1}{3x^2+2x-1}dx$

21 $\int \frac{x^5+x^2+2}{x^3-x}dx$

22 $\int \frac{x^5-4x^3+1}{x^3-4x}dx$

23 $\int \frac{2x^2+3}{x(x-1)^2}dx$

24 $\int \frac{3x^2-x+1}{x^3-x^2}dx$

25 $\int \frac{2x^2-10x+4}{(x+1)(x-3)^2}dx$

26 $\int \frac{2x^2-2x-1}{x^3-x^2}dx$

27 $\int \frac{x^2}{(x+1)^3}dx$

28 $\int \frac{2x^2+3x+3}{(x+1)^3}dx$

29 $\int \frac{2x^2-1}{(4x-1)(x^2+1)}dx$

30 $\int \frac{dx}{x^3+2x}$

31 $\int \frac{x^3+3x^2+x+9}{(x^2+1)(x^2+3)}dx$

32 $\int \frac{x^3+x^2+x+2}{(x^2+1)(x^2+2)}dx$

33 $\int \frac{x^3-2x^2+2x-2}{x^2+1}dx$

34 $\int \frac{x^4+6x^3+10x^2+x}{x^2+6x+10}dx$

## 7.8 – IMPROPAL INTEGRALS

Example 1 Evaluate

$$(a)\ \int_1^{+\infty} \frac{dx}{x^3} \qquad (b)\ \int_1^{+\infty} \frac{dx}{x}$$

Example 2 For what values of p does the integral $\int_1^{+\infty} \frac{dx}{x^p}$ converge?

Example 3 Evaluate $\int_0^{+\infty} (1 - x)e^{-x}\, dx$.

Example 4 Evaluate $\int_{-\infty}^{+\infty} \frac{dx}{1+x^2}$ .

Example 5 Evaluate $\int_0^1 \frac{dx}{\sqrt{1-x}}$ .

Example 6 Evaluate

$$(a)\ \int_1^2 \frac{dx}{1-x} \qquad (b)\ \int_1^4 \frac{dx}{(x-2)^{2/3}}$$

Example 7 Derive the formula for the circumference of a circle of radius r.

## EXERCISE SET 7.8

**1.** In each part, determine whether the integral is improper, and if so, explain why.

$(a)\ \int_1^5 \frac{dx}{x-3}$ $(b)\ \int_1^5 \frac{dx}{x+3}$ $(c)\ \int_0^1 \ln x\, dx$

$(d)\ \int_1^{+\infty} e^{-x}\, dx$ $(e)\ \int_{-\infty}^{+\infty} \frac{dx}{\sqrt[3]{x-1}}$ $(f)\ \int_0^{\pi/4} tanx\, x\, dx$

2. In each part, determine all values of p for which the integral is improper.

$(a)\ \int_0^1 \frac{dx}{x^p}$ $(b)\ \int_1^2 \frac{dx}{x-p}$ $(c)\ \int_0^1 e^{-px}\, dx$

Evaluate the integrals that converge (3–32).

3. $\int_0^{+\infty} e^{-2x}$

4. $\int_{-1}^{+\infty} \frac{x}{1+x^2}\, dx$

5. $\int_3^{+\infty} \frac{2}{x^2-1}\, dx$

6. $\int_0^{+\infty} xe^{-x^2}\, dx$

7. $\int_e^{+\infty} \frac{1}{x\ln^3 x}\, dx$

8. $\int_2^{+\infty} \frac{1}{x\sqrt{\ln x}}\, dx$

9. $\int_{-\infty}^0 \frac{dx}{(2x-1)^3}$

10. $\int_{-\infty}^3 \frac{dx}{x^2+9}$

11. $\int_{-\infty}^0 e^{3x}\, dx$

12. $\int_{-\infty}^0 \frac{e^x\, dx}{3-2e^x}$

13. $\int_{-\infty}^{+\infty} x\, dx$

14. $\int_{-\infty}^{+\infty} \frac{x}{\sqrt{x^2+2}}\, dx$

15. $\int_{-\infty}^{+\infty} \frac{x}{(x^2+3)^2}\, dx$

16. $\int_{-\infty}^{+\infty} \frac{e^{-t}}{1+e^{-2t}}\, dt$

17. $\int_0^4 \frac{dx}{(x-4)^2}$

18. $\int_0^8 \frac{dx}{\sqrt[3]{x}}$

19. $\int_0^{\pi/2} \tan x\, dx$

20. $\int_0^4 \frac{dx}{\sqrt{4-x}}$

21. $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

22. $\int_{-3}^1 \frac{x\, dx}{\sqrt{9-x^2}}$

23. $\int_{\pi/3}^{\pi/2} \frac{\sin x}{\sqrt{1-2\cos x}}\, dx$

24. $\int_0^{\pi/4} \frac{\sec^2 x}{1-\tan x}\, dx$

25. $\int_0^3 \frac{dx}{x-2}$

26. $\int_{-2}^2 \frac{dx}{x^2}$

27. $\int_{-1}^8 x^{-1/3}\, dx$

28. $\int_0^1 \frac{dx}{(x-1)^{2/3}}$

29. $\int_0^{+\infty} \frac{1}{x^2}\,dx$

30. $\int_1^{+\infty} \frac{dx}{x\sqrt{x^2-1}}$

31. $\int_0^1 \frac{dx}{\sqrt{x}(x+1)}$

32. $\int_0^{+\infty} \frac{dx}{\sqrt{x}(x+1)}$

Determine whether the statement is true or false. Explain your answer (33-36).

33. $\int_1^{+\infty} x^{-4/3}\,dx$ converges to 3.

34. If *f* is continuous on [a, +∞) and $lim_{x\to+\infty}\ f(x) = 1$, then $\int_a^{+\infty} f(x)\,dx$ converges.

35. $\int_1^2 \frac{1}{x(x-3)}\,dx$ is an improper integral.

36. $\int_{-1}^1 \frac{1}{x^3}\,dx = 0$

Make the u-substitution and evaluate the resulting definite integral (37-40).

37. $\int_0^{+\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}}\,dx$; $u = \sqrt{x}$ [*Note:* $u \to +\infty$ as $x\to+\infty$.]

38. $\int_{12}^{+\infty} \frac{dx}{\sqrt{x}(x+4)}$; $u = \sqrt{x}$ [*Note:* $u \to +\infty$ as $x\to+\infty$.]

39. $\int_0^{+\infty} \frac{e^{-x}}{\sqrt{1-e^{-x}}}\,dx$; $u = 1 - e^{-x}$ [*Note:* $u \to 1$ as $x\to+\infty$.]

40. $\int_0^{+\infty} \frac{e^{-x}}{\sqrt{1-e^{-2x}}}\,dx$; $u = e^{-x}$

44. In each part, confirm the result with a CAS.

(*a*) $\int_0^{+\infty} \frac{\sin x}{\sqrt{x}}\,dx = \sqrt{\frac{\pi}{2}}$ (*b*) $\int_{-\infty}^{+\infty} e^{-x^2}\,dx = \sqrt{\pi}$ (*c*) $\int_0^1 \frac{lnx}{1+x}\,dx = -\frac{\pi^2}{12}$

45. Find the length of the curve $y = (4 - x^{2/3})^{3/2}$ over the interval [0, 8].

46. Find the length of the curve $y = \sqrt{4 - x^2}$ over the interval [0, 2].

Use L'Hôpital's rule to help evaluate the improper integral(47-48).

47. $\int_0^1 \ln x\,dx$

48. $\int_1^{+\infty} \frac{\ln x}{x^2}\, dx$

49. Find the area of the region between the x-axis and the curve y = $e^{-3x} for\ x \geq 0$.

50. Find the area of the region between the x-axis and the curve y = $\frac{8}{x^2-4} for\ x \geq 4$.

51. Suppose that the region between the x-axis and the curve y = $e^{-x} for\ x \geq 0$ is revolved about the x-axis.

(a) Find the volume of the solid that is generated.

(b) Find the surface area of the solid.

## CHAPTER 7 REVIEW EXERCISE

1. $\int \sqrt{4+9x}\, dx$
2. $\int \frac{1}{\sec \pi x} dx$
3. $\int \sqrt{\cos x} \sin x\, dx$
4. $\int \frac{dx}{x \ln x}$
5. $\int x\, tan^2(x^2) sec^2(x^2)\, dx$
6. $\int_0^9 \frac{\sqrt{x}}{x+9} dx$
7. (a) Evaluate the integral

$$\int \frac{1}{\sqrt{2x - x^2}} dx$$

three ways: using the substitution u = $\sqrt{x}$, using the substitution u = $\sqrt{2-x}$, and

completing the square.

(b) Show that the answers in part (a) are equivalent.

8. Evaluate the integral $\int_0^1 \frac{x^3}{\sqrt{x^2+1}} dx$

(a) using integration by parts

(b) using the substitution u = $\sqrt{x^2+1}$ .

Use integration by parts to evaluate the integral (9-12).

9. $\int x\, e^{-x}\, dx$

10. $\int x \sin 2x\, dx$

11. $\int \ln(2x+3)\, dx$

12. $\int_0^{1/2} tan^{-1}(2x)\, dx$

13. Evaluate $\int 8x^4 \cos 2x\, dx$ using tabular integration by parts.

14. A particle moving along the x-axis has velocity function v(t) = $t^2 e^{-t}$. How far

does the particle travel from time t =0 to t =5?

Evaluate the integral (15-20).

15. $\int sin^2 5\theta\, d\theta$

16. $\int sin^3 2x\, cos^2 2x\, dx$

17. $\int \sin x \cos 2x\, dx$

18. $\int_0^{\pi/6} \sin 2x \cos 4x\, dx$

19. $\int sin^4 2x\, dx$

20. $\int x\, cos^5(x^2)\, dx$

Evaluate the integral by making an appropriate trigonometric substitution(21-26).

21. $\int \frac{x^2}{\sqrt{9-x^2}}\, dx$

22. $\int \frac{dx}{x^2\sqrt{16-x^2}}$

23. $\int \frac{dx}{\sqrt{x^2-1}}$

24. $\int \frac{x^2}{\sqrt{x^2-25}}\, dx$

25. $\int \frac{x^2}{\sqrt{9+x^2}}\, dx$

26. $\int \frac{\sqrt{1+4x^2}}{x}\, dx$

Evaluate the integral using the method of partial fractions(27-32).

27. $\int \frac{dx}{x^2+3x-4}$

28. $\int \frac{dx}{x^2+8x+7}$

29. $\int \frac{x^2+2}{x+2}\, dx$

30. $\int \frac{x^2+x-16}{(x-1)(x-3)^2}\, dx$

31. $\int \frac{x^2}{(x+2)^3}\, dx$

32. $\int \frac{dx}{x^3+x}$

33. Consider the integral $\int \frac{1}{x^3-x}\, dx$.

(a) Evaluate the integral using the substitution x = sec θ . For what values of x is your result valid?

(b) Evaluate the integral using the substitution $x = \sin\theta$ . For what values of x is your result valid?

(c) Evaluate the integral using the method of partial fractions. For what values of x is your result valid?

34. Find the area of the region that is enclosed by the curves $y = (x - 3)/(x^3 + x^2)$, $y = 0$, $x = 1$, and $x = 2$.

Evaluate the integral if it converges (47-50).

47. $\int_0^{+\infty} e^{-x}\, dx$

48. $\int_{-\infty}^{2} \frac{dx}{x^2+4}$

49. $\int_0^9 \frac{dx}{\sqrt{9-x}}$

50. $\int_0^1 \frac{1}{2x-1}\, dx$

**51.** Find the area that is enclosed between the x-axis and the curve $y = (\ln x - 1)/x^2$ for $x \geq e$.

**52.** Find the volume of the solid that is generated when the region between the x-axis and the curve $y = e^{-x}$ for $x \geq 0$ is revolved about the y -axis.

**53.** Find a positive value of a that satisfies the equation

$$\int_0^{+\infty} \frac{1}{x^2+a^2}\, dx = 1$$

**54.** Consider the following methods for evaluating integrals: u-substitution, integration by parts, partial fractions, reduction formulas, and trigonometric substitutions. In each part, state the approach that you would try first to evaluate the integral. If none of them seems appropriate, then say so. You need not evaluate the integral.

(*a*) $\int x \sin x\, dx$

(*b*) $\int \cos x \sin x\, dx$

(*c*) $\int \tan^7 x\, dx$

(*d*) $\int \tan^7 x \sec^2 x\, dx$

(e) $\int \frac{3x^2}{x^3+1}\, dx$

(*f*) $\int \frac{3x^2}{(x+1)^3}\, dx$

(*g*) $\int tan^{-1}x\, dx$

(h) $\int \sqrt{4-x^2}\, dx$

(i) $\int x\sqrt{4-x^2}\, dx$

Evaluate the integral (55-74).

55. $\int \frac{dx}{(3+x^2)^{3/2}}$

56. $\int x \cos 3x \ dx$

57. $\int_0^{\pi/4} tan^7\,\theta \ d\theta$

58. $\int \frac{\cos\theta}{sin^2\theta - 6sin\theta + 12} d\theta$

59. $\int \sin^2 2x \cos^3 2x \ dx$

60. $\int_0^4 \frac{1}{(x-3)^2} \ dx$

61. $\int e^{2x} \cos 3x \ dx$

62. $\int_{-1/\sqrt{2}}^{1/\sqrt{2}} (1-2x^2)^{3/2} \ dx$

63. $\int \frac{dx}{(x-1)(x+2)(x-3)}$

64. $\int_0^{1/3} \frac{dx}{(4-9x^2)^2}$

65. $\int_4^8 \frac{\sqrt{x-4}}{x} \ dx$

66. $\int_0^{\ln 2} \sqrt{e^x - 1} \ dx$

67. $\int \frac{1}{\sqrt{e^x+1}} \ dx$

68. $\int \frac{dx}{x(x^2+x+1)}$

69. $\int_0^{1/2} sin^{-1}x \ dx$

70. $\int tan^5 4x \ sec^4 4x \ dx$

71. $\int \frac{x+3}{\sqrt{x^2+2x+2}} dx$

72. $\int \frac{sec^2\theta}{tan^3\theta - tan^2\theta} d\theta$

73. $\int_a^{+\infty} \frac{x}{(x^2+1)^2} \ dx$

74. $\int_0^{+\infty} \frac{dx}{a^2+b^2x^2}$, a, b > 0

# TRIGONOMETRY ALL GROWN UP

It is time we settled what is known and unknown about trigonometry. If you had Calculus I with me then you have already heard many of these things but for the most part I did not require you remember them. That time is over. I expect you to absorb the material in this section one way or another. I do expect you remember all the trigonometric identities I present in this section, more than that, I expect you learn how to derive similar identities if the need arises. Logically there are a variety of routes to remember this material. I choose the route that allows us to derive as much as possible through essentially algebraic arguments. I call this the imaginary exponential technique. There is nothing really "imaginary" about this, its unfortunate terminology since complex numbers are just as "real" as real numbers. Both real and imaginary numbers are well-defined mathematical objects. There are rules and equations which govern them. Both provide a language which is used to describe a plethora of physical systems (although, I would argue, that is not necessary for them to be sensible mathematical objects)

**9.3.1: What is a complex number?**

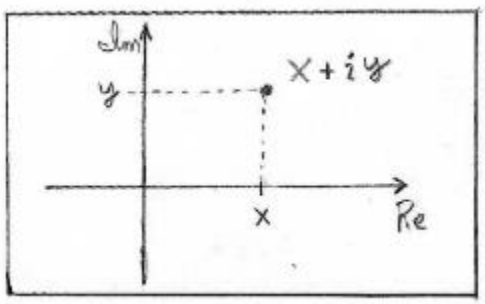
Complex numbers are pairs of real numbers that enjoy a certain rather beautiful multiplication; $(a,b)*(c,d)=(ac-bd,ad+bc)$. This is usually denoted

$$(a+ib)(c+id)=ac+iad+ibc+i^2bd=ac-bd+i(ad+bc)$$

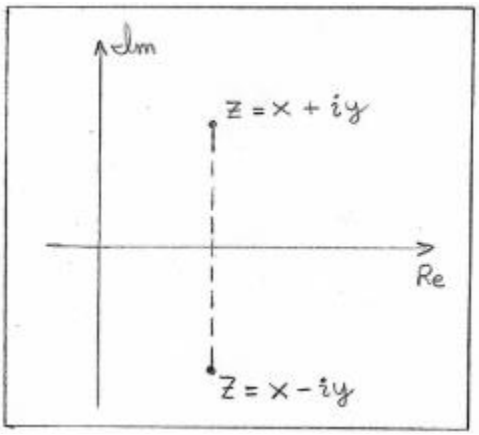
Where the observation $i^2=(0,1)*(0,1)=(-1,0)=-1$. We typically introduce this by saying that $i=\sqrt{-1}$ but this is the same as saying $i^2=-1$. Complex numbers can be added, subtracted, multiplied and divided just the same as real numbers. Complex number have a real and imaginary part,

$$\text{Re}\,(a,b)=\text{Re}\,(a+ib)=a\;\text{Im}\,(a,b)=\text{Im}\,(a+ib)=b$$

In general, if $z\in\mathbb{C}$ then $z=\text{Re}\,(z)+i\text{Im}\,(z)$. It should be emphasized that $\text{Re}\,(z),\text{Im}\,(z)\in\mathbb{R}$ so there is a natural correspondence between complex numbers and the Cartesian Plane $\mathbb{R}^2$; I use this correspondence when I write $(x,y)=x+iy$. This plane is called the complex plane. The $x$-axis is called the real-axis, the $y$-axis is called the imaginary-axis.




Every complex number $z=x+iy$ has a complex-conjugate $z^*=x-iy$. In the complex plane the mapping $z\to z^*$ is a reflection across the $x$-axis.




Recall that any polynomial with real-coefficients can be completely factored over the complex numbers. For example, we usually say that $x^2 + 1$ is an irreducible quadratic. This is true with respect to real numbers, however if we use complex numbers to assist with the factorization then we can factor $x^2 + 1 = (x - i)(x + i)$. Generally, a quadratic polynomial $ax^2 + bx + c$ with $b^2 - 4ac < 0$ is called irreducible because we cannot factor it over the real numbers. Notice that the quadratic formula still makes sense in this case it just gives complex solutions. We can pull an $i = \sqrt{-1}$ out of the square root;

$\sqrt{b^2 - 4ac} = \sqrt{-1(4ac - b^2)} = i\sqrt{4ac - b^2}$ where the quantity $\sqrt{4ac - b^2} \in \mathbb{R}$ since $4ac - b^2 > 0$. If $ax^2 + bx^2 + c = 0$ then it can be shown,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm i\sqrt{4ac - b^2}}{2a} = \alpha \pm i\beta$$

Where I have defined $\text{Re}\,(x) = \alpha = -\frac{b}{2a}$ and $\text{Im}\,(x) = \beta = \frac{\sqrt{4ac-b^2}}{2a}$. The quadratic polynomial factors as follows in this case:

$$x^2 + bx + c = a(x - (\alpha + i\beta))(x - (\alpha + i\beta))$$

The roots $\alpha + i\beta$ and $\alpha - i\beta$ form a conjugate pair. Any polynomial with real coefficients can be completely factored with the help of complex numbers. When an irreducible quadratic appears in the factorization it gives rise to a pair of linear factors whose roots form a conjugate pair. There is much more to say about complex numbers but this little subsection will more than suffice for the purposes of this course.

**9.3.2: What is the complex exponential function?**

We define exp: $\mathbb{C} \to \mathbb{C}$ by the following formula:

$$\exp(z) = \exp(\text{Re}\,(z) + i\text{Im}\,(z)) \equiv e^{\text{Re}\,(z)}(\cos(\text{Im}\,(z)) + i\sin(\text{Im}\,(z)))$$

We can show that this definition yields the following desirable properties:

1.) $\exp(z + w) = \exp(z)\exp(w)$

2.) $e^{\text{Re}\,(z)} = \text{Re}\,(\exp(z))$

3.) $\exp(i\text{Im}\,(z)) = \cos(\text{Im}\,(z)) + i\sin(\text{Im}\,(z))$

4.) $\exp(0) = 1$

5.) $\exp(-z) = \frac{1}{\exp(z)}$

Here $e^{Re(z)}$ denotes the plain-old real exponential function we discussed at length in previous chapters. Essentially, the second condition says that the complex exponential function must reproduce the real exponential function when the input is a complex number with zero imaginary part. Condition 3.) is called Euler's Identity.

Let me show you the proof of 1.). Suppose that $z = x + iy$ and $w = a + ib$ where $x, y, a, b \in \mathbb{R}$. Observe:

$$\begin{aligned}\exp(z+w) &= \exp(x + iy + a + ib)\\ &= \exp(x + a + i(y + b))\end{aligned}$$

$$\begin{aligned}&= e^{x+a}(\cos(y + b) + i\sin(y + b)), \text{ used the definition here}\\ &= e^x e^a(\cos(y)\cos(b) - \sin(y)\sin(b) + i(\sin(y)\cos(b) + \cos(y)\sin(b)))\end{aligned}$$

In the last step we used the adding angles formulas for sine and cosine. These can be derived geometrically. They follow from the law of cosines. On the other hand observe:

$$\begin{aligned}\exp(z)\exp(w) &= \exp(x + iy)\exp(a + ib)\\ &= e^x(\cos(y) + i\sin(y))e^a(\cos(b) + i\sin(b))\\ &= e^x e^a(\cos(y)\cos(b) - \sin(y)\sin(b) + i(\sin(y)\cos(b) + \cos(y)\sin(b)))\end{aligned}$$

Comparing the equations above we verify that $\exp(z + w) = \exp(z)\exp(w)$. I will use the notation $\exp(z) = e^z$ from this point onward.

**9.3.3: Deconstructing sine and cosine**

We can calculate that $e^{\pm i\theta} = \cos(\pm\theta) + i\sin(\pm\theta) = \cos(\theta) \pm i\sin(\theta)$ because cosine is even and sine is odd. If we add and subtract $e^{i\theta} = \cos(\theta) + i\sin(\theta)$ and $e^{-i\theta} = \cos(\theta) - i\sin(\theta)$ then we obtain formulas for sine and cosine in terms of the imaginary exponentials $e^{i\theta}, e^{-i\theta}$ as follows:

$$\cos(\theta) = \frac{1}{2}\left(e^{i\theta} + e^{-i\theta}\right) \quad \sin(\theta) = \frac{1}{2i}\left(e^{i\theta} - e^{-i\theta}\right)$$

Now we can derive pretty much any trigonometric identity you run across.

**Example 9.3.3.1:** Notice this gives us a way to calculate $\int \cos^2(x)dx$

$$\begin{aligned}\cos^2(x) &= \frac{1}{2}\left(e^{ix}+e^{-ix}\right)\frac{1}{2}\left(e^{ix}+e^{-ix}\right)\\ &= \frac{1}{4}\left(\left(e^{ix}\right)^2+2e^{ix}e^{-ix}+\left(e^{-ix}\right)^2\right)\\ &= \frac{1}{4}\left(e^{2ix}+2+e^{-2ix}\right)\\ &= \frac{1}{2}\left[1+\frac{1}{2}\left(e^{2ix}+e^{-2ix}\right)\right]\\ &= \frac{1}{2}(1+\cos(2x)).\end{aligned}$$

**Example 9.3.3.2:** Notice this gives us a way to calculate $\int \sin^2(x)dx$

$$\begin{aligned}\sin^2(x) &= \frac{1}{2i}\left(e^{ix}-e^{-ix}\right)\frac{1}{2i}\left(e^{ix}-e^{-ix}\right)\\ &= \frac{1}{4i^2}\left(\left(e^{ix}\right)^2-2e^{ix}e^{-ix}+\left(e^{-ix}\right)^2\right)\\ &= \frac{1}{-4}\left(e^{2ix}-2+e^{-2ix}\right)\\ &= \frac{1}{2}\left[1-\frac{1}{2}\left(e^{2ix}+e^{-2ix}\right)\right]\\ &= \frac{1}{2}(1-\cos(2x))\end{aligned}$$

Notice that $\sin^2(x)+\cos^2(x) = \frac{1}{2}(1-\cos(2x))+\frac{1}{2}(1+\cos(2x)) = 1$. Another way to look at this is that if you have either of these identities then you could use the pythagorean identity to obtain the other one:

$$\sin^2(x) = 1-\cos^2(x) = 1-\frac{1}{2}(1+\cos(2x)) = \frac{1}{2}(1-\cos(2x)).$$

Let's see another application of the imaginary exponentials.
Example 9.3.3.3: The identity below makes the integration of $\cos(x)\sin(4x)$ easy.

$$\begin{aligned}\cos(x)\sin(4x) &= \frac{1}{2}\left(e^{ix}+e^{-ix}\right)\frac{1}{2i}\left(e^{4ix}-e^{-4ix}\right)\\ &= \frac{1}{4i}\left(\left(e^{ix}e^{4ix}-e^{ix}e^{-4ix}+e^{-ix}e^{4ix}-e^{-ix}e^{-4ix}\right)\right.\\ &= \frac{1}{4i}\left(e^{5ix}-e^{-5ix}+e^{ix}-e^{-ix}\right)\\ &= \frac{1}{2}\frac{1}{2i}\left(e^{5ix}-e^{-5ix}\right)+\frac{1}{2}\frac{1}{2i}\left(e^{ix}-e^{-ix}\right)\\ &= \frac{1}{2}\sin(5x)+\frac{1}{2}\sin(x)\end{aligned}$$

Example 9.3.3.4: The identity below makes the integration of $\cos(ax)\cos(bx)$ easy.

$$\begin{aligned}\cos(ax)\cos(bx) &= \frac{1}{2}\left(e^{aix}+e^{-aix}\right)\frac{1}{2i}\left(e^{bix}-e^{-bix}\right)\\ &= \frac{1}{4}\left(\left(e^{aix}e^{bix}+e^{aix}e^{-bix}+e^{-aix}e^{bix}+e^{-aix}e^{-bix}\right)\right.\\ &= \frac{1}{4}\left(e^{(a+b)ix}+e^{-(a+b)ix}+e^{(a-b)ix}+e^{-(a-b)ix}\right)\\ &= \frac{1}{2}\cos((a+b)x)+\frac{1}{2}\cos((a-b)x).\end{aligned}$$

Let's see another application of the imaginary exponentials.
Example 9.3.3.3: The identity below makes the integration of $\cos(x)\sin(4x)$ easy.

$$\begin{aligned}\cos(x)\sin(4x) &= \frac{1}{2}\left(e^{ix}+e^{-ix}\right)\frac{1}{2i}\left(e^{4ix}-e^{-4ix}\right)\\ &= \frac{1}{4i}\left(\left(e^{ix}e^{4ix}-e^{ix}e^{-4ix}+e^{-ix}e^{4ix}-e^{-ix}e^{-4ix}\right)\right.\\ &= \frac{1}{4i}\left(e^{5ix}-e^{-5ix}+e^{ix}-e^{-ix}\right)\\ &= \frac{1}{2}\frac{1}{2i}\left(e^{5ix}-e^{-5ix}\right)+\frac{1}{2}\frac{1}{2i}\left(e^{ix}-e^{-ix}\right)\\ &= \frac{1}{2}\sin(5x)+\frac{1}{2}\sin(x)\end{aligned}$$

Example 9.3.3.4: The identity below makes the integration of $\cos(ax)\cos(bx)$ easy.

$$\begin{aligned}\cos(ax)\cos(bx) &= \frac{1}{2}\left(e^{aix}+e^{-aix}\right)\frac{1}{2i}\left(e^{bix}-e^{-bix}\right)\\ &= \frac{1}{4}\left(\left(e^{aix}e^{bix}+e^{aix}e^{-bix}+e^{-aix}e^{bix}+e^{-aix}e^{-bix}\right)\right.\\ &= \frac{1}{4}\left(e^{(a+b)ix}+e^{-(a+b)ix}+e^{(a-b)ix}+e^{-(a-b)ix}\right)\\ &= \frac{1}{2}\cos((a+b)x)+\frac{1}{2}\cos((a-b)x).\end{aligned}$$

I hope you can see the idea here. Perhaps you do not yet appreciate why we need to

find trigonometric identities for integration, but I'll fix that in a quiz sometime soon...

insert maniacal laughter here…

**9.3.4:** Identities that we can derive (or memorize)
The identities that follow are either geometrically motivated or can be derived via the methods advertised at length in the last section.
**1.)** $\cos(A+B)=\cos(A)\cos(B)-\sin(A)\sin(B)$
**2.)** $\sin(A+B)=\sin(A)\cos(B)+\cos(A)\sin(B)$
**3.)** $\sin\left(x+\frac{\pi}{2}\right)=\cos(x)$
**4.)** $\cos\left(x-\frac{\pi}{2}\right)=\sin(x)$

**5.)** $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$

**6.)** $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$

**7.)** $\sin(2x) = 2\sin(x)\cos(x)$

**8.)** $\cos(2x) = \cos^2(x) - \sin^2(x)$

**9.)** $\cos(A)\cos(B) = \frac{1}{2}\cos(A - B) + \frac{1}{2}\cos(A + B)$

**10.)** $\sin(A)\sin(B) = \frac{1}{2}\cos(A - B) - \frac{1}{2}\cos(A + B)$

**11.)** $\sin(A)\cos(B) = \frac{1}{2}\sin(A - B) + \frac{1}{2}\sin(A + B)$

I do expect you can derive or recall all of the identities above. As we will see in the next subsection these identities will help us integrate many otherwise intractable integrals.

# MIT Integration Bee

# 2023

## Regular Season 2023

**Problems**:

1 $\int_0^{2\pi} \max(\sin(x), \sin(2x))dx$

2 $\int_0^1 \frac{\frac{x+1}{x+2}}{\frac{x+3}{x+4}} dx$

3 $\int_0^3 \left(\min\left(2x, \frac{5-x}{2}\right) - \max\left(-\frac{x}{2}, 2x - 5\right)\right) dx$

4 $\int \underbrace{1 - \frac{1}{1 - \frac{1}{1 - \frac{1}{1-}}}}_{2023(1-)'} dx$

5 $\int_0^{\pi/2} x\cot x dx$

6 $\int \left(\frac{x^6 + x^4 - x^2 - 1}{x^4}\right) e^{x + 1/x} dx$

7 $\int \frac{dx}{\sqrt{(x+1)^3(x-1)}}$

8 $\int_0^{\pi} x\sin^4(x)dx$

9 $\int \binom{x}{5}^{-1} dx$

10 $\int \frac{\sin(2x)\cos(3x)}{\sin^2 x \cos^3 x} dx$

11 $\int \left(\sqrt{2\log x} + \frac{1}{\sqrt{2\log x}}\right) dx$

12 $\int \frac{\log(\cos(x))}{\cos^2(x)} dx$

13 $\int_0^{\frac{\pi}{2}+1} \sin(x - \sin(x - \sin(x - \cdots))) dx$

14 $\int_0^{100} \lfloor x \rfloor x \lceil x \rceil dx = \frac{100^4 - 100^2}{4}$

15 $\int_{-\infty}^{\infty} \frac{\frac{1}{(x-1)^2} + \frac{3}{(x-3)^2} + \frac{5}{(x-5)^2}}{1 + \left(\frac{1}{x-1} + \frac{1}{(x-3)^3} + \frac{1}{(x-5)^3}\right)^2} dx$

16 $\int_0^{\pi} \sin^2(3x + \cos^4(5x)) dx$

17 $\int_0^5 (-1)^{\lfloor x \rfloor + \lfloor x/\sqrt{2} \rfloor + \lfloor x/\sqrt{3} \rfloor} dx$

18 $\int_0^{\infty} \frac{x+1}{x+2} \cdot \frac{x+3}{x+4} \cdot \frac{x+5}{x+6} \cdots dx$

19 $\int_0^{\frac{\pi}{2}} \frac{\sin(23x)}{\sin(x)} dx$

20 $\int_1^{100} \left(\frac{\lfloor x/2 \rfloor}{\lfloor x \rfloor} + \frac{\lceil x/2 \rceil}{\lceil x \rceil}\right) dx$

**Answers:**

1. $\frac{5}{2}$

2. $1 - \log\left(\frac{81}{64}\right)$

3. 5

4. $x - \log x$

5. $\frac{\pi}{2} \log 2$

6. $\left(x^2 - 2x + 4 - \frac{2}{x} + \frac{1}{x^2}\right) e^{x+1/x}$

7. $\sqrt{\frac{x-1}{x+1}}$

8. $\frac{3\pi^2}{16}$

9. $5\log(x) - 20\log(x-1) + 30\log(x-2) - 20\log(x-3) + 5\log(x-4)$

10. $8\log\sin x - 6\log\tan x$

11. $x\sqrt{2\log x}$

12. $\tan(x)\log(\cos(x)) + \tan(x) - x$

13. $\frac{3}{2}$

14. $24997500$

15. $3\pi$

16. $\frac{\pi}{2}$

17. $8\sqrt{2} + 6\sqrt{3} - 21$

18. $0$

19. $\frac{\pi}{2}$

20. $\frac{197}{2}$

## Quarter Final 2023

**Problems:**

1 $\int_0^1 \frac{x^4(1-x)^2}{1+x^2}dx$

2 $\int \begin{pmatrix}\cos(3x)\cos(5x)\cos(6x)\cos(7x)\\ -\cos(x)\cos(2x)\cos(4x)\cos(8x)\end{pmatrix}dx$

3 $\int_{\sqrt{e}}^{\infty} x^{-\log x}dx$

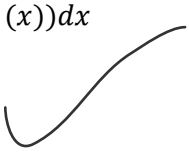
4 $\int \frac{1-2x}{(1+x)^2x^{2/3}}dx$

5 $\lim_{n\to\infty}\left(\frac{1}{n}\int_0^n \cos^2\left(\frac{\pi x^2}{\sqrt{2}}\right)dx\right)$

6 $\int_0^{2^{20}} \sum_{n=0}^{\infty} \left\{\frac{x}{2^n}\right\} dx$

7 $\int_0^{\infty} \operatorname{sech}^2(x + \tan(x))dx$

8 $\int_0^{\pi/2} \frac{dx}{1+\cos x+\sin x}$



9 $\lim_{\varepsilon \to 0^+} \left(\varepsilon^4 \int_0^{\pi/2-\varepsilon} \tan^5(x)dx\right)$

10 $\int_0^1 \left\lfloor \sqrt{1+\frac{1}{x}} \right\rfloor dx$

**Answers:**

1. $\frac{7}{10} - \log 2$
2. $\frac{1}{8}\left(\frac{\sin(21x)}{21} - \frac{\sin(13x)}{13}\right)$
3. $\frac{\sqrt[4]{e\pi^2}}{2}$
4. $\frac{3x^{1/3}}{1+x}$
5. $\frac{1}{2}$
6. $12 \cdot 2^9 = 6144$
7. 1
8. $\log 2$
9. $\frac{1}{4}$
10. $\frac{7}{4}$

## Semi Final 2023

**Problems:**

1 $\int e^{\cos x} \cos(2x + \sin x)dx$

2 $\int_0^1 (9x^9 - x^{50} + 9x^{m9} - x^{900} + 9x^{200} - x^{500} + 9x^{900} - x^{900} + \cdots)dx$

3 $\int_0^{\pi} \frac{2\cos(x)-\cos(2021x)-2\cos(2022x)-\cos(2023x)+2}{1-\cos(2x)}dx$

4 $\int \frac{3\log x-1+2x}{x\log x+x^2+2x^4}dx$

5 $\int (\sqrt{x+1}-\sqrt{x})^{\pi}dx$

6 $\int_{-2}^{2} ((((x^2-2)^2-2)^2-2)^2-2)dx$

7 $\int_0^{\infty} \frac{\tanh(x)}{x\cosh(2x)}dx$

8 $\int \sin(4\arctan(x))dx$

**Answers:**

1. $e^{\cos x}(\sin(x+\sin x)-\sin(\sin x))$
2. $1$
3. $2022\pi$
4. $3\log x-\log(\log x+x+2x^3)$
5. $\frac{1}{2}\left(\frac{(\sqrt{x+1}-\sqrt{x})^{\pi+2}}{\pi+2}-\frac{(\sqrt{x+1}-\sqrt{x})^{\pi-2}}{\pi-2}\right)$
6. $-\frac{8}{255}$
7. $\log 2$
8. $-\frac{4}{1+x^2}-2\log(1+x^2)$

# Final 2023

**Problems**:

1 $\int_0^{\frac{\pi}{2}} \frac{\sqrt[3]{\tan x}}{(\sin x+\cos x)^2}dx$

2 $\int_0^{\pi} \left(\frac{\sin(2x)\sin(3x)\sin(5x)\sin(30x)}{\sin(x)\sin(6x)\sin(10x)\sin(15x)}\right)^2 dx$

3 $\int_{-1/2}^{1/2} \sqrt{x^2+1+\sqrt{x^4+x^2+1}}dx$

4 $\left\lfloor 10^{20}\int_2^{\infty} \frac{x^9}{x^{20}-48x^{10}+575}dx \right\rfloor$

5 $\int_0^1 \left(\sum_{n=1}^{\infty} \frac{\lfloor 2^n x \rfloor}{3^n}\right)^2 dx$

**Answers**:

1. $\frac{2\sqrt{3}\pi}{9}$
2. $7\pi$
3. $\frac{\sqrt{7}}{2\sqrt{2}} + \frac{3}{4\sqrt{2}} \log\left(\frac{\sqrt{7}+2}{\sqrt{3}}\right)$
4. $10^{16} + \frac{10^{10}-1}{3} + \frac{10^4}{5} = 10000003333335333$
5. $\frac{27}{32}$

# 2022

## Regular Season 2022

### Problems

1. $\int_0^{100} \lceil \sqrt{x} \rceil dx = \boxed{715}$
2. $\int \frac{log(1+x)}{x^2} dx = log(x) - \frac{(x+1)log(x+1)}{x}$
3. $\int_{\frac{\pi}{2}-1}^{\frac{\pi}{2}+1} cos(arcsin(arccos(sin((x))))dx = \boxed{\frac{\pi}{2}}$
4. $\int_{-2}^{2} |(x-2)(x-1)x(x+1)(x+2)| dx = \boxed{\frac{19}{3}}$
5. $\int 2020 sin^{2019}(x) cos^{2019}(x) - 8084 sin^{2021}(x) cos^{2021}(x) dx$
   a. $= \boxed{sin^{2020}(x) cos^{2022}(x) - sin^{2022}(x) cos^{2020}(x)}$
6. $\int \frac{3x^3+2x^2+1}{\sqrt[3]{x^3+1}} dx = \boxed{(x+1)(x^3+1)^{\frac{2}{3}}}$
7. $\int \frac{1}{sin^4 x cos^4 x} dx = \boxed{-8cot2x - \frac{8}{3} cot^3 2x}$
8. $\int \frac{x+sinx}{1+cosx} dx = \boxed{x \tan\frac{x}{2}}$
9. $\int sinh^3 x cosh^2 x dx = \boxed{\frac{1}{5}\left(sinh^2 x - \frac{2}{3}\right) cosh^3 x}$
10. $\int 4^x 3^{2^x} = \boxed{\frac{3^{2^x}}{log2}\left(\frac{2^x}{log3} - \frac{1}{(log3)^2}\right)}$
11. $\int \frac{cos(x)-sin(x)}{2+sin(2x)} dx = \boxed{arctan(sin(x)+cos(x))}$

**12.** $\int \frac{sec^2(1+logx)-tan(1+logx)}{x^2}dx = \boxed{\frac{tan(1+logx)}{x}}$

**13.** $\int_0^1 \sqrt{(\frac{1}{x}\ log\ \frac{1}{x})}dx = \boxed{\sqrt{(2\pi)}}$

14. $\sum_{n=2}^{\infty} \int_0^{\infty} \frac{(x-1)x^n}{1+x^n+x^{n+1}+x^{2n+1}}dx = \boxed{\frac{\pi}{2}-1}$

15. $\int_0^{2\pi} \left(1-cos(x)\right)^5 cos(5x)dx = \boxed{-\frac{\pi}{16}}$

16. $\int_0^{10} \lceil x \rceil \left(\max_{k\in\mathbb{Z}\geq 0} \frac{x^k}{k!}\right)dx = \frac{10^{10}}{9!}$

**17.** $\int \frac{4sinx+3\ cosx}{3sinx+4cosx}dx = \boxed{\frac{24x-7log(3sinx+4cosx)}{25}}$

**18.** $\int_{-1}^{1} \sqrt{(4-(1+|x|)^2)} - \left(\sqrt{3}-\sqrt{(4-x^2)}\right)dx = \boxed{2\pi-2\sqrt{3}}$

19. $\int x^2 sin(logx)dx = \frac{x^3}{10}(3sin(logx)-cos(logx))$

**20.** $\int_0^{\infty} (36x^5-12x^6+x^7)e^{(-x)}dx = \boxed{720}$

# Quarter Final 2022

## Problems

**1.** $\int_1^{2022} \frac{\{x\}}{x}dx = \boxed{2021-log\left(\frac{2022^{2021}}{2021!}\right)}$

**2.** $\lim_{n\to\infty} n\int_0^{\frac{\pi}{4}} tan^n(x)dx = \boxed{\frac{1}{2}}$

**3.** $\int_0^{\infty} \frac{x^{1010}}{(1+x)^{2022}}dx = \boxed{\frac{1010!^2}{2021!}}$

4. $\int \arcsin(x)\arccos(x)dx = \boxed{x(\arcsin(x)\arccos(x)+2)+\sqrt{1-x^2}(\arccos(x)-\arcsin(x))}$

5. $max_{\{x_1,x_2,x_3,x_4,x_5,x_6,x_7\}=\{1,2,3,4,5,6,7\}} \int_{\int_{x_1}^{x_2} x_3dx}^{\int_{x_4}^{x_5} x_6dx} x_7dx = \boxed{217}$

**6.** $\lim_{n\to\infty} \sqrt[n]{\int_0^2 (1+6x-7x^2+4x^3-x^4)^n dx} = \boxed{3}$

7. $\int_0^1 \frac{x^4}{\sqrt{(1-x)}}dx = \boxed{\frac{256}{315}}$

8. $\int_0^{\infty} \left\lceil \frac{1}{\lceil x \rceil - x} \right\rceil^{-\lceil x \rceil} dx = \boxed{\frac{\pi^2}{6}-1}$

9. $\lim_{n\to\infty} n\int_0^{\infty} sin\left(\frac{1}{x^n}\right)dx = \boxed{\frac{\pi}{2}}$

10. $\int_0^1 \frac{-x+\sqrt{4-3x^2}}{2}dx = \boxed{\frac{\pi}{3\sqrt{3}}}$

11. $\lim_{n\to\infty} \sqrt{n}\int_{-\frac{1}{2}}^{\frac{1}{2}} (1-3x^2+x^4)^n dx = \boxed{\sqrt{\frac{\pi}{3}}}$

12. $\int_{\frac{1}{2022}}^{2022} \frac{1+x^2}{x^2+x^{2022}} = \boxed{2022-\frac{1}{2022}}$

# Semi Final 2022

## Problems

1. $\int_0^{\infty} \frac{x(e^{-x}+1)}{e^x-1}dx = \boxed{\frac{\pi^2}{3}-1}$

2. $\int_0^{\infty} x^5 e^{-x} \sin(x) dx = \boxed{-15}$
3. $\int_{\frac{1}{2}}^{2} \log\left(\frac{\log\left(x+\frac{1}{x}\right)}{\log\left(x^2-x+\frac{17}{4}\right)}\right) dx = \boxed{-\frac{3}{2}\log 2}$
4. $\int_2^{\frac{5}{2}} \frac{(x^3-3x)^3-3(x^3-3x)}{\sqrt{x^2-4}} dx = \boxed{\frac{2^9-2^{-9}}{9}}$
5. $\int_{-1}^{1} \text{||||} \left| x\left|-\frac{2}{3}\right| - \frac{2}{3^2}\left|-\frac{2}{3^3}\right| - \cdots \right| dx = \boxed{\frac{1}{7}}$
6. $\int \frac{1}{(x^2+1)^3} dx = \boxed{\frac{3}{8}\arctan(x) + \frac{3x^3+5x}{8(x^2+1)^2}}$
7. $\int_{-\infty}^{\infty} \frac{1}{x^2-2x\cot(x)+\csc^2(x)} dx = \boxed{\pi}$
8. $\int_0^{\frac{\pi}{6}} log\left(\sqrt{3}+tanx\right) dx = \boxed{\frac{\pi \log(2)}{6}}$

# Final 2022

**Problems**

1. $\int \sqrt{(\sin(20x) + 3\sin(21x) + \sin(22x))^2 + (\cos(20x) + 3\cos(21x) + \cos(22x))^2}\, dx = \boxed{3x + 2sinx}$
2. $\int_0^{\infty} \frac{e^{-2x} sin(3x)}{x} dx = \boxed{arctan\frac{3}{2}}$
3. $\int_{\mathbf{0}}^{\mathbf{2\pi}} cos(2022x) \frac{sin(10050x)sin(10251x)}{sin(50x)sin(51x)} dx = \boxed{6\pi}$
4. $\int_0^1 x^{\frac{1}{3}}(1-x)^{\frac{2}{3}} dx = \boxed{\frac{2\pi}{9\sqrt{3}}}$
5. $\left\lfloor log \int_{2022}^{\infty} 10^{-x^3}\, dx \right\rfloor = \boxed{-2022^3 - 8}$

# Integration bee 2022

1 $\int \frac{1}{1-e^{-2022x}}\, dx$
2 $\int \frac{\tan^3(1+\ln x)}{x}\, dx$
3 $\int \frac{1}{2-2x+x^2}\, dx$
4 $\int_{-1}^{0} \frac{15165}{2} x\sqrt{1+x} dx$
5 $\int_0^1 (x-3^x)^2\, dx$
6 $\int \frac{x^{12}}{x^2+1}\, dx$
7 $\int \frac{x^2}{(x-1)(x^2+x+1)} dx$
8 $\int \frac{\sin 2x}{1+2\sin^2 x}\, dx$
9 $\int_0^1 (2x-1)\frac{\sqrt{x}+\sqrt{1-x}}{\sqrt{x}-\sqrt{1-x}}\, dx$
10 $\int \frac{2\cos x - 2\sin x}{2+\sin 2x}\, dx$
11 $\int_0^e x^{\frac{1}{\ln x}}\, dx$

12 $\int \sqrt{x\sqrt{x\sqrt{x\ldots}}}\,\mathrm{d}x$

13 $\int_1^2 x^{x^2+1}(2\ln\, x+1)dx$

14 $\int_0^{2022\pi} \sin\,(\sin\, x-x)\mathrm{d}x$

15 $\int_0^1 x\tan^{-1}\,((x-1)(x+1))\mathrm{d}x$

16 $\int \frac{1}{\sqrt{x\sqrt{x}-x^2}}\,\mathrm{d}x$

17 $\int_1^{e^2} \left(\ln\left(x^{\ln\, x-6}\right)+5\right)x^3\,\mathrm{d}x$

18 $\int_0^{\pi/2} \frac{1}{\tan^{\sqrt{2022}}(x)+1}\,\mathrm{d}x$

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19 . $\int_0^1 \sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}\,dx$

20 $\int_0^{\frac{\pi}{2}} \prod_{k=1}^7 \cos\, kxdx$

# Round 2

# Knockout Match 1

1 $\int \sqrt{12-3x^2}dx$

2 $\int \frac{x}{1-5x^2}\sqrt{\frac{2}{1+5x^2}-1}dx$

3 $\int \frac{x^{1010}}{1-x^{21/22}}\,\mathrm{d}x$

4 $\int_{-\infty}^0 x^{3xdzz^{lx}}\mathrm{dr}$

5 $\int_0^1 2^{[h_h 2]}dr$

# Knockout Match 2

6 $\int \frac{8x^3}{1+x^5}\,\mathrm{d}x$

7 $\int_{-\pi/4}^{x/4} \frac{1}{1-\sin\, x}\,\mathrm{d}x$

8 $\int_{-1}^{\frac{3}{2}} \frac{\cos\, x(1+\tan^{-1}\, x)}{2-\cos^2\, x}dx$

9 $\iint_2^{\log_2 5} \frac{2^{2x+1}-9\times 2^x-6}{4^x-2^x-6}dx$

10 Define $f(x)$ as the $[x]$ th decinal place of $x$. Find $\int_1^{202n} f(x)\mathrm{d}x$.

# Knockout Match 3

11 $\int_0^\infty \frac{1}{\left(x+\frac{1}{x}\right)^2}dx$

12 $\int_{\frac{1}{b}}^1 \sum_{k=0}^\infty \left(x^k\left((k+1)\ln\, x+x^{k^2+k}\right)dx\right.$

13 $\int_0^\infty \frac{1}{(1+x^{2(x2)})(1+x^2)}dx$

14 $\int_0^\infty \sum_{k=[x]}^\infty \frac{2^{-k}}{k+1}dx$

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## Extra Questions

15 $\iint \frac{\sin^3 x}{\sqrt{\cos x}} dx$

16 $\int_0^1 \frac{(\sin^{-1} x)^2}{\sqrt{1-x^2}} dx$

17 $\iint \ln(1+\sqrt{x}) dx$

# File from 2013

1 $\int \log(x^2) - 2\log(2x) dx = -x\log(4) + C$

2 $\int_{-1}^{3} e^{|x|} dx - e^3 + e - 2$

3 $\int \frac{(\log x)(\cos x) - (\sin x)(1/x)}{(\log x)^2} dx = \frac{\sin x}{\log x} + C$

4 $\int_1^{11} x^3 - 3x^2 + 3x - 1 dx = 2500$

5 $\int_0^2 \sqrt{12 - 3x^2} dx = \pi\sqrt{3}$

6 $\int_0^6 x + (x-3)^7 + \sin(x-3) dx = 1$

7 $\int \sin x \sqrt{1 + \tan^2 x} dx = -\log \cos x + C$

8 $\int \frac{x^5 - x^3 + x^2 - 1}{x^4 - x^3 + x - 1} dx = \frac{x^2}{2} + x + C$

9 $\int_0^1 \log x dx = -1$

10 $\int \frac{1}{1-e^{-x}} dx = \log(1 - e^x) + C$

11 $\int_0^x \sin^2 x \cos^2 x \, dx = \pi/8$

12 $\int_0^{441} \frac{\pi \sin(\pi\sqrt{x})}{\sqrt{x}} dx = 4$

13 $\tan^2 \int x \, dx = -x + \tan x + C$

14 $\int_0^{256} (x - \lfloor x \rfloor)^2 dx = 256/3$

15 $\int e^{\sqrt[4]{x}} dx - e^{\sqrt[4]{x}} \left(4x^{3/4} - 12\sqrt{x} + 24\sqrt[4]{x} - 24\right) + C$

16 $\int \cos x \cot x \, dx = \cos x - \log \cos(x/2) + \log \sin(x/2) + C$

17 $\int 2\log x + (\log x)^2 dx = x(\log x)^2 + \mathrm{C}$

18 $\int \frac{x^3}{1+x^2} dx = \frac{x^2}{2} - \frac{1}{2}\log(x^2 + 1) + C$

19 $\int \frac{1}{2 - 2x + x^2} dx = -\tan^{-1}(1 - x) + C$

20 $\int \sin x \log(\sin x) \, dx = \cos(x) + \log\left(\tan\left(\frac{x}{2}\right)\right) - \cos(x)\log(\sin(x)) + C$

21 $\int \frac{x}{1-x^4} dx = \frac{1}{4}(\log(1 + x^2) - \log(1 - x^2)) + C$

22 $\int \sqrt{12 - 3x^2} dx = \frac{\sqrt{3}}{2}\left(x\sqrt{4 - x^2} + 4\sin^{-1}\left(\frac{x}{2}\right)\right) + \mathrm{C}$

23 $\sec^5 \int x \tan^3 x \, dx - \frac{1}{35}\sec^5(x)(5\sec^2(x) - 7)$

24 $\int_{-3/4}^{x/4} \frac{1}{1 - \sin x} dx = 2$

# MIT Integration Bee Qualifying Exam, 21 January 2014

1 $\int_1^e \log(x^2) dx = 2$

2 $\int_{-9}^{9} \sin(\sqrt[3]{x}) dx = 0 = \frac{1}{2}x^2 \arctan(x) - \frac{x}{2} + \frac{1}{2}\arctan(x)$

3 $\int_0^{\infty} \frac{d}{dx}\left[e^{1+x-x^2}\right] dx = -e$

4 $\int_0^2 \sqrt{x+\sqrt{x+\sqrt{x+\cdots}}}\,dx = \frac{19}{6}$

5 $\int \sqrt{x}e^{\sqrt{x}}dx = 2e^{\sqrt{x}}(x-2\sqrt{x}+2)$

6 $\int \sin(2x)\cos(3x)\,dx = \frac{\cos x}{2} - \frac{\cos 5x}{10}$

7 $\int_0^{2\pi} 1+2\sin x \mid dx = \frac{2\pi}{3} + 4\sqrt{3}$

8 $\int x(1-x)^{2014}dx = \frac{(1-x)^{2016}}{2016} - \frac{(1-x)^{2015}}{2015}$

9 $\int \operatorname{arcsinh}(x) = x\operatorname{arcsinh}(x) - \sqrt{x^2+1}$

10 $\int_{-1}^{0} \frac{x^2}{x-1}dx = \frac{1}{2} - \log 2$

11 $x\arctan\int x\,dx$

12 $\int \frac{dx}{x^2-15x-2014} = \frac{1}{91}\log\left|\frac{x-53}{x+38}\right|$

13 $\int e^x[\log(1+x^2) - 2(1+x)\arctan x]dx = e^x[\log(1+x^2) - 2x\arctan x]$

14 $\int (\arcsin x)^2 dx = 2\sqrt{1-x^2}\arcsin x - 2x + x(\arcsin x)^2$

15 $\int \frac{\sqrt{x^2-1}}{x}dx = \sqrt{x^2-1} - \arctan\sqrt{x^2-1}$

16 $\int x\sec^2(4x)\,dx - \frac{x\tan(4x)}{4} + \frac{\log(\cos(4x))}{16}$

17 $\int \frac{2}{6-11x+6x^2-x^3}dx = 2\log(2-x) - \log(x^2-4x+3)$

18 $\int_0^1 \frac{1}{[1-\log_2(1-x)]}dx = \log 2$

19 $\int_0^{1/\sqrt{3}} \sqrt{x+\sqrt{x^2+1}}dx = \frac{2}{3}$

20 $\int_0^{5\pi/2} \frac{dx}{2+\cos x} = \frac{7\sqrt{3}\pi}{9}$

# MIT Integration Bee Qualifying Exam, 20 January 2015

1 $\int (\cos^4 x - \sin^4 x)dx = \frac{1}{2}\sin 2x$

2 $\int \frac{x}{\sqrt{2+4x}}dx = \frac{1}{3}(x-1)\sqrt{x+\frac{1}{2}}$

3 $\int_0^8 \frac{\cos\sqrt{x}}{\sqrt{x}}dx = 2\sin\left(2\sqrt{2}\right) = \frac{1}{28} - \frac{3}{64}\log\left(\frac{21}{5}\right)$

4 $\int \sec x dx = \operatorname{sech}^{-1}\cos x = \log(1+\log x) + \log x(\log(1+\log x) - 1)$

5 $\int_0^{\frac{x}{2}} \frac{e^{\sin x}}{\tan x \csc x}dx = e - 1$

6 $\int_1^e x(\log x)^2 dx = \frac{e^2}{4} - \frac{1}{4}$

7 $\int \frac{1}{5+4\sqrt{x}+x}dx = \log\left(x+4\sqrt{x}+5\right) - 4\tan^{-1}\left(\sqrt{x}+2\right)$

8 $\int (2015)^x dx = \frac{(2015)^x}{\log(2015)}$

9 $\int_0^2 \frac{x}{(x-3)(x+5)^2}dx$

10 $\frac{\log(1+\log x)}{x}dx$

11 $\int \sqrt{\csc x - \sin x}\,dx = 2^{\frac{3}{4}}$

12 $\int \frac{1}{\sqrt{x^2+25}}dx = \log\left|x+\sqrt{x^2+25}\right|$

13 $\int_2^e \frac{\log^2 x-1}{x\log^2 x}dx = -\frac{(\log 2-1)^2}{\log 2}$

14 $e^{3x}\arctan\int e^x dx = \frac{e^{3x}\arctan(e^x)}{3} - \frac{e^{2x}}{6} + \frac{\log(1+e^{2x})}{6}$

15 $\int_0^{2\pi} \frac{1}{\sin^4 x+\cos^4 x}dx = 2\pi\sqrt{2}$

16 $\int \frac{1+e^x}{1-e^x}dx - x - 2lo\,g|1-e^x|$

17 $\tan^4\int x\,dx = \frac{\tan^3 x}{3} - ta\,n\,x + x = 2 + \frac{9}{4}lo\,g\,3 - \frac{3}{4}lo\,g\,5$

18 $\int si\,n\,x\tan^2 x\,dx = co\,s\,x + \frac{1}{co\,s\,x}$

19 $\int \frac{x+1}{x^2+2x+3}dx = \frac{1}{2}\log(x^2+2x+3)$

# MIT Integration Bee Qualifying Exam (Ans),10 January 2016

[1] $\int \tanh x dx = \log(\cosh x)$

[2] $\int_4^4 |x^2-x|dx - 113$

[3] $\int_1^e \text{lot}(\sqrt{x})\,dx = \frac{1}{2}$

[4] $\int \frac{\log(\log(x))}{x\log(x)}dx = \frac{1}{2}\log(\log(x))^2$

[6] $\int_0^{\frac{\pi}{3}} \frac{dx}{1+\tan^2(x)} = \frac{\pi}{6} + \frac{\sqrt{3}}{8}$

[7] $\int_{-27}^{27} arcsin\left(\frac{x^{1/3}}{3}\right)dx = 0$

[8] $\int_{50}^{100} [\log_2 x]dx = 286$

[9] $\int (e^x\cos x - e^x\sin x)dx = e^x\cos x$

[10] $\int_0^{\infty} x^3 e^{-x^2}dr = \frac{1}{2}$

[11] $\int \left((2e^{x^2}x+1)\cos x - (e^{x^2}+x)\sin x\right)dx = (x+e^{x^2})\cos x$

[12] $\int \sin\left(\sin(x)\right)\cos(\sin(x))\cos(x)dx = \frac{1}{2}\sin^2(\sin(x))$

[13] $\int \left(\frac{\cos(x)+\sin(x)}{x^2} + \frac{\sin(x)-\cos(x)}{x}\right)dx = -\frac{\sin x+\cos x}{x}$

[14] $\int x^3\sqrt{x^2+1}dx = \frac{1}{15}(x^2+1)^{3/2}(3x^2-2)$

[15] $\int \frac{x}{x^4+x^2+1}dx = \frac{1}{\sqrt{3}}\arctan\left(\frac{2x^2+1}{\sqrt{3}}\right)$

[16] $\int e^{e^{2016x+6048x}}dx = \frac{1}{2016}(e^{4032x} - 2e^{2016x} + 2)e^{e^{2016x}}$

[17] $\int_{\pi/3}^{pi/2} \frac{1-\cos x}{\sin x}dx = \log(3/2)$

[18] $\int \frac{dx}{1-x+x^2-x^3} = \frac{1}{2}\left(\arctan(x) - \log(1-x) + \log\sqrt{1+x^2}\right)$

[19] $\int_0^{\infty} \frac{dx}{2+\cosh x} = \frac{\log(2+\sqrt{3})}{\sqrt{3}}$

# File from 2017

1. Integration of $\frac{x^2}{\sqrt{x^3+2}}dx = \frac{2\sqrt{x^3+2}}{3}$

2. $\int_1^\infty \frac{\log\ x}{x^2}dx = 1$
3. $\int \text{sech}\ (x)dx = 2\arctan\ (e^x)$
4. $\int x^3 e^{x^2}dx = \frac{1}{2}(x^2-1)e^{x^2}$
5. $\int_1^2 \frac{1}{x\sqrt{x^2-1}}dx = \pi/3$
6. $\int_1^\infty \frac{dx}{x(x^2+1)} = \frac{\log\ 2}{2}$
7. $\int \cosh^{-1}\ xdx = x\cosh^{-1}\ x - \sqrt{x^2-1}$
8. $\int_{-\infty}^\infty e^{-2x^2-5x-3}dx = e^{1/8}\sqrt{\pi/2}$
9. $\int \sin\ \sqrt{x}dx = 2\sin\ \sqrt{x} - 2\sqrt{x}\cos\ \sqrt{x}$
10. $\int_0^\infty \frac{dx}{(x+1/x)^2} = \frac{\pi}{4}$
11. $\int \frac{(2+x)e^{-x}}{x^3}dx = -\frac{e^{-x}}{x^2}$
12. $\int_0^1 \frac{dx}{\sqrt{x(1-x)}} = \pi$
13. $\int_0^\infty \frac{\tanh\ (x)}{\exp\ (x)}dx = \frac{\pi}{2} - 1$
14. $\int_0^{\frac{\pi}{2}} \sqrt{\sin\ (x)+1}dx = 2$
15. $\lim_{n\to\infty} I_n$, where $I_1 = \int_0^1 \frac{dx}{1+\sqrt{x}}$, $I_2 = \int_0^1 \frac{dx}{1+\frac{1}{1+\sqrt{x}}}$, ... $= \frac{\sqrt{5}-1}{2}$
16. $\int_{-\infty}^\infty \frac{\sin^2\ (x+\pi/4)}{e^{x^2}}dx = \frac{\sqrt{\pi}}{2}$
17. $\int_{-\infty}^\infty 3x^2(x^3+1)^2 e^{-x^6-2x^3}dx = \frac{e\sqrt{\pi}}{2}$
18. $\int_0^{\pi/2} \frac{dx}{1+\tan^{2017}\ x} = \pi/4$
19. $\int e^{2x}\cos\ (3x)dx = \frac{1}{13}e^{2x}(3\sin\ (3x) + 2\cos\ (3x))$
20. $\int (\cos\ (x))^{\cos\ (x)+1}\tan\ (x)\big(1 + \log\ (\cos\ (x))dx = -\cos\ (x)^{\cos\ (x)}$

# MIT Integration Bee Qualifying Exam (Ans), 23 January 2018

1. $\int \frac{e^x}{e^x+2}dx = \log\ (e^x+2)$
2. $\int \sqrt{x\cdot\sqrt[3]{x\cdot\sqrt[4]{x\cdot\sqrt[3]{x\cdots}}}}dx = \frac{x^{e-1}}{e-1}$
3. $\int_0^{2018\pi} |\sin\ (2018x)|dx = 4036$
4. $\int \frac{dx}{\tan\ x + \cot\ x} = \frac{1}{2}\sin^2\ (x)$

5. $\int \frac{x^5}{2+x^{12}}dx = \frac{1}{6\sqrt{2}}\arctan\left(x^6/\sqrt{2}\right)$
6. $\int (\cos\ x\cosh\ x + \sin\ x\sinh\ x)dx = \sin(x)\cosh(x)$
7. $\int \frac{e^x + \cos\ x}{e^x + \sin\ x}dx = \log(e^x + \sin\ x)$
8. $\int \sin(\cos(\sin\ x))\sin(\sin\ x)\cos\ xdx = \cos(\cos(\sin(x))$
9. $\int \frac{dx}{1+\sin(x)} = \tan\ x - \sec\ x$
10. $\int \frac{\cos\ x}{1-\cos(2x)}dx = -\frac{1}{2}\csc\ x$
11. $\int e^x(1/x + \log\ x)dx = e^2\log\ x$
12. $\int \tanh^2(x)dx = x - \tanh\ x$
13. $\int \frac{2017x^{2016} + 2018x^{2017}}{1 + x^{5054} + 2x^{4035} + x^{4036}}dx = \arctan(x^{2018} + x^{2017})$
14. $\int \frac{\sin(2x) - \sin^2\ x}{\cos(2x) - \cos^2\ x}dx = x - 2\log(\sin\ x)$
15. $\int \frac{dx}{x^{\frac{41}{15}} \cdot x^{\frac{16}{15}} + x^{\frac{5}{2x}}} = \frac{25}{16}\arctan\left(x^{\frac{46}{212}}\right)$
16. $\int_0^{\pi/2} \frac{\cos(x)}{2-\cos^2(x)}dx = \frac{\pi}{4}$
17. $\int \frac{dx}{(1+x^2)^{3/2}} = \frac{x}{\sqrt{1+x^2}}$
18. $\int \frac{dx}{\sqrt{x\sqrt{x} - x^2}} = 4\arcsin(\sqrt[4]{x})$
19. $\int \frac{x-1}{x + x^2\log\ x}dx = \log(1/x + \log\ x)$
20. $\int \csc(x)\sec(x)dx = \log(\tan(x))$

# MIT Integration Bee Qualifying Exam (Ans), 29 January 2019

1 $\int_0^{2\pi} \tan(\cos(x))dx = 0$

2 $\int \frac{x+1}{x(x+\log\ x)}dx = \log(x + \log\ x)$

3 $\int e^{x+e^x} + e^{x-e^x}dx = e^{e^x} - e^{-e^x}$

4 $\int_{-1/2}^{1/2} \frac{dx}{1-x^2} = \log\ 3$

5 $\int_0^2 2^{\log\ x}dx = \frac{2^{1+\log\ 2}}{1+\log\ 2}$

6 $\int_{-2\pi}^{2\pi} (\cos\ 3x + \sin\ 2x)(-\sin\ 2019x + \cos\ 3x)dx = 2\pi$

7 $\int \cos\ x \cdot \cos\ \sin\ x \cdot \cos\ \sin\ \sin\ xdx = \sin\ \sin\ \sin\ x$

8 $\int_0^{\infty} \frac{e^{-\frac{2019}{4t^2}}}{t^2}dt = \sqrt{\frac{\pi}{2019}}$

9 $\int \sin(\sqrt{x})dx = 2\sin(\sqrt{x}) - 2\sqrt{x}\cos(\sqrt{x})$

10 $\int_0^1 \frac{\sqrt{x}}{1+x}dx = 2 - \frac{\pi}{2}$

11 $\int_0^{2\pi} \cos(x)\cos(2x)\cos(3x)dx = \frac{\pi}{2}$

12 $\lim_{n\to\infty} \int_{-\infty}^{\infty} e^{-x^{2n}}dx = 2$

13 $\int_0^e x^{\frac{1}{\log_1}}dx = e^2$

14 $\int_0^{\pi/100} \frac{\sin(20x)+\sin(19x)}{\cos(20x)+\cos(19x)}dx = -\frac{2}{39}\log\cos\frac{39\pi}{200}$

15 $\int (e^x\cos^2(x) + e^x\cos(x)\sin(x) - e^x\sin^2(x))dx = e^x\cos(x)\sin(x)$

16 $\int_0^{\pi/2} \frac{\sin x}{\sin(x+\pi/4)}dx = \frac{\pi}{2\sqrt{2}}$

17 $\int \frac{dx}{x+\sqrt[3]{x}} = \frac{3}{2}\log\left(1+x^{2/3}\right)$

18 $\int_0^2 x^{x^2+1}(2\log(x)+1)dx = 15$

19 $\int \frac{2x^3-1}{x(x^3+1)}dx = \log(x^3+1) - \log x$

20 $\int \cos(\arctan(x))dx = \sinh^{-1} x = \log\left(\sqrt{x^2+1}+x\right)$

# MIT Integration Bee Qualifying Exam (Ans), 21 January 2020

1 $\int \frac{\log(2x)}{x\log x}dx = \log x + (\log 2)\log\log x$

2 $\int_0^\infty \frac{1}{e^x+1}dx = \log 2$

3 $\int_e^{e^e} \frac{\log x \cdot \log(\log x)}{x}dx = \frac{1}{4}(1+e^2)$

4 $\int_0^1 \log\left(\frac{1+x}{1-x}\right)dx = \log 4$

5 $\int \frac{1}{x^2+(x-1)^2}dx = \arctan(2x-1)$

6 $\int \sqrt{x\sqrt{x\sqrt{x\cdots}}}dx = x^2/2$

7 $\int \sin^4 x\cos^4 x(\cos x + \sin x)(\cos x - \sin x)dx = \frac{1}{5}\sin^5(x)\cos^5(x)$

8 $\int \log(x^2+1)dx = x\log(x^2+1) + 2\arctan(x) - 2x$

9 $\int_0^{2\pi} \cos^{2020}(x)dx = 2^{-2019}\pi\binom{2020}{1010}$

10 $\int \frac{2x+1}{2x^2+2x+1}dx = \frac{1}{2}\ln(2x^2+2x+1)$

11 $\int_{1/\sqrt{2}}^1 \frac{\arcsin(x)}{x^3}dx = 1/2$

12 $\int_0^{\pi/2} \sin(2x)\cos(\cos(x))dx = 2(\cos(1) + \sin(1) - 1)$

13 $\int_0^{2\pi} \sin(\sin(x) - x)dx = 0$

14 $\int \frac{1}{x-1} + \frac{\sum_{k=0}^{2018}(k+1)x^k}{\sum_{k=0}^{2019}x^k}dx = \log(1 - x^{2020})$

15 $\int_0^{\frac{\pi}{2}} \frac{1}{\tan^{\sqrt{2020}}(x) + 1}dx = \pi/4$

16 $\int x(1-x)^{2020}dx = \frac{(1-x)^{2022}}{2022} - \frac{(1-x)^{2021}}{2021}$

17 $\int \frac{\sec^4(x)\tan(x)}{\sec^4(x) + 4}dx = \frac{1}{4}\log(\sec^4(x) + 4)$

18 $\int x^{2x}(2\log(x) + 2)dx = x^{2x}$

19 $\int_0^1 \sqrt{1-x^2}dx = \pi/4$

20 $\int_0^{\infty} x^5 e^{-x^4}dx = \sqrt{\pi}/8$

# MIT Integration Bee Qualifying Exam (Ans), 18 January 2022

1 $\int \frac{1 + \cos x}{x + \sin x}dx = \log(x + \sin x)$

2 $\int_1^{\sqrt{3}} \frac{\arctan x + \operatorname{arccot} x}{x}dx = \frac{\pi}{4}\log(3)$

3 $\int_0^{2022} x^2 - \lfloor x \rfloor \lceil x \rceil dx = 674$

4 $\int \frac{\sinh(x)}{\cosh(x) - \sinh(x)}dx = -\frac{1}{2}x + \frac{1}{4}e^{2x}$

5 $\int \frac{x}{\sqrt{x-1} + \sqrt{x+1}}dx = \frac{(x+1)^{\frac{5}{2}}}{5} - \frac{(x+1)^{\frac{3}{2}}}{3} - \frac{(x-1)^{\frac{5}{2}}}{5} - \frac{(x-1)^{\frac{3}{2}}}{3}$

6 $\int_0^{\pi} \cos(x + \cos x)dx = 0$

7 $\int x^3\sin(x^2)dx = \frac{\sin(x^2) - x^2\cos(x^2)}{2}$

8 $\int \frac{x}{1-x^4}dx = \frac{1}{4}\log\left(\frac{1+x^2}{1-x^2}\right)$

9 $\int \frac{1}{\cosh^2 x}dx = \tanh(x)$

10 $\int_0^1 e^{e^x} - e^{e^x - x}dx = e^{e-1} - e$

11 $\lim_{n\to\infty} \int_0^3 \underbrace{\sin\left(\frac{\pi}{3}\sin\left(\frac{\pi}{3}\sin\left(\ldots\sin\left(\frac{\pi}{3}x\right)\ldots\right)\right)\right)}_{n \text{ sin's}}dx = \frac{3}{2}$

12 $\int_0^1 \sqrt{1 - \sqrt{x}}dx = \frac{8}{15}$

13 $\int \frac{x^3}{1 + x + \frac{x^2}{2} + \frac{x^3}{6}}dx = 6\left(x - \log\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)\right)$

14 $\int \sin(x + \sin(x)) - \sin(x - \sin(x))dx = -2\cos(\sin(x))$

15 $\int \tan^4(x)\sec^3(x) + \tan^2(x)\sec^5(x)dx = \frac{1}{3}\tan^3(x)\sec^3(x)$

16 $\int (1 + \log x)\log\log x dx = x\log x\log\log x - x$

17 $\int \frac{1}{1+\sin x} + \frac{1}{1+\cos x} + \frac{1}{1+\tan x} + \frac{1}{1+\cot x} + \frac{1}{1+\sec x} + \frac{1}{1+\csc x} dx =$

18 $\int \frac{1}{\sqrt{x-x^2}} dx = 2\arcsin(\sqrt{x})$

19 $\int_0^{\frac{1}{2}} \sum_{n=0}^{\infty} \binom{n+3}{n} x^n dx = \frac{7}{3}$

20 $\int \frac{1}{1+\cos^2 x} dx = \frac{1}{\sqrt{2}} \arctan\left(\frac{\tan(x)}{\sqrt{2}}\right)$

# MIT Integration Bee Qualifying Exam (Ans), 24 January 2023

1 $\int x^{\frac{1}{\log x}} dx = ex$

2 $\int \operatorname{sech}(x) dx = 2\arctan(e^x)$

3 $\int \frac{e^x}{(1+e^x)\log(1+e^x)} dx = \log\left(\log(1+e^x)\right)$

4 $\int (1 + x + x^2 + x^3 + x^4)(1 - x + x^2 - x^3 + x^4) dx = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \frac{x^9}{9}$

5 $\int_0^4 \binom{x}{5} dx = 0$

6 $\int (x + \sin(x) + x\cos(x) + \sin(x)\cos(x)) dx = \frac{(x+\sin(x))^2}{2}$

7 $\int (\sin^2(x) + \cos^2(x) + \tan^2(x) + \cot^2(x) + \sec^2(x) + \csc^2(x)) dx$
$= 2\tan(x) - 2\cot(x) - x$

8 $\int_0^{2\pi} \lfloor 2023\sin(x) \rfloor dx = -\pi$

9 $\int (2\log x + 1) e^{(\log x)^2} dx = x e^{(\log x)^2}$

10 $\int \left((1-x)^3 + (x-x^2)^3 + (x^2-1)^3 - 3(1-x)(x-x^2)(x^2-1)\right) dx =$

11 $\int_{-2023}^{2023} \underbrace{|||||x| - 1| - 1| \cdots | - 1|}_{2023(-1)'s} dx = 2023$

12 $\int (\sin^6 x + \cos^6 x + 3\sin^2 x\cos^2 x) dx = x$

13 $\int (x + e + 1) x^e e^x dx = x^{e+1} e^x$

14 $\int_0^1 \left( \frac{x^2}{2-x^2} + \sqrt{\frac{2x}{x+1}} \right) dx =$

15 $\int \frac{1 + 2x^{2022}}{x + x^{2023}} dx = \frac{1}{2022} \log(x^{2022} + x^{4044})$

16 $\int (3\sin(20x)\cos(23x) + 20\sin(43x)) dx = \sin(20x)\sin(23x)$

17 $\int_0^1 \prod_{k=0}^{\infty} \left( \frac{1}{1+x^{2^k}} \right) dx = \frac{1}{2}$

18 $\int \frac{\sin x}{2e^x + \cos x + \sin x} dx = \frac{x - \log(2e^x + \cos x + \sin x)}{2}$

19 $\int \frac{\log(x/\pi)}{(\log x)^{\log(\pi e)}} dx = \frac{x}{(\log x)^{\log \pi}}$

20 $\int_{-3/2}^{-1/2}(x^5+5x^4+10x^3+8x^2+x)dx = \frac{5}{6}$

**Gamma Beta Functions**

1. $\Gamma(n) = \int_0^\infty x^{n-1}e^{-x}dx\,;\ n>0$
2. $\Gamma(n+1) = n\Gamma(n) = n!$
3. $\Gamma(1) = 1, \Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$
4. $\Gamma(n)\Gamma(1-n) = \frac{\pi}{\sin n\pi}\ ;\ 0<n<1$
5. $\int_0^{\frac{\pi}{2}} \sin^p \quad \theta \cos^q \quad \theta d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$
6. $\mathrm{B}(m,n) = \mathrm{B}(n,m)$
7. $\mathrm{B}(m,n) = \int_0^1 x^{m-1}(1-x)^{n-1}\,dx\,;\ m,n>0$
8. $\mathrm{B}(m,n) = 2\int_0^{\frac{\pi}{2}} \sin^{2m-1} \quad \theta \cos^{2n-1} \quad \theta d\theta\,;\ m,n>0$
9. $\mathrm{B}(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$
10. $\int_0^\infty \frac{y^{n-1}}{(1+y)^{m+n}}dy = B(m,n)$
11. **Reduction formula for $\Gamma n$ : $\Gamma(n+1) = n\Gamma n$**
12. $\int_0^\infty e^{-kx}x^{n-1}dx = \frac{\Gamma n}{k^n}$
13. $\beta(m,n) = 2\int_0^{\frac{\pi}{2}} \sin^{2m-1}\ \theta \cos^{2n-1}\ \theta d\theta$
14. $\int_0^{\frac{\pi}{2}} \sin^p\ \theta \cos^q\ \theta d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\Gamma\left(\frac{p+q+2}{2}\right)}$
15. $\Gamma m\Gamma\left(m+\frac{1}{2}\right) = \frac{\sqrt{\pi}\cdot\Gamma(2m)}{2^{2m-1}}, m>0$
16. . $\Gamma n\Gamma(1-n) = \frac{\pi}{\sin\ n\pi}, 0<n<1$
17.

**A.** Evaluate in terms of gamma function

1. $\int_0^4 x^{\frac{3}{2}}(4-x)^{\frac{5}{2}}\,dx$
2. $\int_0^b y^5\,\sqrt{b^2-y^2}dy$
3. $\int_0^\infty e^{-x^2}\,dx$
4. $\int_0^\infty x^5\,e^{-4x}dx$
5. $\int_0^\infty x^6\,e^{-3x}dx$
6. $\int_0^\infty x^5\,e^{-x^2}\,dx$
7. $\int_0^\infty x^9\,e^{-x^2}\,dx$
8. $\int_0^\infty \quad \sqrt{x}e^{-x^2}\,dx$
9. $\int_0^1 \frac{x^3}{\sqrt{1-x^3}}\,dx$
10. $\int_0^1 \frac{1}{\sqrt{\ln\left(\frac{1}{x}\right)}}\,dx$
11. $\int_0^1 \frac{1}{\sqrt{x\ln\left(\frac{1}{x}\right)}}\,dx$
12. $\int_0^1 \left(1-\frac{1}{x}\right)^{\frac{1}{3}}\,dx$

**B.** Evaluate in terms of beta function

1. $\int_0^1 \frac{x^2}{\sqrt{1-x}}\,dx$
2. $\int_0^1 x^7(1-x)^3\,dx$
3. $\int_0^1 \frac{1}{\sqrt{1-x^3}}\,dx$
4. $\int_0^1 x^3\,\sqrt{1-x}dx$
5. $\int_0^1 x^{\frac{5}{2}}(1-x)^{\frac{3}{2}}\,dx$
6. $\int_0^a y^7\,\sqrt{a^4-y^4}dy$

7. $\int_0^4 y^3 \sqrt{64-y^3}dy$

8. $\int_0^1 x^2(1-x^3)^{\frac{3}{2}}dx$

9. $\int_0^\infty \frac{1}{1+x^4}dx$

**C.** Evaluate the following integrals

1. $\int_0^\pi \sin^5\theta \cos^4\theta\, d\theta$

2. $\int_0^\pi \sin^6\theta \cos^7\theta\, d\theta$

3. $\int_0^{\frac{\pi}{6}} \sin^2 6\theta \cos^4 3\theta\, d\theta$

4. $\int_0^{\frac{\pi}{4}} \sin^2 4\theta \cos^3 2\theta\, d\theta$

5. $\int_0^{\frac{\pi}{2}} \sin^4\theta \cos^2\theta\, d\theta$

6. $\int_0^{\frac{\pi}{8}} \sin^2 8\theta \cos^4 4\theta\, d\theta$

**Example 3.** Evaluate $i.\int_0^{\frac{\pi}{2}} \sin^3 x \cos^{\frac{5}{2}} x dx$ ii. $\int_0^{\frac{\pi}{2}} \sin^{10} x dx$ iii. $\int_0^{\frac{\pi}{2}} \sqrt{\tan\theta} + \sqrt{\sec\theta} d\theta$

$$iv \cdot \int_0^2 x^{m-1}(2-x)^{n-1}dx \text{ v. } \int_0^\pi \sin^2\theta(1+\cos\theta)^4 d\theta \; vi \cdot \int_0^1 x^{m-1}\left(\log\frac{1}{x}\right)^{n-1} dx$$

**Example 4.** Prove that $i.\beta(m,n) = \beta(m+1,n) + \beta(m,n+1)$

ii. $\frac{\beta(m+1,n)}{\beta(m,n)} = \frac{m}{m+n}$

iii. $\beta\left(m,\frac{1}{2}\right) = 2^{2m-1}\beta(m,m)$

**Example 5.** Express the following integrals in terms of Beta function

i. $\int_0^1 x^m(1-x^2)^n dx$ ii. $\int_0^1 \frac{x^2}{\sqrt{1-x^5}}dx$

**Example 6.** Prove that $i.\ \Gamma\left(\frac{3}{2}-x\right)\Gamma\left(\frac{3}{2}+x\right) = \left(\frac{1}{4}-x^2\right)\pi.\sec\pi x$

**Example 7.** Express the integral $\int_0^1 x^m(1-x^n)^p dx$ in terms of gamma function and hence evaluate $\int_0^1 x^{\frac{3}{2}}\left(1-x^{\frac{1}{2}}\right)^{\frac{1}{2}} dx$

**Example 8.** Evaluate *i*. $\int_0^\infty x^{n-1}\cos\ axdx$ ii. $\int_0^\infty x^{n-1}\sin\ axdx$

**Exercise**

1 Show that $\int_0^{\frac{\pi}{2}}\tan^n\ xdx = \frac{\pi}{2}\sec\left(\frac{n\pi}{2}\right), 0 < n < 1$

2 Given that $\Gamma\left(\frac{8}{5}\right) = 0.8935$, find the values of $\Gamma\left(-\frac{5}{2}\right)$ and $\Gamma\left(-\frac{12}{5}\right)$

3 Find i. $\beta\left(\frac{3}{2},\frac{1}{2}\right)$ ii. $\beta\left(\frac{4}{3},\frac{5}{3}\right)$

4 Evaluate $\int_0^1 x^2(1-x^2)^4 dx$

5 Prove that $\Gamma n = 2\int_0^\infty e^{-x^2}x^{2n-1}dx$

6 Prove that for $a, b > 0$
i. $\int_0^\infty x^{n-1}e^{-ax}\cos\ bxdx = \frac{\Gamma n}{(a^2+b^2)^{n/2}}\cos\left(n\tan^{-1}\frac{b}{a}\right)$
ii. $\int_0^\infty x^{n-1}e^{-ax}\sin\ bxdx = \frac{\Gamma n}{(a^2+b^2)^{n/2}}\sin\left(n\tan^{-1}\frac{b}{a}\right)$

7 Show that $\frac{\beta(m+1,n)}{m} = \frac{\beta(m,n)}{m+n+1}$

8 Show that $\int_0^{\frac{\pi}{2}}\frac{1}{\sqrt{\sin\ x}}dx \cdot \int_0^{\frac{\pi}{2}}\sqrt{\sin\ x}dx = \pi$

9 Show that $\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}dx = \beta(m,n)$

10 Prove that $\beta\left(m,\frac{1}{2}\right) = 2^{2m-1}\cdot\beta(m,m)$

**Answers**

2 $-\frac{8}{15}\sqrt{\pi}, -1.108$

3 $i\cdot\frac{\pi}{2}$ ii. $\frac{2\pi}{9\sqrt{3}}$

4 $\frac{128}{3465}$

1. $\int_0^\infty x^4 e^{-x} x^{n-1} dx$
2. $\int_0^\infty x^{\frac{3}{2}} e^{-x} dx$
3. $\int_0^\infty x^{\frac{5}{2}} e^{-x} dx$
4. $\int_0^1 x^2 (1-x)^6 dx$
5. $\int_0^1 \left[\frac{1}{\sqrt[3]{[x^2(1-x)]}}\right] dx$
6. $\int_0^1 \left[\frac{1}{\sqrt[4]{[x^3(1-x)]}}\right] dx$
7. $\int_0^1 \sqrt{x} \cdot (1-x) dx$
8. $\int_0^1 \sqrt{x^5} \cdot (1-x) dx$
9. $\int_0^\infty x^6 e^{-2x} dx$
10. $\int_0^\infty \sqrt{x}. e^{-x\wedge 3} dx$
11. $\int_0^\infty x^m e^{-kx\wedge n} dx$
12. $\int_0^2 \frac{x^2}{\sqrt{(2-x)}}. dx$
13. $\int_0^a x^4 . \sqrt{(a^2 - x^2)} . dx$
14. $\int_0^2 x . \sqrt{(8 - x^3)} . dx$
15. $\int_0^\infty \frac{dx}{1+x^4}$
16. a. $\int_0^{\frac{\pi}{2}} sin^3 . cos^2 x . dx$

    b. $\int_0^{\frac{\pi}{2}} sin^4 . cos^5 x . dx$

17. $a. \int_0^{\frac{\pi}{2}} sin^6 x.dx$

b. $\int_0^{\frac{\pi}{2}} cos^6 x.dx$

18. $a. \int_0^{\pi} cos^4 x dx$

b. $\int_0^{2\pi} sin^8 x dx$

19. $a. \int_0^{\infty} x^6.e^{-2x} dx$

b. $\int_0^{\infty} \sqrt{x}.e^{-x^3} dx$

20. $\int_0^{\infty} x^m e^{-kx\wedge n} dx$

21. $\int_0^2 \frac{x^2}{\sqrt{(2-x)}} dx$

# Plane Curves and Area

1 Determine the area below $f(x) = 8x - 2x^2$ and above the x-axis.

2 Determine the area above $f(x) = 3x^2 + 6x - 9$ and below the x-axis.

3 Determine the area to the right of $g(y) = y^2 + 4y - 5$ and to the left of the $y$-axis.

4 Determine the area to the left of $g(y) = -4y^2 + 24y - 20$ and to the right of the $y$-axis.

5 Determine the area below $f(x) = 10 - 2x^2$ and above the line $y = 3$.

6 Determine the area above $f(x) = x^2 + 2x + 3$ and below the line $y = 11$.

7 Determine the area to the right of $g(y) = y^2 + 2y - 4$ and to the left of the line $x = -1$.

8 Determine the area to the left of $g(y) = 2 + 4y - y^2$ and to the right of the line $x = -1$.

**For problems 9 - 26 determine the area of the region bounded by the given set of curves.**

9 $y = x^3 + 2, y = 1$ and $x = 2$.

10 $y = x^2 - 6x + 10$ and $y = 5$.

11 $y = x^2 - 6x + 10, x = 1, x = 5$ and the x-axis.

12 $x = y^2 + 2y + 4$ and $x = 4$.

13 $y = 5 - \sqrt{x}, x = 1, x = 4$ and the $x$-axis.

*14* $x = \boldsymbol{e}^{y}, x = 1, y = 1$ *and* $y = 2$.

*15* $x = 4y - y^2$ *and the* $y$*-axis.*

*16* $y = x^2 + 2x + 4, y = 3x + 6, x = -3$ *and* $x = 3$.

*17* $x = 6y - y^2, x = 2y, y = -2$ *and* $y = 5$.

*18* $y = x^2 + 8, y = 3x^2, x = -3$ *and* $x = 4$.

*19* $x = y^2, x = y^3$ *and* $y = 2$.

20 $y = \frac{7}{x}, y = \frac{1}{x} - 3, x = -1$ and $x = -4$.

21 $y = 2x^2 + 1, y = 7 - x, x = 4$ and the $y$-axis.

22 $y = \sin\left(\frac{1}{2}x\right), y = 3 + \cos(2x), x = 0$ and $x = \frac{\pi}{4}$.

23 23. $x = \sqrt{2y + 6}, x = y - 1, y = 1$ and $y = 6$.

24 $y = 2 - \mathbf{e}^{2-x}, y = x^2 - 4x + 7, x = 3$ and the y-axis. Note: These functions do not intersect.

25 $y = \mathbf{e}^{2x-1}, y = \mathbf{e}^{5-x}, x = 0$ and $x = 3$.

26 $x = \cos(\pi y), x = 3, y = 0$ and $y = 4$.

# Solids of Revolution / Method of Rings

For problems 1 - 16 use the method disks/rings to determine the volume of the solid obtained by rotating the region bounded by the given curves about the given axis.

1 Rotate the region bounded by $y = 2x^2, y = 8$ and the $y$- axis about the $y$-axis.

2 Rotate the region bounded by $y = 2x^2, y = 8$ and the $y$- axis about the $x$-axis.

3 Rotate the region bounded by $y = 2x^2, x = 2$ and the $x$-axis about the $x$-axis.

4 Rotate the region bounded by $y = 2x^2, x = 2$ and the $x$-axis about the $y$-axis.

5 Rotate the region bounded by $x = y^3, x = 8$ and the $x$-ais about the $x$-axis.

6 Rotate the region bounded by $x = y^3, x = 8$ and the $x$-axis about the $y$-axis.

7 Rotate the region bounded by $x = y^3, y = 2$ and the $y$ -axis about the $x$-axis.

8 Rotate the region bounded by $x = y^3, y = 2$ and the $y$-axis about the $y$-axis.

9 Rotate the region bounded by $y = \frac{1}{x^2}, y = 9, x = -2, x = -\frac{1}{3}$ about the $y$-axis.

10 Rotate the region bounded by $y = \frac{1}{x^2}, y = 9, x = -2, x = -\frac{1}{3}$ about the $x$-axis.

11 Rotate the region bounded by $y = 4 + 3\mathbf{e}^{-x}, y = 2, x = \frac{1}{2}$ and $x = 3$ about the $x$-axis.

12 Rotate the region bounded by $x = 5 - y^2$ and $x = 4$ about the $y$-axis.

13 Rotate the region bounded by $y = 6 - 2x, y = 3 + x$ and $x = 3$ about the $x$-axis.

14 Rotate the region bounded by $y = 6 - 2x, y = 3 + x$ and $y = 6$ about the $y$-axis.

15 Rotate the region bounded by $y = x^2 - 2x + 4$ and $y = x + 14$ about the $x$-axis.

16 Rotate the region bounded by $x = (y - 3)^2$ and $x = 16$ about the $y$-axis.

17 Use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by $y = 2x^2, y = 8$ and the $y$-axis about the
(a) line $x = 3$
(b) line $x = -2$
(c) line $y = 11$
(d) line $y = -4$

18 Use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by $x = y^2 - 6y + 9$ and $x = -y^2 + 6y - 1$ about the
(a) line $x = 10$
(b) line $x = -3$ 19. Use the method of disks/rings to determine the volume of the solid obtained by rotating the triangle with vertices $(3,2)$, $(7,2)$ and $(7,14)$ about the
(a) line $x = 12$
(b) line $x = 2$
(c) line $x = -1$
(d) line $y = 14$
(e) line $y = 1$
(f) line $y = -3$

19 Use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by $y = 4 + 3e^{-x}, y = 2, x = \frac{1}{2}$ and $x = 3$ about the
(a) line $y = 7$
(b) line $y = 1$
(c) line $y = -3$

20 Use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by $x = 3 + y^2$ and $x = 2y + 11$ about the
(a) line $x = 23$
(b) line $x = 2$
(c) line $x = -1$

21 Use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by $y = 5 + \sqrt{x}, y = 5$ and $x = 4$ about the
(a) line $y = 8$

(b) line $y = 2$
(c) line $y = -2$

22 Use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by $y = 10 - 2x, y = x + 1$ and $y = 7$ about the
(a) line $x = 8$
(b) line $x = 1$
(c) line $x = -4$

23 Use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by $y = -x^2 - 2x - 5$ and $y = 2x - 17$ about the
(a) line $y = 3$
(b) line $y = -1$
(c) line $y = -34$

24 Use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by $x = -2y^2 - 3$ and $x = -5$ about the
(a) line $x = 4$
(b) line $x = -2$
(c) line $x = -9$

**25 Solids of Revolution / Method of Cylinders**

26 For problems 1 - 14 use the method cylinders to determine the volume of the solid obtained by rotating the region bounded by the given curves about the given axis.

27 Rotate the region bounded by $y = 2x^2, y = 8$ and the $y$-axis about the $y$-axis.

28 Rotate the region bounded by $y = 2x^2, y = 8$ and the $y$-axis about the $x$-axis.

29 Rotate the region bounded by $y = 2x^2, x = 2$ and the $x$-axis about the $x$-axis.

30 Rotate the region bounded by $y = 2x^2, x = 2$ and the $x$-axis about the $y$-axis.

31 Rotate the region bounded by $x = y^3, x = 8$ and the $x$-axis about the $x$-axis.

32 Rotate the region bounded by $x = y^3, x = 8$ and the $x$-axis about the $y$-axis.

33 Rotate the region bounded by $x = y^3, y = 2$ and the $y$-axis about the $x$-axis.

34 Rotate the region bounded by $x = y^3, y = 2$ and the $y$-axis about the $y$-axis.

35 Rotate the region bounded by $y = \frac{1}{x}, y = \frac{1}{3}$ and $x = \frac{1}{2}$ about the $y$-axis.

36 Rotate the region bounded by $y = \frac{1}{x}, y = \frac{1}{3}$ and $x = \frac{1}{2}$ about the $x$-axis.

37 Rotate the region bounded by $y = 6 - 2x, y = 3 + x$ and $x = 3$ about the $y$-axis.

38 Rotate the region bounded by $y = 6 - 2x, y = 3 + x$ and $y = 6$ about the $x$-axis.

39 Rotate the region bounded by $y = x^2 - 6x + 11$ and $y = 6$ about the $y$-axis.

40 Rotate the region bounded by $x = y^2 - 8y + 19$ and $x = 2y + 3$ about the $x$-axis.

41 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $y = 2x^2, y = 8$ and the $y$-axis about the
(a) line $x = 3$
(b) line $x = -2$
(c) line $y = 11$
(d) line $y = -4$

42 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $x = y^3, x = 8$ and the $x$-axis about the
(a) line $x = 10$
(b) line $x = -3$
(c) line $y = 3$
(d) line $y = -4$

43 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $x = y^2 - 6y + 9$ and $x = -y^2 + 6y - 1$ about the
(a) line $y = 7$
(b) line $y = -2$ 18. Use the method of cylinders to determine the volume of the solid obtained by rotating the triangle with vertices $(3,2), (7,2)$ and $(7,14)$ about the
(a) line $x = 12$
(b) line $x = 2$
(c) line $x = -1$
(d) line $y = 14$
(e) line $y = 1$
(f) line $y = -3$

1 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $y = 5 + \sqrt{x - 3}, y = 5$ and $x = 4$ about the
(a) line $x = 9$
(b) line $x = 2$
(c) line $x = -1$

2 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $y = -x^2 - 10x + 6$ and $y = 2x + 26$ about the
(a) line $x = 2$
(b) line $x = -1$
(c) line $x == 14$

3 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $x = y^2 - 10y + 27$ and $x = 11$ about the
(a) line $y = 10$
(b) line $y = 1$
(c) line $y = -3$

4 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $y = 2x^2 + 1, y = 7 - x, x = 3$ and $x = \frac{3}{2}$ about the

(a) line $x = 6$
(b) line $x = 1$
(c) line $x = -2$

5 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $y = 4 + 3e^{-x}, y = 2, x = \frac{1}{2}$ and $x = 3$ about the
(a) line $x = 5$
(b) line $x = \frac{1}{4}$
(c) line $x = -1$

6 Use the method of cylinders to determine the volume of the solid obtained by rotating the region bounded by $x = y^2 - 8y + 19$ and $x = 2y + 3$ about the
(a) line $y = 9$
(b) line $y = 1$
(c) line $y = -3$

# More Volume Problems

1 Use the method of finding volume from this section to determine the volume of a sphere of radius $r$.

2 Find the volume of the solid whose base is the region bounded by $x = 2 - y^2$and $x = y^2 - 2$and whose cross-sections are squares with the base perpendicular to the $y$-axis. See figure below to see a sketch of the cross-sections.

**8.1 Arc Length**

1 Set up, but do not evaluate, an integral for the length of $y = 14 - 9x, -22 \le y \le 31$ using,
(a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
(b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

2 Set up, but do not evaluate, an integral for the length of $x = \mathbf{e}^{2y}, -1 \le y \le 0$ using,
(a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
(b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

3 Set up, but do not evaluate, an integral for the length of $y = \tan(2x), 0 \le x \le \frac{\pi}{3}$ using,
(a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
(b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

4 Set up, but do not evaluate, an integral for the length of $\frac{x^2}{16} + 9y^2 = 1$.

5 For $x = 6y + 1, -2 \leq y \leq 8$
(a) Use an integral to find the length of the curve.
(b) Verify your answer from part (a) geometrically.

6 Determine the length of $y = \frac{4}{3}x + 2, 0 \leq x \leq 9$.

7 Determine the length of $y = (8x + 3)^{\frac{3}{2}}, 11^{\frac{3}{2}} \leq y \leq 27^{\frac{3}{2}}$.

8 Determine the length of $x = (10 - 2y)^{\frac{3}{2}}, -1 \leq y \leq 2$.

9 Determine the length of $x = 2 + (y - 1)^2, 2 \leq y \leq 5$.

**10** Determine the length of $y = (3x + 2)^2, 1 \leq x \leq 4$.

# Surface Area

11 Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $y = 7x + 2, -5 \leq y \leq 0$ about the $x$-axis using.
(a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
(b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

12 Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $y = 1 + 2x^5, 0 \leq x \leq 1$ about the $x$-axis using.
(a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
(b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

13 Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $x = \mathbf{e}^{2y}, -1 \leq y \leq 0$ about the $y$-axis using.
(a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
(b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

14 Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $y = \cos\left(\frac{1}{2}x\right), 0 \leq x \leq \pi$ about
(a) the $x$-axis
(b) the $y$-axis

15 Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $x = \sqrt{3 + 7y}, 0 \leq y \leq 1$ about
(a) the $x$-axis
(b) the $y$-axis

16 Find the surface area of the object obtained by rotating $y = \frac{1}{4}\sqrt{6x+2}, \frac{\sqrt{2}}{2} \le y \le \frac{\sqrt{5}}{2}$ about the $x$-axis.

17 Find the surface area of the object obtained by rotating $y = 4 - x, 1 \le x \le 6$ about the $y$-axis.

18 Find the surface area of the object obtained by rotating $x = 2y + 5, -1 \le x \le 2$ about the $y$-axis.

19 Find the surface area of the object obtained by rotating $x = 1 - y^2, 0 \le y \le 3$ about the $x$-axis.

20 Find the surface area of the object obtained by rotating $x = \mathrm{e}^{2y}, -1 \le y \le 0$ about the $y$-axis.

21 Find for the surface area of the object obtained by rotating $y = \cos\left(\frac{1}{2}x\right), 0 \le x \le \pi$ about the $x$-axis.

# Center of Mass

Find the eared of make for each of the following region

22 The region bounded by $y = x^2 + x - 2$ and the x-axis

23 Find the region with vertices $(-2-2), (4,-2)$ and (4,4).

24 The region bounded by $y - (2) - 2\}^3$ ard $y = 1$

25 The region bounded by $y = \cos(x)$ and the x axis between $-\frac{x}{2} \le x \le \frac{a}{2}$

**Parametric Equations and Curves**

**For problems 1 - 9 eliminate the parameter for the given set of parametric equations, sketch the graph of the parametric curve and give any limits that might exist on x and y.**

1. $x = 2 + t \quad ; \quad y = t^2 - 4t + 7$

2. $x = 2 + t \quad ; \quad y = t^2 - 4t + 7 \quad ; (-3 \le t \le 1)$

3. $x = 1 - t^2 \quad ; \quad \mathrm{y} = 3 + 2\mathrm{t}$

4. $x = 1 - t^2 \quad ; \quad \mathrm{y} = 3 + 2\mathrm{t}\,; \quad -2 \le \mathrm{t} \le 3$

5. $x = \frac{1}{2}\sqrt{t} \quad ; \quad \mathrm{y} = \mathrm{t} - \sqrt{t} - 6; \quad t \ge 0$

6. $x = \frac{1}{2}\sqrt{t}\ ; \quad \mathrm{y} = \mathrm{t} - \sqrt{t} - 6\,; \quad 8 \le t \le 20$

7. $x = -6\cos(4t)\ ; \quad x = -6\sin(4t); \quad -\frac{\pi}{4} \le t \le \frac{\pi}{8}$

$x = 1 - 3\sin\left(\frac{1}{2}t\right); \quad y = 4\cos\left(\frac{1}{2}t\right)$

8. $x = 6 - 7\mathbf{e}^{-2t} \qquad y = 4 + 3\mathbf{e}^{-2t}$

**9. Answer each of the questions about the following set of parametric equations**

$$x = 3\cos(at) \qquad y = 3\sin(at) \qquad 0 \le t \le 2\pi$$

(a) Sketch the graph of the parametric curve for a = 1.

(b) Sketch the graph of the parametric curve for a = 6.

(c) Sketch the graph of the parametric curve for $a = \frac{1}{5}$.

(d) In general, for a > 0, how does the value of a affect the graph of the parametric curve?

**For problems 11 - 21 the path of a particle is given by the set of parametric equations. Completely describe the path of the particle. To completely describe the path of the particle you will need to provide the following information**.

I. A sketch of the parametric curve (including direction of motion) based on the equation you get by eliminating the parameter.

II. Limits on x and y.

III. A range of t's for a single trace of the parametric curve.

IV. The number of traces of the curve the particle makes if an overall range of t's is provided in the problem.

10. $x = 6\cos\left(\frac{1}{3}t\right) \qquad y = 2 + \sin\left(\frac{1}{3}t\right) \qquad 0 \le t \le 75\pi$

11. $x = 7 - 3\sin(2t) \quad y = 4 + 2\cos(2t)$

12. $x = 6\cos^2(3t) \qquad y = 2 - 3\sin^2(3t) \qquad -\frac{5}{6}\pi \le t \le 3\pi$

13. $x = \sqrt{2 + \cos^2\left(\frac{7}{4}t\right)} \qquad y = \frac{1}{3}\sin\left(\frac{7}{4}t\right)$

14. $x = 6 - \sin^3(4t) \quad y = 2\sin(4t) - 127 \qquad \pi \le t \le 201\pi$

15. $x = 3 + \cos\left(\frac{1}{6}t\right) \quad y = 4 + \cos^2\left(\frac{1}{6}t\right) \qquad -90\pi \le t \le 216\pi$

16. $x = e^{-4t} \qquad y = 2e^{12t}$

17. $x = 1 + e^{3t} \qquad y = e^{6t} \qquad -1 \le t \le 6$

18. $x = 1 - \ln(t) \qquad y = [[\ln(t)]]^2 \qquad t > 0$

19. $x = \cos\left(\frac{1}{2}t\right) \quad y = \sec\left(\frac{1}{2}t\right) \qquad -\pi < t < \pi$

20. $x = \sin(2t)$ $y = \sin^2(2t) - 4\sin(2t)$ $-\frac{91}{4}\pi \le t \le \frac{17}{4}\pi$

**For problems 22 - 27 write down a set of parametric equations for the given equation that meets the given extra conditions (if any).**

21. $x = ain(3 - y^2) + \csc^2(y)$

22. $y = \frac{6\cos(x)-8}{x^2+9x}$

23. $x^2 + y^2 = 100$ and the parametric curve resulting from the parametric equations should be at (0, 10) when t = 0 and the curve should have a clockwise rotation.

24. $x^2 + y^2 = 100$ and the parametric curve resulting from the parametric equations should be at (0, 10) when t = 0 and the curve should have a counterclockwise rotation.

25. $\frac{x^2}{25} + y^2 = 1$ and the parametric curve resulting from the parametric equations should be at (−5, 0) when t = 0 and the curve should have a counterclockwise rotation.

26. $\frac{x^2}{25} + y^2 = 1$ and the parametric curve resulting from the parametric equations should be at (−5, 0) when t = 0 and the curve should have a clockwise rotation

27. Eliminate the parameter for the following set of parametric equations and identify the resulting equation.

$$x = h + a\cos(\omega t) \qquad y = k + b\sin(\omega t)$$

**Tangents with Parametric Equations**

**For problems 1 - 3 compute $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for the given set of parametric equations**.

**1.** $x = 7t^2 - 9t$ $y = t^6 + 2t^2$

**2.** $x = \tan(2t) - 12$ $y = 3\sin(2t) + \sec(2t) + 4t$

**3.** $x = \ln(3t^2) + 8t$ $y = \ln(t^4) - 6\ln(t^2)$

**For problems 4 - 7 find the equation of the tangent line(s) to the given set of parametric equations at the given point.**

**4.** $x = t^3 + \cos(\pi t)$ $y = 4t + \sin(2t + 6)$ at $t = -3$

**5.** $x = t^2 + 2t - 1$ $y = t^3 + 7t^2 + 8t$ at $t = 1$

**6.** $x = 6 - \mathbf{e}^{t^3-9t} y = t^3 + 3t^2 - 18t + 2$ at (5,2)

**7.** $x = t^2 + 5t - 6$ $y = t^2 + 2t - 8$ at $(-6,7)$

# Area with Parametric Equations

**For problems 1 - 3 determine the area of the region below the parametric curve given by the set of parametric equations. For each problem you may assume that each curve traces out exactly once from right to left for the given range of t. For these problems you should only use the given parametric equations to determine the answer.**

1. $x = t^2 + 5t - 1 \qquad y = 40 - t^2 \qquad -2 \le t \le 5$

2. $x = 3\cos^2(t) - \sin^2(t) \qquad y = 6 + \cos(t) \qquad -\frac{\pi}{2} \le t \le 0$

3. $x = e^{\frac{1}{4}t} - 2 \qquad y = 4 + e^{\frac{1}{4}t} - e^{\frac{1}{2}t} \qquad -6 \le t \le 1$

# Arc Length with Parametric Equations

**For all the problems in this section you should only use the given parametric equations to determine the answer. For problems 1 - 5 determine the length of the parametric curve given by the set of parametric equations. For these problems you may assume that the curve traces out exactly once for the given range of t's.**

1. $x = 3 + 9t \qquad y = 10 - 15t \qquad -5 \le t \le 8$

2. $x = 6(3 + t)^{\frac{3}{2}} \qquad y = -3t^{\frac{3}{2}} \qquad -2 \le t \le 1$

3. $x = 4t^2 - 3 \qquad y = 3t \qquad 0 \le t \le 5$

4. $x = 3 + t \qquad y = 6 + (t - 1)^2 \qquad 1 \le t \le 3$

5. $x = t^2 - 1 \qquad y = t^4 + 5 \qquad 0 \le t \le 1$

**For problems 6 and 7 a particle travels along a path defined by the following set of parametric equations. Determine the total distance the particle travels and compare this to the length of the parametric curve itself.**

6. $x = 6\cos^2(3t) \qquad y = 2 - 3\sin^2(3t) \qquad -\frac{5\pi}{6} \le t \le 3\pi$

7. $x = 3 + \cos\left(\frac{1}{6}t\right) \qquad y = 4 + \cos^2\left(\frac{1}{6}t\right) \qquad -90\pi \le t \le 216\pi$

**For problems 8 - 10 set up, but do not evaluate, an integral that gives the length of the parametric curve given by the set of parametric equations. For these problems you may assume that the curve traces out exactly once for the given range of t's.**

8. $x = t\cos(2t) \qquad y = \sin(3t) \qquad 2 \le t \le 3$

9. $x = 1 - \sin\left(1 + \sqrt{t}\right) \qquad y = \sin(e^{-t}) \qquad 1 \le t \le 4$

10. $x = \ln(t + 2) \qquad y = \frac{1}{t+7} \qquad -1 \le t \le 2$

# Surface Area with Parametric Equations

**For all the problems in this section you should only use the given parametric equations to determine the answer. For problems1 - 4 determine the surface area of the object obtained by rotating the parametric curve about the given axis. For these problems you may assume that the curve traces out exactly once for the given range of t's.**

1. Rotate $x = t^2 - 3 \quad y = 2 + t^2 \quad 0 \le t \le 5$ about the x-axis.
2. Rotate $x = -8t \quad y = 6 + t^2 \quad -3 \le t \le 0$ about the y-axis.
3. Rotate $x = t^2 \quad y = t^4 - 2 \quad 0 \le t \le 2$ about the y-axis.
4. Rotate $x = 2 + t \quad y = 4e^{-\frac{1}{2}t} \quad -1 \le t \le 2$ about the x-axis.

**For problems 5 - 7 set up, but do not evaluate, an integral that gives the surface area of the object obtained by rotating the parametric curve about the given axis. For these problems you may assume that the curve traces out exactly once for the given range of t's**

5. Rotate $x = 2 + e^{\cos(t)} \quad y = 1 + t^2 \quad -2 \le t \le 0$ about the x-axis.
6. Rotate $x = cos^2(t) \quad y = 2\cos(2t) - \sin(t) \quad 0 \le t \le 1$ about the y-axis.
7. Rotate $x = t^2 \quad y = \ln(3 + e^{-t}) \quad 0 \le t \le 2$ about the x-axis

# Polar Coordinates

1. For the point with polar coordinates $\left(-9, \frac{3\pi}{7}\right)$ determine three different sets of coordinates for the same point all of which have angles different from $\frac{3\pi}{7}$ and are in the range $-2\pi \le \theta \le 2\pi$.
2. For the point with polar coordinates $\left(7, -\frac{2\pi}{3}\right)$ determine three different sets of coordinates for the same point all of which have angles different from $-\frac{2\pi}{3}$ and are in the range $-2\pi \le \theta \le 2\pi$.
3. The polar coordinates of a point are (14, 2.48). Determine the Cartesian coordinates for the point.
4. The polar coordinates of a point are $\left(-\frac{3}{10}, -5.29\right)$. Determine the Cartesian coordinates or the point.
5. The Cartesian coordinate of a point are $(-3, 5)$. Determine a set of polar coordinates for the point.
6. The Cartesian coordinate of a point are $(4, -7)$. Determine a set of polar coordinates for the point.
7. The Cartesian coordinate of a point are $(-3, -12)$. Determine a set of polar coordinates for the p**.**

**For problems 8 and 9 find the values of t that will have horizontal or vertical tangent lines for the given set of parametric equations**

**8.** $x = t^3 - 5t^2 + t + 1 \qquad y = t^4 + 8t^3 + 3t^2$

**9.** $x = 7t^2 - e^{2-t^2} \qquad y = 10\sin\left(\frac{1}{2}t\right) - 1$

**For problems 10 - 13 convert the given equation into an equation in terms of Cartesian coordinate.**

**10.** $r - \frac{8\sin(\theta)}{r} = 2\cos(\theta)$

**11.** $r^3 \csc(\theta) = 5\cos(\theta) - 6$

**12.** $8 - r = r^2 \sin(2\theta)$

**13.** $r = 2a\cos(\theta) + 2b\sin(\theta)$

**For problems 14 - 27 sketch the graph of the given polar equation.**

1. $-7 = r\sin(\theta)$
2. $\theta = \frac{5\pi}{7}$
3. $\theta = -\frac{9\pi}{5}$
4. $r\cos(\theta) = 4$
5. $r = 6\sin(\theta)$
6. $r = 100$
7. $r = 24\cos(\theta)$
8. $r = -15\sin(\theta)$
9. $r = 4 + 12\cos(\theta)$
10. $r = 7 - 7\sin(\theta)$
11. $r = 1 + 3\sin(\theta)$
12. $r = 5 - 4\cos(\theta)$
13. $r = 8 + 3\sin(\theta)$
14. $r = 1 - \cos(\theta)$

**For problems 8 and 9 convert the given equation into an equation in terms of polar coordinates**.

8. $7x^2y + 8y = 3 - 6x^2 - 6y^2$

9. $\frac{7y}{x^2+y^2-8x} = 9 + y^2$

# Tangents with Polar Coordinates

1. Find the tangent line to $r = \theta\sin(3\theta)$ at $\theta = \frac{\pi}{2}$.

2. Find the tangent line to $r = \cos(2\theta)$ at $\theta = -\frac{\pi}{4}$.

3. Find the tangent line to $r = \cos(\theta^2 - \theta)$ at $\theta = \pi$.

# Area with Polar Coordinates

1. Find the area inside the inner loop of $r = 3 + 10\sin(\theta)$.

2. Find the area inside the inner loop of $r = 5 + 12\cos(\theta)$.

3. Find the area inside the graph of $r = 8 + \cos(\theta)$ and to the right of the y-axis.

4. Find the area inside the graph of $r = 5 - 4\sin(\theta)$ and the below the x-axis.

5. Find the area that is inside $r = 4$ and outside $r = 4 - 2\sin(\theta)$.

6. Find the area that is inside $r = 7 - 3\cos(\theta)$ and outside $r = 4$.

7. Find the area that is inside $r = 6 + 6\cos(\theta)$ and outside $r = 4 - 3\cos(\theta)$.

8. Find the area that is inside $r = 4 + 2\sin(\theta)$ and outside $r = 5 - \sin(\theta)$.

9. Find the area that is inside $r = 5 - \sin(\theta)$ and outside $r = 4 + 2\sin(\theta)$.

10. Find the area that is inside both $r = 6 - 4\sin(\theta)$ and $r = 5$.

11. Find the area that is inside both $r = 3 + 2\cos(\theta)$ and $r = 3 - \cos(\theta)$.

# Arc Length with Polar Coordinates

For problems 1 - 3 determine the length of the given polar curve. For these problems you may assume that the curve traces out exactly once for the given range of $\theta$.

1 $r = \frac{1}{\cos(\theta)}, 0 \le \theta \le \frac{\pi}{3}$

2 $r = \theta^2, 0 \le \theta \le 3\pi$

3 $r = 6\cos(\theta) - 3\sin(\theta), 0 \le \theta \le \pi$

For problems 4 - 6 set up, but do not evaluate, an integral that gives the length of the given polar curve. For these problems you may assume that the curve traces out exactly once for the given range of $\theta$.

4 $r = \sin(\theta^2), 0 \le \theta \le \pi$

5 $r = \cos(1 + \sin(\theta)), 0 \le \theta \le 2\pi$

6 $r = \mathbf{e}^{-\frac{1}{4}\theta}\cos(\theta), 0 \le \theta \le 3\pi$

## Surface Area with Polar Coordinates

For problems 1 - 4 set up, but do not evaluate, an integral that gives the surface area of the curve rotated about the given axis. For these problems you may assume that the curve traces out exactly once for the given range of $\theta$.

1 $r = \cos\left(\mathbf{e}^{-\frac{1}{4}\theta}\right), 0 \le \theta \le \frac{\pi}{2}$ rotated about the $x$-axis.

2 $r = \theta\sin(\theta), 0 \le \theta \le \frac{\pi}{2}$ rotated about the $y$-axis.

3 $r = \cos(\theta)\sin(2\theta), 0 \le \theta \le \frac{\pi}{6}$ rotated about the $x$-axis.

4 $r = \theta + \sin(\theta), \frac{\pi}{2} \le \theta \le \pi$ rotated about the $y$-axis.

## Average Function Value

For problems 1&2 determine $f_{\text{avg}}$ for the function on the given interval.

1 $f(x) = 8x - 3 + 5\mathrm{e}^{2-x}$ on $[0,2]$

2 $f(x) = \cos(2x) - \sin\left(\frac{x}{2}\right)$ on $\left[-\frac{\pi}{2}, \pi\right]$

3 Find $f_{\text{avg}}$ for $f(x) = 4x^2 - x + 5$ on $[-2,3]$ and determine the value(s) of $c$ in $[-2,3]$ for which $f(c) = f_{\text{avg}}$

## Area Between Curves

1 Determine the area below $f(x) = 3 + 2x - x^2$ and above the $x$-axis.

2 Determine the area to the left of $g(y) = 3 - y^2$ and to the right of $x = -1$.
For problems 3 - 11 determine the area of the region bounded by the given set of curves.

3 $y = x^2 + 2, y = \sin(x), x = -1$ and $x = 2$

4 $y = \frac{8}{x}, y = 2x$ and $x = 4$

5 $x = 3 + y^2, x = 2 - y^2, y = 1$ and $y = -2$

6 $x = y^2 - y - 6$ and $x = 2y + 4$

7 $y = x\sqrt{x^2 + 1}, y = \mathrm{e}^{-\frac{1}{2}x}, x = -3$ and the $y$-axis.

8 $y = 4x + 3, y = 6 - x - 2x^2, x = -4$ and $x = 2$

9 $y = \frac{1}{x+2}, y = (x+2)^2, x = -\frac{3}{2}, x = 1$

10 $x = y^2 + 1, x = 5, y = -3$ and $y = 3$

11 $x = \mathrm{e}^{1+2y}, x = \mathrm{e}^{1-y}, y = -2$ and $y = 1$

# Solids of Revolution / Method of Rings

For each of the following problems use the method of disks/rings to determine the volume of the solid obtained by rotating the region bounded by the given curves about the given axis.

1 Rotate the region bounded by $y = \sqrt{x}, y = 3$ and the $y$-axis about the $y$-axis.

2 Rotate the region bounded by $y = 7 - x^2, x = -2, x = 2$ and the $x$-axis about the $x$-axis.

3 Rotate the region bounded by $x = y^2 - 6y + 10$ and $x = 5$ about the $y$-axis.

4 Rotate the region bounded by $y = 2x^2$ and $y = x^3$ about the $x$-axis.

5 Rotate the region bounded by $y = 6\mathrm{e}^{-2x}$ and $y = 6 + 4x - 2x^2$ between $x = 0$ and $x = 1$ about the line $y = -2$.

6 Rotate the region bounded by $y = 10 - 6x + x^2, y = -10 + 6x - x^2, x = 1$ and $x = 5$ about the line $y = 8$.

7 Rotate the region bounded by $x = y^2 - 4$ and $x = 6 - 3y$ about the line $x = 24$.

8 Rotate the region bounded by $y = 2x + 1, x = 4$ and $y = 3$ about the line $x = -4$.

# Arc Length

1. Set up, but do not evaluate, an integral for the length of $y = \sqrt{x + 2}, 1 \le x \le 7$ using,
   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

2. Set up, but do not evaluate, an integral for the length of $x = \cos(y), 0 \le x \le \frac{1}{2}$ using,
   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$
   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

3. Determine the length of $y = 7(6 + x)^{\frac{3}{2}}, 189 \le y \le 875$.

4. Determine the length of $x = 4(3 + y)^2, 1 \le y \le 4$.

# Surface Area

1. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $x = \sqrt{y + 5}, \sqrt{5} \le x \le 3$ about the $y$-axis using,

(a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$

(b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

2. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $y = \sin(2x), 0 \le x \le \frac{\pi}{8}$ about the $x$-axis using,

   (a) $ds = \sqrt{1 + \left[\frac{dy}{dx}\right]^2}\, dx$

   (b) $ds = \sqrt{1 + \left[\frac{dx}{dy}\right]^2}\, dy$

3. Set up, but do not evaluate, an integral for the surface area of the object obtained by rotating $y = x^3 + 4, 1 \le x \le 5$ about the given axis. You can use either ds.
   (a) the $x$-axis
   (b) the $y$-axis

4. Find the surface area of the object obtained by rotating $y = 4 + 3x^2, 1 \le x \le 2$ about the $y$-axis.
5. Find the surface area of the object obtained by rotating $y = \sin(2x), 0 \le x \le \frac{\pi}{8}$ about the $x$-axis.

# Parametric Equations and Curves

For problems 1 - 6 eliminate the parameter for the given set of parametric equations, sketch the graph of the parametric curve and give any limits that might exist on $x$ and $y$.

1 $x = 4 - 2t\ y = 3 + 6t - 4t^2$

2 $x = 4 - 2t\ y = 3 + 6t - 4t^2\ 0 \le t \le 3$

3 $x = \sqrt{t+1}\ y = \frac{1}{t+1}\ t > -1$

4 $x = 3\sin(t)\ y = -4\cos(t)\ 0 \le t \le 2\pi$

5 $x = 3\sin(2t)\ y = -4\cos(2t)\ 0 \le t \le 2\pi$

6 $x = 3\sin\left(\frac{1}{3}t\right)\ y = -4\cos\left(\frac{1}{3}t\right)\ 0 \le t \le 2\pi$

For problems 1 - 6 eliminate the parameter for the given set of parametric equations, sketch the graph of the parametric curve and give any limits that might exist on $x$ and $y$.

7 $x = 4 - 2t\ y = 3 + 6t - 4t^2$

8 $x = 4 - 2t\ y = 3 + 6t - 4t^2\ 0 \le t \le 3$

9 $x = \sqrt{t+1}\ y = \frac{1}{t+1}\ t > -1$

10 $x = 3\sin(t)\ y = -4\cos(t)\ 0 \le t \le 2\pi$

11 $x = 3\sin(2t)\ y = -4\cos(2t)\ 0 \le t \le 2\pi$

12 $x = 3\sin\left(\frac{1}{3}t\right)\ y = -4\cos\left(\frac{1}{3}t\right)\ 0 \le t \le 2\pi$

For problems 7 - 11 the path of a particle is given by the set of parametric equations. Completely describe the path of the particle. To completely describe the path of the particle you will need to provide the following information. (i) A sketch of the parametric curve (including direction of motion) based on the equation you get by eliminating the parameter.

(ii) Limits on $x$ and $y$.

(iii) A range of $t$ 's for a single trace of the parametric curve.

(iv) The number of traces of the curve the particle makes if an overall range of $t$ 's is provided in the problem.

7 $x = 3 - 2\cos(3t)\ y = 1 + 4\sin(3t)$

8 $x = 4\sin\left(\frac{1}{4}t\right)\ y = 1 - 2\cos^2\left(\frac{1}{4}t\right)\ -52\pi \le t \le 34\pi$

9 $x = \sqrt{4 + \cos\left(\frac{5}{2}t\right)}\ y = 1 + \frac{1}{3}\cos\left(\frac{5}{2}t\right)\ -48\pi \le t \le 2\pi$

10 $x = 2\mathbf{e}^t\ y = \cos(1 + \mathbf{e}^{3t})\ 0 \le t \le \frac{3}{4}$

11 $x = \frac{1}{2}\mathbf{e}^{-3t}\ y = \mathbf{e}^{-6t} + 2\mathbf{e}^{-3t} - 8$

# Tangents with Parametric Equations

**For problems 1 and 2 compute $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for the given set of parametric equations.**

1 $x = 4t^3 - t^2 + 7t\ y = t^4 - 6$

2 $x = e^{-7t} + 2\ y = 6e^{2t} + e^{-3t} - 4t$

**For problems 3 and 4 find the equation of the tangent line(s) to the given set of parametric equations at the given point**.

3. $x = 2cos(3t) - 4sin(3t)\ y = 3tan(6t)$ at $t = \frac{\pi}{2}$

4. $x = t^2 - 2t - 11\ y = t(t-4)^3 - 3t^2(t-4)^2 + 7$ at $(-3,7)$

5. Find the values of $t$ that will have horizontal or vertical tangent lines for the following set of parametric equations. $x = t^5 - 7t^4 - 3t^3\ y = 2\cos(3t) + 4t$

# Area with Parametric Equations

For problems 1 and 2 determine the area of the region below the parametric curve given by the set of parametric equations. For each problem, you may assume that each curve traces out exactly once from left to right for the given range of $t$. For these problems, you should only use the given parametric equations to determine the answer.

1 $x = 4t^3 - t^2\ y = t^4 + 2t^2\ 1 \le t \le 3$
2 $x = 3 - \cos^3(t)\ y = 4 + \sin(t)\ 0 \le t \le \pi$

# Arc Length with Parametric Equations

For all the problems in this section, you should only use the given parametric equations to determine the answer.

For problems 1 and 2 determine the length of the parametric curve given by the set of parametric equations. For these problems, you may assume that the curve traces out exactly once for the given range of t's.

1 $x = 8t^2\ y = 3 + (8 - t)^2\ 0 \le t \le 4$
2 $x = 3t + 1\ y = 4 - t^2\ -2 \le t \le 0$
3 A particle travels along a path defined by the following set of parametric equations. Determine the total distance the particle travels and compare this to the length of the parametric curve itself.

$$x = 4\sin\left(\frac{1}{4}t\right)\ y = 1 - 2\cos^2\left(\frac{1}{4}t\right)\ -52\pi \le t \le 34\pi$$

For problems 4 and 5 set up, but do not evaluate, an integral that gives the length of the parametric curve given by the set of parametric equations. For these problems, you may assume that the curve traces out exactly once for the given range of t's.

4 $x = 2 + t^2\ y = e^t\sin(2t)\ 0 \le t \le 3148/242$
5 $x = \cos^3(2t)\ y = \sin(1 - t^2)\ -\frac{3}{2} \le t \le 0$

# Surface Area with Parametric Equations

For problems 1 - 3 determine the surface area of the object obtained by ro- tating the parametric curve about the given axis. For these problems, you may assume that the curve traces out exactly once for the given range of $t$

1. Rotate $x = 3 + 2t\ y = 9 - 3t\ 1 \le t \le 4$ about the $y$-axis.

2. Rotate $x = 9 + 2t^2\ y = 4t\ 0 \le t \le 2$ about the $x$-axis.

3. Rotate $x = 3\cos(\pi t)\ y = 5t + 2\ 0 \le t \le \frac{1}{2}$ about the $y$-axis.

For problems 4 and 5 set up, but do not evaluate, an integral that gives the surface area of the object obtained by rotating the parametric curve about the given axis. For these problems, you may assume that the curve traces out exactly once for the given range of $t$ s.

4 Rotate $x = 1 + \ln(5 + t^2)$ $y = 2t - 2t^2$ $0 \le t \le 2$ about the $x$-axis.

5 Rotate $x = 1 + 3t^2$ $y = \sin(2t)\cos\left(\frac{1}{4}t\right)$ $0 \le t \le \frac{1}{2}$ about the $y$-axis.

# Polar Coordinates

1 For the point with polar coordinates $\left(2, \frac{\pi}{4}\right)$ determine three different sets of coordinates for the same point all of which have angles different from $\frac{\pi}{7}$ and are in the range $-2\text{t} \le 0 \le 2$ r.

2 The polar coordinates of a point are $(-5,023)$. Determine the Cartesian coordinates for the point

3 The Cartesian coordinate of a point are $(2, -6)$. Determine a set of polar coordinates for the plot.

4 The Cartesian coordinate of a point is $\{-8,1)$. Determine a set of polar coordinates for the point.

For problems 5 and 6 convert the given equation into an equation in sets of polar coordinates.

5 $\frac{4}{3x^2+3y^2} = 6 - xy$

6 $x^2 = \frac{4x}{y} - 3y^2 + 2$

For problems 7 and 8 convert the given equation into an equation in terms of Cartesian coordinates.

7 $6\cos^2 \sin(\theta) = 4 - \cos$

8 $\frac{2}{r} = \sin(\theta) - \sec(\theta)$

For problems $9 - 16$ sketch the graph of the okven polar equation.

9 $\cos(\theta) = \frac{6}{r}$

10 $= -\frac{\pi}{3}$

11 $r = -14\cos(\theta)$

12 $r = 7$

13 $r = 9\sin(i)$ 14. $r = 8 + 8\cos(\theta)$

14 $r = 5 - 2\sin(\theta)$

15 $r = 4 - 9\sin(\theta)$

# Arc Length with Polar Coordinates

Determine the length of the following polar curve. You may assume that the curve traces out exactly once for the given range of $\theta$.

$$r = -4\sin(\theta), 0 \leq \theta \leq \pi$$

**For problems 2 and 3 set up, but do not evaluate, an integral that gives the length of the given polar curve. For these problems, you may assume that the curve traces out exactly once for the given range of $\boldsymbol{\theta}$.**

2. $r = \theta\cos(\theta), 0 \leq \theta \leq \pi$

3. $r = \cos(2\theta) + \sin(3\theta), 0 \leq \theta \leq 2\pi$

# Surface Area with Polar Coordinates

**For problems 1 and 2 set up, but do not evaluate, an integral that gives the surface area of the curve rotated about the given axis. For these problems, you may assume that the curve traces out exactly once for the given range of $\boldsymbol{\theta}$.**

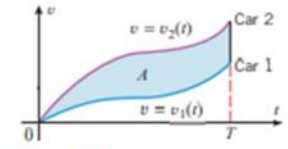
1. $r = 5 - 4\sin(\theta), 0 \leq \theta \leq \pi$ rotated about the $x$-axis.

2. $r = \cos^2(\theta), -\frac{\pi}{6} \leq \theta \leq \frac{\pi}{6}$ rotated about the $y$-axis

**Example1:** Find the area of the region bounded above by y = x + 6, bounded below by y = $x^2$ and bounded on the sides by the lines x = 0 and x = 2.

**Example 2:** Find the area of the region that is enclosed between the curves y = $x^2$and y = x + 6

**Example 3:** Figure 6.1.8 shows velocity versus time curves for two race cars that move along a straight track, starting from rest at the same time. Give a physical interpretation of the area A between the curves over the interval 0 ≤ t ≤ T .



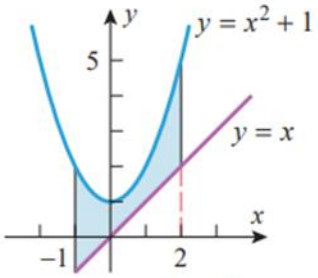
▲ Figure 6.1.8

**Example 4:** Find the area of the region enclosed by x = $y^2$ and y = x − 2.

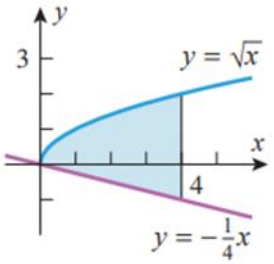
**Example 5:** Find the area of the region enclosed by x = $y^2$ and y = x − 2, integrating with respect to y.

1–4 Find the area of the shaded region

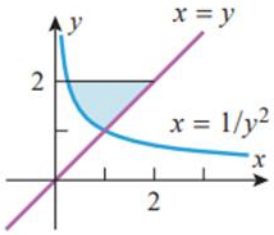
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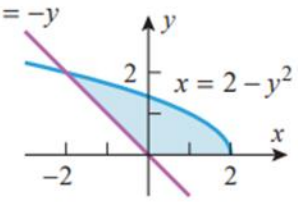

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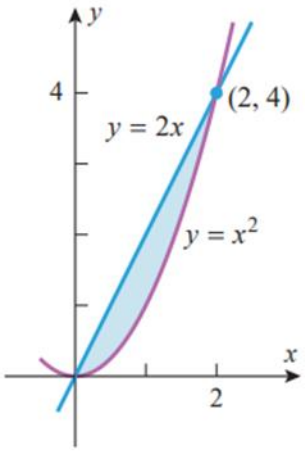



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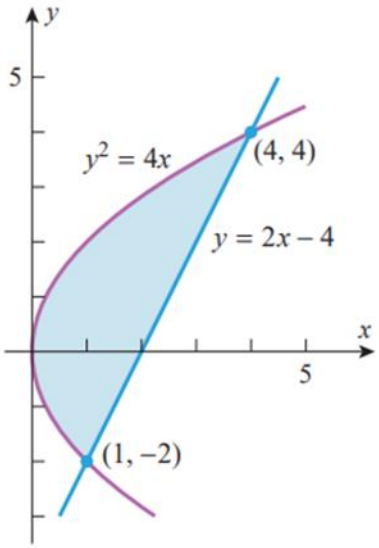



**5–6 Find the area of the shaded region by (a) integrating with respect to x and (b) integrating with respect to y.**

**5.**




**6.**




7–18 Sketch the region enclosed by the curves and find its area.

7. $y = x^2$, $y = \sqrt{x}$, $x = 1/4$ , $x = 1$

8. $y = x^3 - 4x$, $y = 0$, $x = 0$, $x = 2$

9. $y = \cos 2x$, $y = 0$, $x = \pi/4$, $x = \pi/2$

10. $y = \sec^2 x$, $y = 2$, $x = -\pi/4$, $x = \pi/4$

11. $x = \sin y$, $x = 0$, $y = \pi/4$, $y = 3\pi/4$

12. $x^2 = y, x = y - 2$

13. $y = e^x, y = e^{2x}, x = 0, x = \ln 2$

14. $x = 1/y, x = 0, y = 1, y = e$

15. $y = \frac{2}{1+x^2}, y = |x|$

16. $y = \frac{1}{\sqrt{1-x^2}}, y = 2$

17. $y = 2 + |x - 1|, y = -\frac{1}{5}x + 7$

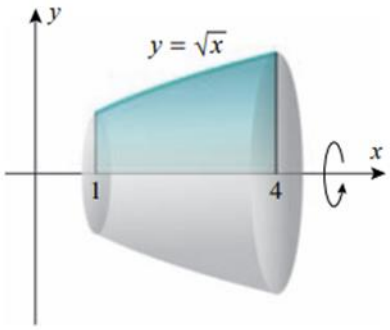
18. $y = x, y = 4x, y = -x + 2$

19-26 Use a graphing utility, where helpful, to find the area of the region enclosed by the curves.

19. $y = x^3 - 4x^2 + 3x, y = 0$
20. $y = x^3 - 2x^2,\ y = 2x^2 - 3x$
21. $y = \sin x, y = \cos x, x = 0, x = 2\pi$
22. $y = x^3 - 4x, y = 0$
23. $x = y^3 - y, x = 0$
24. $x = y^3 - 4y^2 + 3y, x = y^2 - y$
25. $y = xe^{x^2}, y = 2|x|$
26. $y = \frac{1}{x\sqrt{1-(\ln x)^2}}, y = \frac{3}{x}$

**Example 1:** Derive the formula for the volume of a right pyramid whose altitude is h and whose base is a square with sides of length a.

**Example 2 :** Find the volume of the solid that is obtained when the region under the curve y = √x over the interval [1, 4] is revolved about the x-axis.




▲ Figure 6.2.10

**Example 03:** Derive the formula for the volume of a sphere of radius r.

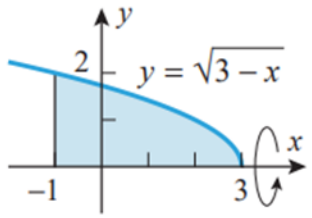
**Example 04:** Find the volume of the solid generated when the region between the graphs of the equations f(x) = 1/2 + $x^2$, and g(x) = x over the interval [0, 2] is revolved about the x-axis

**Example 05:** Find the volume of the solid generated when the region enclosed by y = √x, y=2, and x = 0 is revolved about the y-axis.

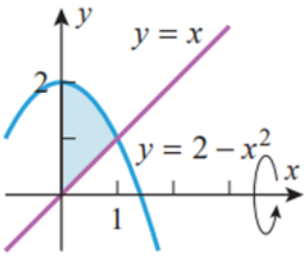
**Example 06:** Find the volume of the solid generated when the region under the curve y = $x^2$ over the interval [0, 2] is rotated about the line y = −1.

**Find the volume of the solid that results when the shaded region is revolved about the indicated axis(1-8) .**

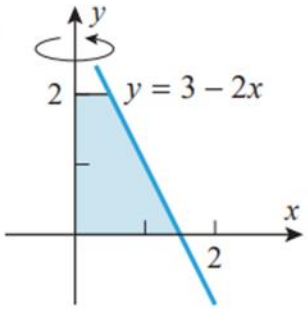
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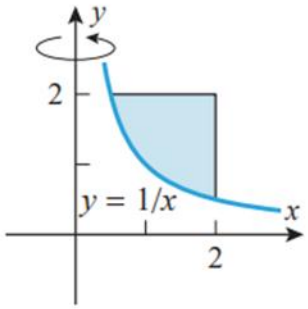

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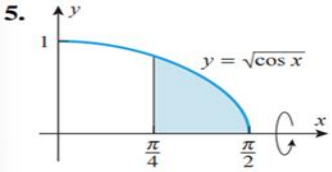

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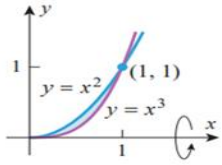

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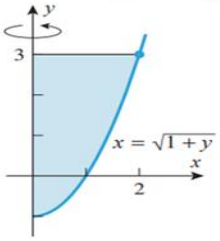

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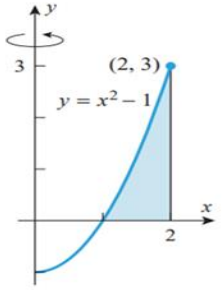

**6.**




**7.**




**8.**



**11-18 Find the volume of the solid that results when the region enclosed by the given curves is revolved about the $x$-axis.**

11. $y = \sqrt{25 - x^2}, y = 3$

12. $y = 9 - x^2, y = 0$

13. $x = \sqrt{y}, x = y/4$

14. $y = \sin\ x, y = \cos\ x, x = 0, x = \pi/4$

[Hint: Use the identity $\cos\ 2x = \cos^2\ x - \sin^2\ x$.]

15. $y = e^x, y = 0, x = 0, x = \ln\ 3$

16. $y = e^{-2x}, y = 0, x = 0, x = 1$

17. $y = \frac{1}{\sqrt{4+x^2}}, x = -2, x = 2, y = 0$

18. $y = \frac{e^{3x}}{\sqrt{1+e^{6x}}}, x = 0, x = 1, y = 0$

19. Find the volume of the solid whose base is the region bounded between the curve $y = x^3$ and the $y$-axis from $y = 0$ to $y = 1$ and whose cross sections taken perpendicular to the $y$-axis are squares.

20. Find the volume of the solid whose base is the region enclosed between the curve $x = 1 - y^2$ and the $y$-axis and whose cross sections taken perpendicular to the $y$-axis are squares.

**21-26 Find the volume of the solid that results when the region enclosed by the given curves is revolved about the $y$-axis.**

21. $x = \csc\ y, y = \pi/4, y = 3\pi/4, x = 0$

22. $y = x^2, x = y^2$

23. x =$y^2$, x = y + 2

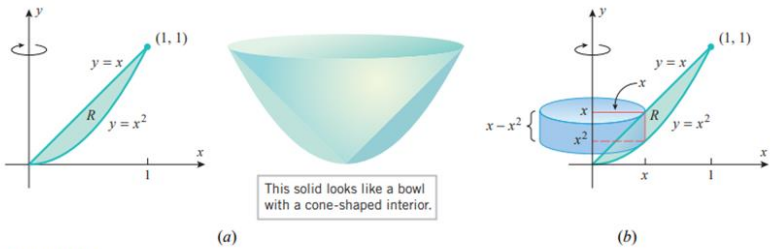
24. x = 1 − $y^2$, x = 2 + $y^2$, y = −1, y = 1

25. y = ln x, x = 0, y = 0, y = 1

26. y = $\sqrt{\frac{1-x^2}{x^2}}$ (x > 0), x = 0, y = 0, y = 2

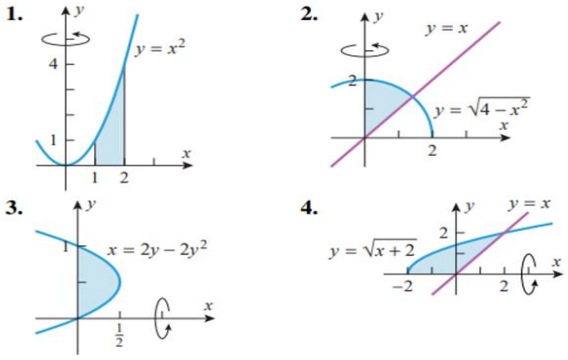
**Example1:** Use cylindrical shells to find the volume of the solid generated when the region enclosed between y = √x, x = 1, x = 4, and the x-axis is revolved about the y-axis.

**Example 2:** Use cylindrical shells to find the volume of the solid generated when the region R in the first quadrant enclosed between y = x and y =$x^2$, is revolved about the y-axis (Figure 6.3.8a).




▲ Figure 6.3.8

**1–4 Use cylindrical shells to find the volume of the solid generated when the shaded region is revolved about the indicated axis.**




5. y = $x^2$, x = 1, y = 0

6. y = √x, x = 4, x = 9, y = 0

7. y = 1/x, y = 0, x = 1, x = 3

8. y = cos($x^2$), x = 0, x = 1 2 √π, y = 0

9. y = 2x − 1, y = −2x + 3, x = 2

10. y = 2x − $x^2$, y = 0

11. y = $\frac{1}{x^2+1}$, x = 0, x = 1, y = 0

12. y = $e^{x^2}$, x = 1, x = √3, y = 0

**13–15 Use cylindrical shells to find the volume of the solid generated when the region enclosed by the given curves is revolved about the x-axis.**

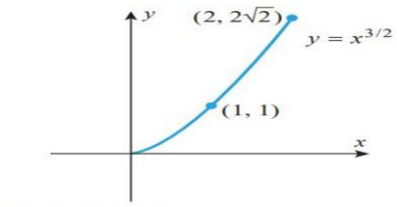
13. $y^2$ = x, y = 1, x = 0

14. x = 2y, y = 2, y = 3, x = 0

15. y = $x^2$, x = 1, y = 0

# LENGTH OF A PLANE CURVE

**Example 1:** Find the arc length of the curve y = $x^{3/2}$ from (1, 1) to (2, 2√2 ) (Figure 6.4.4):




▲ Figure 6.4.4

**Example 2:** From the arc length of $y = sin\ x$ from $x = 0$ to $x = \pi$ is given by the integral

$$L = \int_0^{\pi} \sqrt{1 + (cos\ x)^2}\, dx$$

N.B This integral cannot be evaluated in terms of elementary functions; however, using a calculating utility with a numerical integration capability yields the approximation $L \approx 3.8202$.

**Example 3:** Find the arc length of the following curve.

1 $y = 3x^{3/2} - 1$ from $x = 0$ to $x = 1$

2 $x = \frac{1}{3}(y^2 + 2)^{3/2}$ from $y = 0$ to $y = 1$

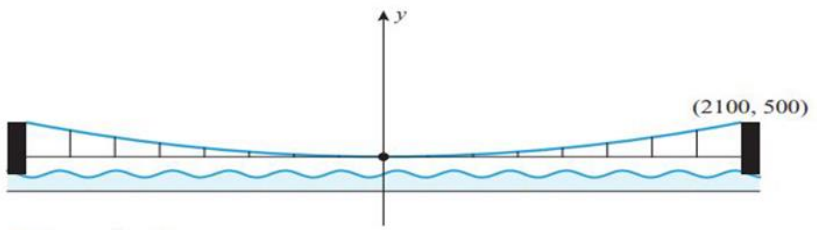
3 $y = x^{2/3}$ from $x = 1$ to $x = 8$

4 $y = (x^6 + 8)/(16x^2)$ from $x = 2$ to $x = 3$

5 $24xy = y^4 + 48$ from $y = 2$ to $y = 4$

6 $x = \frac{1}{8}y^4 + \frac{1}{4}y^{-2}$ from $y = 1$ to $y = 4$

**Example 4:**




▲ **Figure Ex-23**

The central span of the Golden Gate Bridge in California is $4200ft$ long and is suspended from cables that rise $500ft$ above the roadway on either side. Approximately how long is the portion of a cable that lies between the support towers on one side of the roadway? [Hint: As suggested by the accompanying figure, assume the cable is modeled by a parabola $y = ax^2$ that passes through the point (2100,500). Use a CAS or a calculating utility with a numerical integration capability to approximate the length of the cable. Round your answer to the nearest foot.]

# AREA OF A SURFACE OF REVOLUTION

**Example 1:** Find the area of the surface that is generated by revolving the portion of the curve y = $x^3$ between x = 0 and x = 1 about the x-axis.

**Example 2:** Find the area of the surface that is generated by revolving the portion of the curve y = $x^2$ between x = 1 and x = 2 about the y-axis.

**1–4 Find the area of the surface generated by revolving the given curve about the x-axis.**

1. $y = 7x, 0 \le x \le 1$

2. $y = \sqrt{x}, 1 \le x \le 4$

3. $y = \sqrt{4 - x^2}, -1 \le x \le 1$

4. $x = \sqrt[3]{y}, 1 \le y \le 8$

**5–8 Find the area of the surface generated by revolving the given curve about the y-axis.**

5. $x = 9y + 1, 0 \le y \le 2$

6. $x = y^3, 0 \le y \le 1$

7. $x = \sqrt{9 - y^2}, -2 \le y \le 2$

8. $x = 2\sqrt{1 - y}, -1 \le y \le 0$

**9–12 Use a CAS to find the exact area of the surface generated by revolving the curve about the stated axis.**

9. $y = \sqrt{x} - \frac{1}{3}x^{3/2}, 1 \le x \le 3; x$-axis

10. $y = \frac{1}{3}x^3 + \frac{1}{4}x^{-1}, 1 \le x \le 2; x$-axis

11. $8xy^2 = 2y^6 + 1, 1 \le y \le 2$; y-axis

12. $x = \sqrt{16 - y}, \ 0 \le y \le 15$; y-axis

**13–16 Use a CAS or a calculating utility with a numerical integration capability to approximate the area of the surface generated by revolving the curve about the stated axis. Round your answer to two decimal places.**

13. $y = sin\ x, 0 \le x \le \pi; x$-axis

14. $x = tan\ y, 0 \le y \le \pi/4.$

15. $y = e^x\ , 0 \le x \le 1;\ x - axis$

16. $y = e^x, 1 \le y \le e;\ y - axis$

# Double Integrals

The definite integral of a continuous function f of one variable on an interval [a,b] is defined as

1. $\int_a^b f(x)dx = \lim_{n\to\infty}[f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_n)\Delta x]$

where $\Delta x = (b - a)/n$ and $x_1, x_2, \ldots, x_n$ are the endpoints of the subintervals of $[a, b]$ with width $\Delta x$. We saw that if $f(x)$ is a positive function, then $\int_a^b f(x)dx$ can be interpreted as an area and we used $\int_a^b f(x)dx$ to compute average values.

In a similar way we will show here how to define the double integral of a function of two variables $f(x, y)$ on a rectangle. We will see how to interpret it as a volume if $f(x, y)$ is a positive function and how to use it to calculate average values.

## Double Integrals over Rectangles

We start with a function $f(x, y)$ whose domain is a rectangle

$$R = \{(x, y) \mid a \leqslant x \leqslant b, c \leqslant y \leqslant d\}$$

If we divide the interval $[a, b]$ into $m$ subintervals of equal width $\Delta x = (b - a)/m$ and we divide the interval $[c, d]$ into $n$ subintervals of equal width $\Delta y = (d - c)/n$, then, as shown in Figure 1, $R$ is divided into $mn$ subrectangles each with area $\Delta A = \Delta x \Delta y$. The upper right corner of a typical subrectangle has coordinates $(x_i, y_j)$.

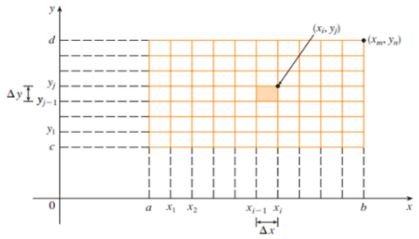


Figure 1: Dividing R into subrectangles

By analogy with Equation 1 we define the double integral of $f$ over the rectangle $R$ as a limit of double Riemann sums:

**2.** $\iint_R f(x, y)\,dA = \lim_{m,n\to\infty} [f(x_1, y_1)\Delta A + f(x_1, y_2)\Delta A + \cdots + f(x_m, y_n)\Delta A]$

There are a total of mn terms in the Riemann sum in Definition 2, one for each of the mn sub-rectangles in Figure 1. If the limit exists, f is called integrable.

For a positive function we can interpret $f(x_i, y_j)$. as the height of a thin rectangular column with base area $\Delta A$ and volume $f(x_i, y_j)\ \Delta A$. (See Figure 2.) So the Riemann sum in Definition 2 can be interpreted as the sum of volumes of columns (see Figure 3) and this sum is an approximation to the volume of the solid that lies under the graph of the surface z = $f(x, y)$ and above the rectangle R.

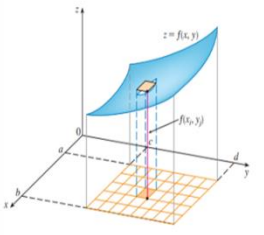


FIGURE 2

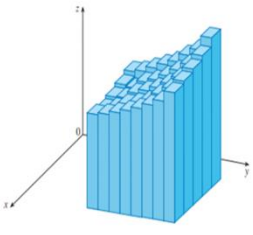


FIGURE 3

As m and n become large in Definition 2 and Figures 2 and 3, the approximation becomes closer and closer to the actual volume, so we define the volume of the solid to be the value of the double integral.

If $f(x,y) \geqslant 0$, the volume of the solid that lies under the surface $z = f(x,y)$ and above the rectangle $R$ is

$$V = \iint_R f(x,y)dA$$

## Iterated Integrals

It's very difficult to evaluate a double integral using Definition 2 directly, so now we show how to express a double integral as an iterated integral, which can then be evaluated by calculating two single integrals. Suppose that $f$ is a function of two variables that is integrable on the rectangle $R = [a,b] \times [c,d]$ . We use the notation $\int_c^d f(x,y)dy$ to mean

that $x$ is held fixed (and treated as a constant) and $f(x,y)$ is integrated with respect to $y$ from $y = c\ to\ y = d$ . This procedure is called partial integration with respect to $y$. (Notice its similarity to partial differentiation.) Now $\int_c^d f(x,y)dy$ is a number that depends on the value of $x$, so it defines a function of $x$ :

$$A(x) = \int_c^d f(x,y)dy$$

If we now integrate the function $A$ with respect to $x$ from $x = a$ to $x = b$, we get

$$\int_a^b A(x)dx = \int_a^b \left[\int_c^d f(x,y)dy\right]dx$$

The integral on the right side of Equation 7 is called an iterated integral. Usually, the brackets are omitted. Thus

$$\int_a^b \int_c^d f(x,y)dydx = \int_a^b \left[\int_c^d f(x,y)dy\right]dx$$

means that we first integrate with respect to $y$ from $c$ to $d$ and then with respect to $x$ from $a$ to $b$. Similarly, the iterated integral

$$\int_c^d \int_a^b f(x,y)dxdy = \int_c^d \left[\int_a^b f(x,y)dx\right]dy$$

**EXAMPLE 1: Evaluate the iterated integrals.**

$(\boldsymbol{a})\ \int_0^3 \int_1^2 \boldsymbol{x^2 y dy dx}$ Answer: $\frac{3}{2}\boldsymbol{x^2}$

(b) $\int_1^2 \int_0^3 \boldsymbol{x^2 y dx dy}$ Answer: $\frac{27}{2}$

[ If $\boldsymbol{R} = \{(\boldsymbol{x}, \boldsymbol{y}) \mid \boldsymbol{a} \leqslant \boldsymbol{x} \leqslant \boldsymbol{b}, \boldsymbol{c} \leqslant \boldsymbol{y} \leqslant \boldsymbol{d}\}$, then

$$\iint_R \boldsymbol{f(x,y)dA} = \int_a^b \int_c^d \boldsymbol{f(x,y)dydx} = \int_c^d \int_a^b \boldsymbol{f(x,y)dxdy}$$

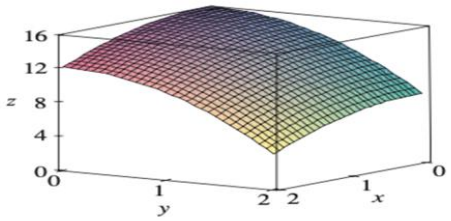



**FIGURE 4**

**EXAMPLE 2 :** Find the volume of the solid $S$ that is enclosed by a paraboloid. $x^2 + 2y^2 + z = 16$, the planes $x = 2$ and $y = 2$, and the three coordinate planes. **Answer: 48**

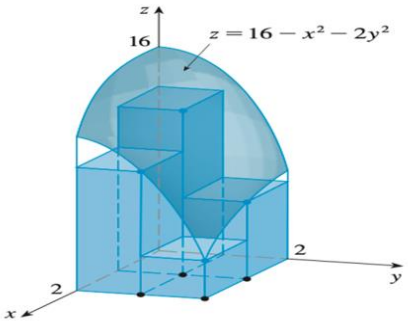



**FIGURE 5**

**NOTE:** Figure 5 illustrates the definitions of volume and the double integral by showing how the solid in Example 2 is approximated by the four columns in the Riemann sum with $m = n = 2$. Figure 6 shows how the columns become better approximations to the volume as $m$ and $n$ increase.

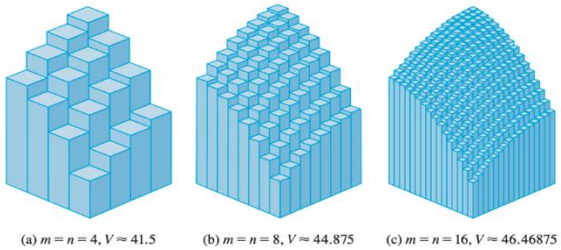



**Figure 6:** The Riemann sum approximations to the volume under $z = 16 - x^2 - 2y^2$ become more accurate as m and n increases

# Double Integrals over More General Regions

What happens if we need to integrate a function $f(x, y)$ over a region $D$ that is not a rectangle? Suppose, for instance, that the domain $D$ of $f$ lies between the graphs of two continuous functions of :

$$\boldsymbol{D = \{(x, y) \mid a \leqslant x \leqslant b, g_1(x) \leqslant y \leqslant g_2(x)\}}$$

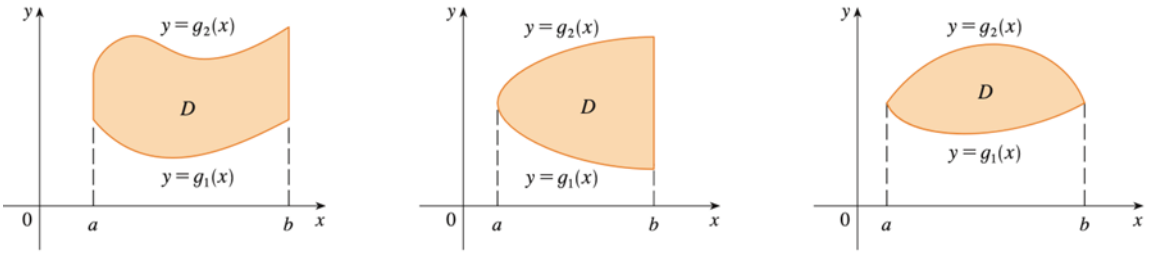


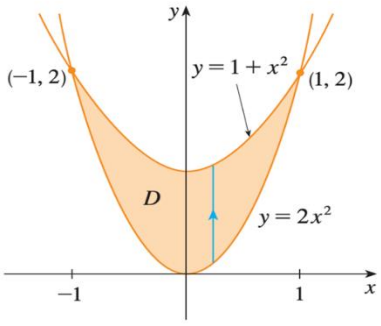

Figure 7 shows three examples of such regions

The double integral of $f$ over $D$, $\iint_D f(x, y)dA$, can be defined by a limit similar to the one in Definition 2 and it can be evaluated as an iterated integral similar to the one in Theorem 6:

$$\boldsymbol{\iint_D f(x, y)dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y)dydx}$$

**EXAMPLE 3:** Evaluate $\iint_D (x + 2y)dA$, where $D$ is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.

Answer: $\frac{32}{15}$




**NOTE:** When we set up a double integral as in Example 3 , it is essential to draw a diagram. Often it is helpful to draw a vertical arrow as in Figure 8. Then the limits of integration for the inner integral can be read from the diagram as follows: The arrow starts at the lower boundary $y = g_1(x)$ , which gives the lower limit in the integral, and the arrow ends at the upper boundary $y = g_2(x)$ , which gives the upper limit of integration.

## Average Value

Recall from Section 6.2 that the average value of a function *f* of one variable defined on an interval [a, b] is

$$f_{ave} = \frac{1}{b-a}\int_a^b f(x)dx$$

In a similar fashion we define the average value of a function f of two variables defined on a rectangle R to be

$$f_{ave} = \frac{1}{A(R)}\iint_R f(x,y)dA$$

where A(R) is the area of **R.**

If $f(x,y) \geq 0$ the equation

$$f_{ave} \times A(R) = \iint_R f(x,y)dA$$

says that the box with base R and heigh $f_{ave}$ has the same volume as the solid that lies under the graph of f. [ If $z = f(x,y)$ describes a mountainous region and you chop off the tops of the mountains at height $f_{ave}$ , then you can use them to fill in the valleys so that the region becomes completely flat. See Figure 9.]

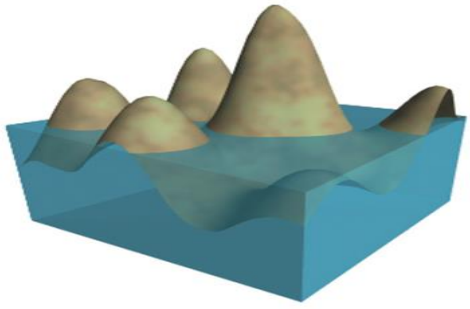


**FIGURE 9**

**EXAMPLE 4** : A manufacturer has modeled its output by a Cobb-Douglas production function

$$P(L,K) = 70L^{0.6}K^{0.4}$$

where $L$ is the number of monthly labor hours and $K$ is the monthly capital investment (in units of \$1000 ). If $L$ varies roughly evenly from 5000 to 6000 and monthly capital investment varies evenly between \$20,000 and \$30,000, find the average monthly output. **Answer:** The average monthly output is about 44,427 units.

**1-2 Find $\int_0^5 f(x,y)dx$ and $\int_0^1 f(x,y)dy$.**

**1** $f(x,y) = 12x^2y^3$

**2** $f(x,y) = y + xe^y$

**3-12 Calculate the iterated integral.**

3. $\int_1^3 \int_0^1 (1 + 4xy)dxdy$
4. $\int_0^1 \int_1^2 (4x^3 - 9x^2y^2)dydx$
5. $\int_0^2 \int_0^1 (2x + y)^8 dxdy$
6. $\int_0^1 \int_1^2 \frac{xe^x}{y} dydx$
7. $\int_1^4 \int_1^2 \left(\frac{x}{y} + \frac{y}{x}\right) dydx$
8. $\int_0^1 \int_0^1 \sqrt{s+t} dsdt$
9. $\int_0^4 \int_0^{\sqrt{y}} xy^2 dxdy$
10. $\int_0^1 \int_{2x}^2 (x - y)dydx$
11. $\int_0^1 \int_{x^2}^x (1 + 2y)dydx$
12. $\int_0^2 \int_y^{2y} xydxdy$

**13-20 Calculate the double integral.**

13. $\iint_R (6x^2y^3 - 5y^4)dA,\ R = \{(x,y) \mid 0 \leqslant x \leqslant 3, 0 \leqslant y \leqslant 1\}$
14. $\iint_R (y + xy^{-2})dA,\ R = \{(x,y) \mid 0 \leqslant x \leqslant 2, 1 \leqslant y \leqslant 2\}$
15. $\iint_R xye^{x^2y} dA,\ R = \{(x,y) \mid 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 2\}$
16. $\iint_R \frac{x}{1+xy} dA,\ R = \{(x,y) \mid 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 1\}$
17. $\iint_R xy^2 dA,\ D$ is the triangular region with vertices $(0,0)$, $(1,0)$, and $(1,1)$
18. $\iint_D \frac{y}{x^5+1} dA,\ D = \{(x,y) \mid 0 \leqslant x \leqslant 1, 0 \leqslant y \leqslant x^2\}$
19. $\iint_D x^3 dA,\ D = \{(x,y) \mid 1 \leqslant x \leqslant e, 0 \leqslant y \leqslant \ln x\}$
20. $\iint_D (x + y)dA,\ D$ is bounded by $y = \sqrt{x}$ and $y = x^2$
21. Find the volume of the solid that lies under the plane $3x + 2y + z = 12$ and above the rectangle $R = \{(x,y) \mid 0 \leqslant x \leqslant 1, -2 \leqslant y \leqslant 3\}$.
22. Find the volume of the solid that lies under the surface $z = 4 + x^2 - y^2$ and above the square $R = \{(x,y) \mid -1 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 2\}$.

23. Find the volume of the solid that lies under the surface $z = xy$ and above the triangle with vertices $(1,1), (4,1)$, and $(1,2)$.

24. Find the volume of the solid that is enclosed by the coordinate planes and the plane $3x + 2y + z = 6$.

25-27 Find the average value of $f$ over the given region.

25. $f(x,y) = x^2y,\ R$ is the rectangle with vertices $(-1,0), (-1,5), (1,5), (1,0)$

26. $f(x,y) = e^y\sqrt{x+e^y},\ R = \{(x,y) \mid 0 \leqslant x \leqslant 4, 0 \leqslant y \leqslant 1\}$

27. $f(x,y) = xy,\ D$ is the triangle with vertices $(0,0), (1,0)$, and $(1,3)$

28. A company models its monthly production by the function

$$\boldsymbol{P(x,y) = 200x^{3/4}y^{1/4}}$$

where $x$ is the number of workers and $y$ is the monthly operating budget in thousand of dollars. The company uses between 50 and 60 workers and its operating budget varies from $\$40{,}000$ to $\$50{,}000$ per month. Estimate the average monthly output.

29. The state of Colorado is in the shape of a rectangle that measures 388 miles west to east and 276 miles south to north. Suppose the function

$$\boldsymbol{f(x,y) = 4.6 + 0.02x - 0.01y + 0.0001xy}$$

approximates the snowfall, in inches, left during a storm at a location $x$ miles east and $y$ miles north of the southwest corner of the state. According to the model, what was the average snowfall for the entire state during the storm?

30. Researchers assessed the level of airborne pollution, in parts per million ( ppm), created by a manufacturing facility throughout a nearby rectangular plot of farmland. The boundaries of the land run 4 miles from east to west and 2 miles from south to north. The level of pollution, measured in parts per million (ppm), at a location $x$ miles west and $y$ miles north of the southeast corner of the plot was modeled by $f(x,y) = 2.7e^{-0.1x-0.4y}$. Use the model to find the average pollution level over the entire plot of land.

# Double Integrals over General Regions

1 $\iint_D f(x,y) + g(x,y)dA = \iint_D f(x,y)dA + \iint_D g(x,y)dA$

2 $\iint_D cf(x,y)dA = c\iint_D f(x,y)dA$, where $c$ is any constant.

3 If the region $D$ can be split into two separate regions $D_1$ and $D_2$ then the integral can be written as

$$\iint_D f(x,y)dA = \iint_{D_1} f(x,y)dA + \iint_{D_2} f(x,y)dA$$

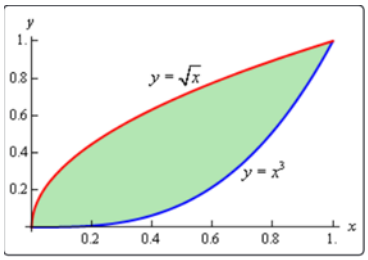
4 $V = \iint_D f(x,y)dA$

5 $Area\ of\ D = \iint_D dA$

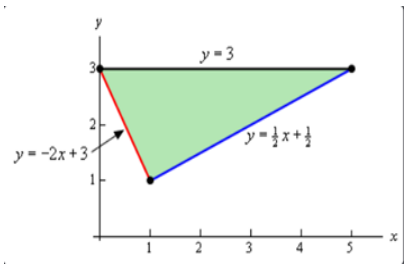
**Example 1 :** Evaluate each of the following integrals over the given region $D$.

(a) $\iint_D e^{\frac{x}{y}}\, dA\ , D = \{(x,y) \mid 1 \le y \le 2, y \le x \le y^3\}$

(b) $\iint_D (4xy - y^3)dA, D$ is the region bounded by $y = \sqrt{x}$ and $y = x^3$.

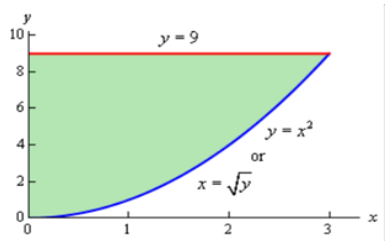



(c) $\iint_D (6x^2 - 40y)\, dA, D$ is the triangle with vertices $(0,3), (1,1)$, and $(5,3)$.

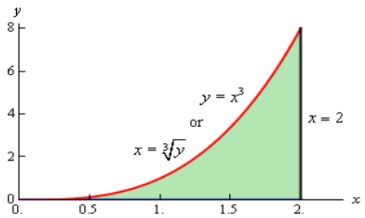



**Example 2:** Evaluate the following integrals by first reversing the order of integration.

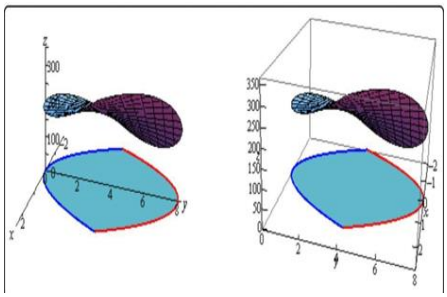
(a) $\int_0^3 \int_{x^2}^9 x^3 e^{y^3}\, dy\, dx$

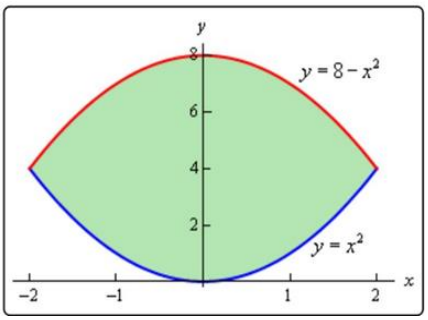



(b) $\int_0^8 \int_{\sqrt[3]{y}}^2 \sqrt{x^4 + 1}\, dxdy$

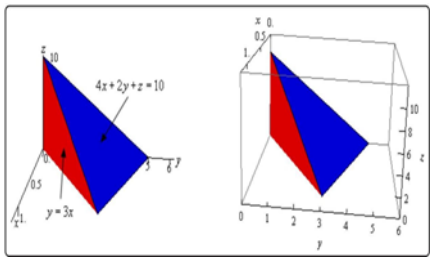


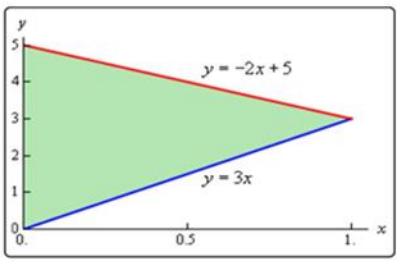
**Example 3:** Find the volume of the solid that lies below the surface given by $z = 16xy + 200$ and lies above the region in the $xy$-plane bounded by $y = x^2$ and $y = 8 - x^2$.






**Example 4:** Find the volume of the solid enclosed by the planes $4x + 2y + z = 10, y = 3x, z = 0, x = 0$.

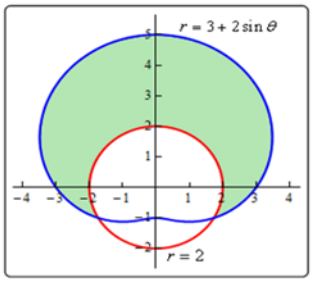


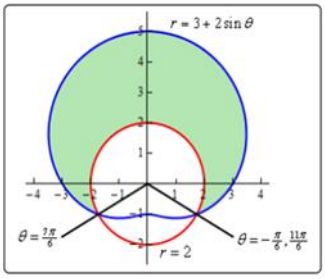




## Double Integrals in Polar Coordinates

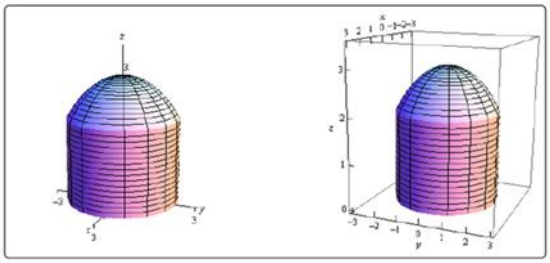
**Example 1:** Evaluate the following integrals by converting them into polar coordinates. (a) $\iint_D 2xydA, D$ is the portion of the region between the circles of radius 2 and radius 5 centered at the origin that lies in the first quadrant. (b) $\iint_D e^{x^2+y^2} dA, D$ is the unit disk centered at the origin.

**Example 2**: Determine the area of the region that lies inside $r = 3 + 2sin\,(\theta)$ and outside $r = 2$.

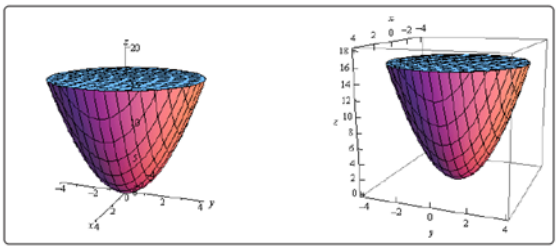


**Example 3:** Determine the volume of the region that lies under the sphere $x^2 + y^2 + z^2 = 9$, above the plane $z = 0$ and inside the cylinder $x^2 + y^2 = 5$.



**Example 4:** Find the volume of the region that lies inside $z = x^2 + y^2$ and below the plane $z = 16$.



**Example** **5**

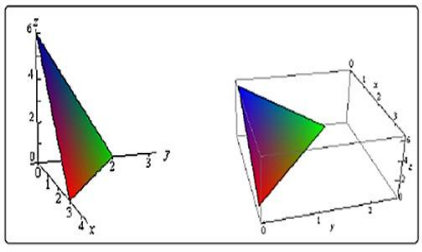
Evaluate the following integral by first converting to polar coordinates.

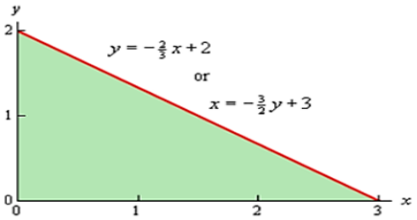
$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{0} cos\,(x^2 + y^2)\,dy\,dx$$

# Triple Integrals

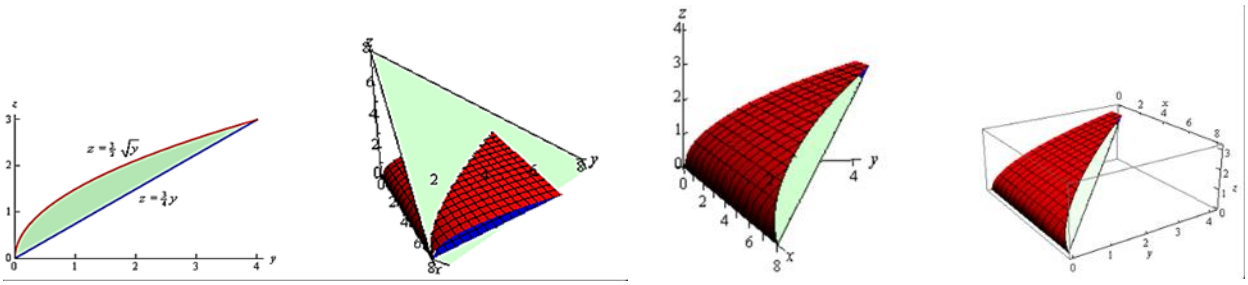
**Example 1**: Evaluate the integral $\iiint_B 8xyzdV\,, B = [2,3] \times [1,2] \times [0,1]$

**Example 2**: Evaluate $\iiint_E 2xdV$ where $E$ is the region under the plane $2x + 3y + z = 6$ that lies in the first octant.

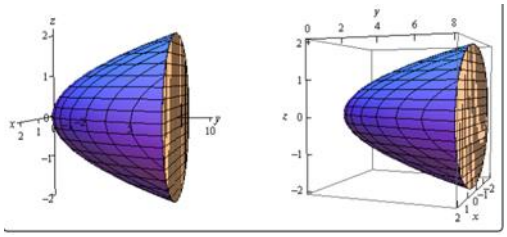





**Example 3:** Determine the volume of the region that lies behind the plane $x+y+z=8$ and in front of the region in the $yz$ -plane that is bounded by $z=\frac{3}{2}\sqrt{y}$ and $z=\frac{3}{4}y$.



**Example 4:** Evaluate $\iiint_E \sqrt{3x^2 + 3z^2}$ dV where $E$ is the solid bounded by $y=2x^2 + 2z^2$ and the plane $y=8$.



## Triple Integrals in Cylindrical Co- ordinates

**Example 1:** Evaluate $\iiint_E y\,dV$ where E is the region that lies below the plane z = x + 2 above the xy-plane and between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$.

**Example 2:** Convert $\int_{-1}^{1} \int_{0}^{\sqrt{1-y^2}} \int_{x^2+y^2}^{\sqrt{x^2+y^2}} (xyz)\,dzdydx$ into an integral in cylindrical coordinates.

## Triple Integrals in Spherical Coordinates

**Example 1:** Evaluate $\iiint_E 16z\,dV$ where $E$ is the upper half of the sphere $x^2+y^2+z^2=1$.

**Example 2:** Evaluate $\iiint_E zx\,dV$ where $E$ is inside both $x^2+y^2+z^2=4$ and the cone (pointing upward) that makes an angle of $\frac{\pi}{3}$ with the negative $z$ -axis and has $x \leq 0$.

**Example 3:** Convert $\int_{0}^{3} \int_{0}^{\sqrt{9-y^2}} \int_{\sqrt{x^2+y^2}}^{\sqrt{18-x^2-y^2}} (x^2 + y^2 + z^2)\,dzdydx$ into spherical coordinates.

## Introduction to Differential Equations:

1. Determine the order, unknown function, and the independent variable in each of the following differential equations:

(a) $y''' - 5xy' = e^x + 1$

$(b)\ t\ddot{y} + t^2\dot{y} - (\sin\ t)\sqrt{y} = t^2 - t + 1$

(c) $s^2\frac{d^2t}{ds^2} + st\frac{dt}{ds} = s$

(d) $5\left(\frac{d^4b}{dp^4}\right)^5 + 7\left(\frac{db}{dp}\right)^{10} + b^7 - b^5 = p$

2. Determine the order, unknown function, and the independent variable in each of the following differential equations:

(a) $y\frac{d^2x}{dy^2} = y^2 + 1$

(b) $y\left(\frac{dx}{dy}\right)^2 = x^2 + 1$

(c) $2\ddot{x} + 3\dot{x} - 5x = 0$

(d)$17y^{(4)} - t^6y^{(2)} - 4.2y^5 = 3\cos t$

3. Determine whether $y(x) = 2e^{-x} + xe^{-x}$ is a solution of $y'' + 2y' + y = 0$.

4. Show that $y = \frac{1}{x^2-1}$ is a solution of $y' + 2xy^2 = 0$ on I = (-1,1) but not on any larger interval containing I.

5. Write the differential equation $y(yy'1) = x$ in differential form.

6. Determine if the following differential equations are linear:

(a) $y' = (\sin x)y + e^x$ (b) $y' = x\sin y + e^x$ (c) y' = 5 (d) $y' = y^2 + x$

(e) $y' + xy^5 = 0$ (f) xy′ + y = $\sqrt{y}$ (g) $y' + xy = e^xy$ (h) $y' + \frac{x}{y} = 0$

**Solution: (6)**

(a) The equation is linear; here $(x) = -\sin x \text{ and } q(x) = e^x$ .

(b) The equation is not linear because of the term $\sin y$.

(c) The equation is linear; here p(x) = 0 and q(x) = 5.

(d) The equation is not linear because of the term $y^2$ .

(e) The equation is not linear because of the $y^5$ term.

(f) The equation is not linear because of the y1/2 term.

(g) The equation is linear. Rewrite it as $y' + (x - e^x)y = 0 \text{ with } p(x) = x - e^x \text{ and } q(x) = 0$.

(h) The equation is not linear because of the 1/y term

7. Determine if the following differential equations are homogeneous:

(a) $y' = \frac{y+x}{x}$ (b) $y' = \frac{y^2}{x}$ (c) $y' = \frac{2xye^{x}/y}{x^2 + y^2 \sin\frac{x}{y}}$ (d) $y' = \frac{x^2+y}{x^3}$

## Separable Variables :

1. Determine if the following differential equations are separable:

(*a*) $\sin x\, dx + y^2 dy = 0$ $(b)xy^2dx - x^2y^2dy = 0$ $(c)(1 + xy)dx + y\, dy = 0$

## Solution:

(a) The differential equation is separable; here $M(x,y) = A(x) = sin\, x\ and\ N(x,y) = B(y) = y^2$.

(*b*) The equation is not separable in its present form, since $M(x,y) = xy^2$ is not a function of $x$ alone. But if we divide both sides of the equation by $x^2y^2$, we obtain the equation $(1/x)dx + (_1)dy = 0$, which is separable. Here, $A(x) = \frac{1}{x} and\ B(y) = -1$.

(*c*) The equation is not separable, since $M(x,y) = 1 + xy$, which is not a function of $x$ alone.

2. $y' = \frac{xy^2}{x^2y+y^3}$; $xy^2dx - (x^2y + y^3)dy = 0$

3. $y' = 2xy + x$; $\left(2\, xye_x^{-x^2} + xe^{-x^2}\right)dx - e^{-x^2}\, dy = 0$

4. Solve $xdx - y^2dy = 0$.

5. Solve $\frac{dy}{dx} = \frac{x^2+2}{y}$.

6. Solve $\frac{dy}{dx} = x^2 - 2x + 2$.

7. Solve the differential equation of $y' = \frac{2y^4+x^4}{xy^3}$ by using Eqs. (4.9) and (4.10).

- $x = yu$ [4.9]
- $\frac{dy}{dx} = u + y\frac{du}{dy}$ [4.10]

8. Solve $y' = \frac{x^2+y^2}{xy}$

Solve the given differential equations or initial-value problems.

9. $(t^2 + 1)dt + (y^2 + y)dy = 0$.

10. $\sin x\, dx + y\, dy = 0$; $y(0) = -2$.

11. $y' = \frac{x^2y-y}{y+1}$; $y(3) = -1$.

# Exact Equation :

1. Determine whether the differential equation $2xydx + (1 + x^2)dy = 0$ is exact. This equation has the form of Eq. (5.l) with $M(x, y) = 2xy$ and $N(x, y) = 1 + x^2$. Since $\partial M/\partial y = \partial N/\partial x = 2x$, the differential equation is exact.

2. Determine whether the differential equation

$$(x + \sin\ y)dx + (x\cos\ y - 2y)dy = 0$$

is exact.
Here $M(x, y) = x + \sin\ y$ and $N(x, y) = x\cos\ y - 2y$. Thus, $\partial M/\partial y = \partial N/\partial x = \cos\ y$, and the differential equation is exact.

3. Solve $y' = \frac{2+ye^{xy}}{2y-xe^{xy}}$.

Rewriting this equation in differential form, we obtain

$$(2 + ye^{xy})dx + (xe^{xy} - 2y)dy = 0$$

Here, $M(x, y) = 2 + ye^{xy}$ and $N(x, y) = xe^{xy} - 2y$ and, since $\partial M/\partial y = \partial N/\partial x = e^{xy} + xye^{xy}$, the differential equation is exact. Substituting $M(x, y)$ into (5.4), we find $\partial g/\partial x = 2 + ye^{xy}$; then integrating with respect to $x$, we obtain

$$\int \frac{\partial g}{\partial x}dx = \int\ [2 + ye^{xy}]dx$$
$$g(x, y) = 2x + e^{xy} + h(y)$$

To find $h(y)$, first differentiate (I) with respect to $y$, obtaining $\partial g/\partial y = xe^{xy} + h'(y)$; then substitute this result along with $N(x, y)$ into (5.5) to obtain

$$xe^{xy} + h'(y) = xe^{xy} - 2y \text{ or } h'(y) = -2y$$

It follows that $h(y) = -y^2 + c_1$. Substituting this $h(y)$ into (l), we obtain

$$g(x, y) = 2x + e^{xy} - y^2 + c_1$$

The solution to the differential equation is given implicitly by (5.6) as

$$2x + e^{xy} - y^2 = c_2\ (c_2 = c - c_1)$$

5.20. Solve $y' = \frac{3yx^2}{x^3+2y^4}$

Rewriting this equation in differential form, we have

$$(3yx^2)dx + (-x^3 - 2y^4)dy = 0$$

which is not exact. Furthermore, no integrating factor is immediately apparent. We can, however, rearrange this equation as

$$x^2(3ydx - xdy) - 2y^4dy = 0$$

The group in parentheses is of the form $aydx + bxdy$, where $a = 3$ and $b = -1$, which has an integrating factor $x^2y^{-2}$. Since the expression in parentheses is already multiplied by $x^2$, we try an integrating factor of the form $I(x, y) = y^{-2}$. Multiplying ( $l$ ) by $y^{-2}$, we have

$$x^2y^{-2}(3ydx - xdy) - 2y^2dy = 0$$

which can be simplified (see Table 5-1) to

$$d(x^3y^{-1}) = 2y^2dy$$

Integrating both sides of (2), we obtain

$$x^3y^{-1} = \frac{2}{3}y^3 + c$$

as the solution in implicit form.

4. Convert $y' = 2xy - x$ into an exact differential equation. Rewriting this equation in differential form, we have

$$(-2xy + x)dx + dy = 0$$

Here $M(x, y) = -2xy + x$ and $N(x, y) = 1$. Since

$$\frac{\partial M}{\partial y} = -2x \text{ and } \frac{\partial N}{\partial x} = 0$$

are not equal, (I) is not exact. But

$$\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{(-2x) - (0)}{1} = -2x$$

is a function of $x$ alone. Using Eq. (5.8), we have $I(x, y) = e^{\int -2xdx} = e^{-x^2}$ as an integrating factor. Multiplying (l) by $e^{-x^2}$, we obtain

$$\left(-2xye^{-x^2} + xe^{-x^2}\right)dx + e^{-x^2}dy = 0$$

Which is exact.

5. $(\cos\ x + x\cos\ t)dt + (\sin\ t - t\sin\ x)dx = 0$

6. $(y + x^4y^2)dx + xdy = 0$

7. $2xydx + y^2dy = 0$

# Modelling with 1st and 2nd Order DEs :

1. A glass of lemonade with a temperature of 40◦F is left to sit in a room whose temperature is a constant 70◦F. Using a principle of physics called Newton's Law of Cooling, one can show that if the temperature of the lemonade reaches 52◦F in 1 hour, then the temperature T of the lemonade as a function of the elapsed time t is modeled by the equation

$$T = 70 = 30e^{-0.5t}$$

where T is in degrees Fahrenheit and t is in hours. The graph of this equation, shown in Figure 5.8.3, conforms to our everyday experience that the temperature of the lemonade gradually approaches the temperature of the room. Find the average temperature $T_{ave}$ of the lemonade over the first 5 hours.

2. A spring exerts a force of 5 N when stretched 1 m beyond its natural length.

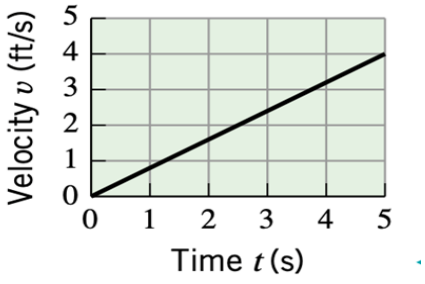
   (a) Find the spring constant $k$.

   (b) How much work is required to stretch the spring 1.8 m beyond its natural length?

3. A space probe of mass m = 5.00 × 104 kg travels in deep space subjected only to the force of its own engine. Starting at a time when the speed of the probe is v = 1.10 × 104 m/s, the engine is fired continuously over a distance of 2.50 × 106 m with a constant force of 4.00 × 105 N in the direction of motion. What is the final speed of the probe?

4. For the variable force $F(x)$ in Exercise 2, consider the distance $d$ for which the work done by the force on the particle when the particle moves from $x = 0$ to $x =$ d is half of the work done when the particle moves from $x = 0$ to $x = 5$. By inspecting the graph of $F$, is $d$ more or less than 2.5? Explain, and then find the exact value of d.

Suppose that a variable force F (x ) is applied in the pos- itive x-direction so that an object moves from x = a to x = b. Relate the work done by the force on the object and the average value of F o A constant force of 10 lb in the positive x-direction is applied to a particle whose velocity versus time curve is shown in the accompanying figure. Find the work done by the force on the particle from time t = 0 to t = 5.ver [a, b], and illustrate this relationship graphically.



5. A spring exerts a force of 6 N when it is stretched from its natural length of 4 m to a length of $4\frac{1}{2}$ m. Find the work required to stretch the spring from its natural length to a length of 6 m.

6. A spring exerts a force of 100 N when it is stretched 0.2 m beyond its natural length. How much work is required to stretch the spring 0.8 m beyond its natural length?

7. A spring whose natural length is 15 cm exerts a force of 45 N when stretched to a length of 20 cm.

   (a) Find the spring constant (in newtons/meter).

   (b) Find the work that is done in stretching the spring 3 cm beyond its natural length.

   (c) Find the work done in stretching the spring from a length of 20 cm to a length of 25 cm.

8. Assume that 10 ft·lb of work is required to stretch a spring 1 ft beyond its natural length. What is the spring constant?

9. A cylindrical tank of radius 5 ft and height 9 ft is two-thirds filled with water. Find the work required to pump all the water over the upper rim.

10. Solve Exercise 14 assuming that the tank is half-filled with water.

11. A cone-shaped water reservoir is 20 ft in diameter across the top and 15 ft deep. If the reservoir is filled to a depth of 10 ft, how much work is required to pump all the water to the top of the reservoir?

12. The vat shown in the accompanying figure contains water to a depth of 2 m. Find the work required to pump all the water to the top of the vat. [Use 9810 N/m3 as the weight density of water.]

13.The cylindrical tank shown in the accompanying figure is filled with a liquid weighing 50 lb/ft3. Find the work required to pump all the liquid to a level 1 ft above the top of the tank.

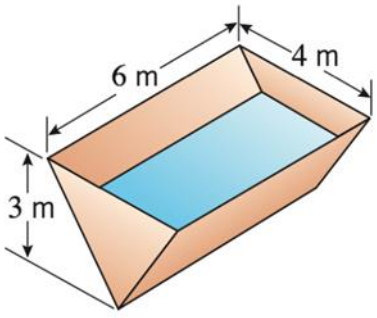



Figure Ex-17

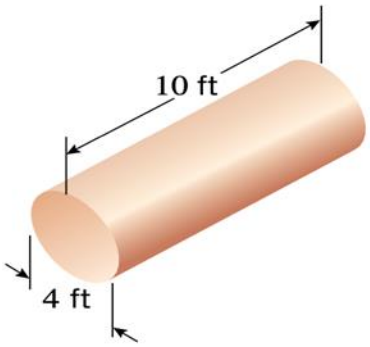



Figure Ex-18

14. A swimming pool is built in the shape of a rectangular par- allelepiped 10 ft deep, 15 ft wide, and 20 ft long.

(a) If the pool is filled to1 ft below the top, how much work is required to pump all the water into a drain at the top edge of the pool?
(b) A one-horsepower motor can do 550 ft·lb of work per second. What size motor is required to empty the pool in 1 hour?

15. How much work is required to fill the swimming pool in Exercise 19 to 1 ft below the top if the water is pumped in through an opening located at the bottom of the pool?

16. A 100 ft length of steel chain weighing 15 lb/ft is dangling from a pulley. How much work is required to wind the chain onto the pulley?

17. A 3 lb bucket containing 20 lb of water is hanging at the end of a 20 ft rope that weighs 4 oz/ft. The other end of the rope is attached to a pulley. How much work is required to wind the length of rope onto the pulley, assuming that the rope is wound onto the pulley at a rate of 2 ft/s and that as the bucket is being lifted, water leaks from the bucket at a rate of 0.5 lb/s?

18. A rocket weighing 3 tons is filled with 40 tons of liquid fuel. In the initial part of the flight, fuel is burned off at a constant rate of 2 tons per 1000 ft of vertical height. How much work in foot-tons (ft·ton) is done lifting the rocket 3000 ft?
19. It follows from Coulomb's law in physics that two like elec- trostatic charges repel each other with a force inversely proportional to the square of the distance between them. Suppose that two charges A and B repel with a force of k newtons when they are positioned at points A(−a, 0) and B(a, 0), where a is measured in meters. Find the work W required to move charge A along the x-axis to the origin if charge B remains stationary.
20. It follows from Newton's Law of Universal Gravitation that the gravitational force exerted by the Earth on an object above the Earth's surface varies inversely as the square of its distance from the Earth's center. Thus, an object's weight $w(x)$ is related to its distance $x$ from the Earth's center by a formula of the form
21. $w(x) = \frac{k}{x^2}$
22. where $k$ is a constant of proportionality that depends on the mass of the object.
(a) Use this fact and the assumption that the Earth is a sphere of radius 4000mi to obtain the formula for $w(x)$ in Example 4.
(b) Find a formula for the weight $w(x)$ of a satellite that is $x$ mi from the Earth's surface if its weight on Earth is 6000 lb.
(c) How much work is required to lift the satellite from the surface of the Earth to an orbital position that is 1000 mi high?
23. The formula $w(x) = k/x^2$ in Exercise 25 is applicable to all celestial bodies. Assuming that the Moon is a sphere of radius 1080mi, find the force that the Moon exerts on an astronaut who is $x$mi from the surface of the Moon if her weight on the Moon's surface is 20 lb.
(b) How much work is required to lift the astronaut to a point that is 10.8mi above the Moon's surface?

24. The world's first commercial high-speed magnetic levitation (MAGLEV) train, a 30 km double-track project connecting Shanghai, China, to Pudong International Airport, began full revenue service in 2003. Suppose that a MAGLEV train has a mass $m = 4.00 \times 10^5$ kg and that starting at a time when the train has a speed of $20\ \mathrm{m/s}$ the engine applies a force of $6.40 \times 10^5$ N in the direction of motion over a distance of $3.00 \times 10^3$ m. Use the work-energy relationship (6) to find the final speed of the train.
25. Assume that a Mars probe of mass $m = 2.00 \times 10^3$ kg is subjected only to the force of its own engine. Starting at a time when the speed of the probe is $v = 1.00 \times 10^4$ m/s, the engine is fired continuously over a distance of $1.50 \times 10^5$ m with a constant force of $2.00 \times 10^5$ N in the direction of motion. Use the work-energy relationship (6) to find the final speed of the probe.
26. On August 10, 1972 a meteorite with an estimated mass of $4 \times 10^6$ kg and an estimated speed of $15\ \mathrm{km/s}$ skipped across the atmosphere above the western United States and Canada but fortunately did not hit the Earth.